

Capital Markets Union and the Reallocation of Portfolio Capital in Europe: A Gravity-Based Simulation of Growth and Home Bias



Jesper Liedholm (176609)

Frans Nilsson (176413)

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Supervisor: Natalia Khorunzhina

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1 Abstract

Europe has substantial private savings, yet its capital markets remain fragmented along national lines. This thesis studies whether the Capital Markets Union can improve the allocation of European portfolio capital by reducing cross-border investment frictions. The analysis combines a financial gravity model with a counterfactual simulation of intra-EU capital market integration. Using bilateral portfolio investment data for 29 European countries, with 2019 as the base year, the thesis estimates how financial-development distance, informational distance, linguistic distance, and geographic distance shape cross-border portfolio holdings. The estimated gravity coefficients are then translated into bilateral friction wedges and embedded in a portfolio allocation model with endogenous return adjustment.

The Capital Markets Union is modelled as a reduction in selected intra-EU frictions. The results show that all four bilateral frictions are negatively associated with portfolio investment, supporting the interpretation that financial, informational, linguistic, and geographic barriers continue to shape capital allocation in Europe. In the counterfactual simulations, CMU reduces domestic portfolio shares and reallocates capital across European countries. Aggregate EU output rises across most of the policy space, with an average model-implied gain of 1.69 percent and a maximum gain of approximately 3.94 percent when both hard and informational frictions are fully reduced. However, the gains are unevenly distributed, with smaller and less capital-rich economies benefiting most in percentage terms, while some larger economies experience negative model-implied effects.

The central conclusion is that CMU operates primarily as a reallocation policy rather than a capital-creation policy. Its effectiveness depends not only on reducing regulatory and financial-development barriers, but also on lowering informational frictions that prevent capital from moving toward higher-value uses. The results should be interpreted as structured counterfactuals rather than forecasts, given the static nature of the model, the portfolio-capital mapping, and limitations in measuring informational frictions.

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2 Introduction

2.1 Motivation and Background

Europe has the capital to finance its own future. The problem is not a lack of savings, but that too much European capital remains trapped inside fragmented national markets. In a world of geopolitical instability, weak European productivity growth, and rising competition from the United States and China, this is no longer only a financial-market problem. It is a competitiveness problem for Europe’s future (Draghi, 2024). If Europe cannot move its own savings efficiently across borders, it risks underfunding the firms, technologies, infrastructure, and strategic investments needed to remain economically competitive and independent.

This diagnosis is central in Draghi (2024), which argues that Europe must mobilise private capital at a much larger scale to finance innovation, defence, the green transition, digital infrastructure, and long-run productivity growth. As Draghi puts it, “to unlock private capital, the EU must build a genuine Capital Markets Union (CMU)” (Draghi, 2024). This is also consistent with the European Commission’s CMU agenda, which emphasises that European savings should be channelled more effectively into productive investment and cross-border market financing (European Commission, 2015a; European Commission, 2020). The promise of CMU is therefore not that it creates new savings, but that it allows existing savings to move more freely to where they can be used best.

This thesis studies that promise quantitatively. It asks whether reducing cross-border portfolio frictions inside the EU can lower home bias, reallocate capital across member states, and raise European output within the model. This is directly relevant for European policy institutions because CMU is not only a legal or financial architecture project; it is a potential mechanism for turning Europe’s fragmented national savings pools into a deeper continental investment market.

2.2 Problem Statement and Research Gap

Despite decades of financial integration reforms, portfolio allocation in Europe remains strongly national. The key missing piece is a policy-relevant counterfactual that traces the full mechanism from bilateral frictions to portfolio allocation and model-implied output. Existing evidence shows that investors display

strong home bias and that cross-border frictions matter, but it does not show how much capital would move if specific frictions were reduced under CMU, which countries would gain or lose, or whether the resulting allocation would raise EU output within the model. It is also unclear whether formal financial-development barriers and informational barriers have the same economic effect. This thesis addresses that gap by estimating the frictions empirically, converting them into bilateral investment wedges, and simulating how separate reductions in hard and informational frictions change portfolio shares, simulated capital allocation, and model-implied output.

2.3 Research Question

The main research question is:

How would a CMU-style reduction in intra-EU portfolio frictions affect home bias, simulated cross-border capital allocation, and model-implied output across European countries?

To answer this, the thesis addresses three sub-questions:

We first examine which bilateral frictions are most strongly associated with observed portfolio holdings in the gravity estimation. We then analyse how a reduction in intra-EU frictions changes domestic portfolio shares and the simulated distribution of portfolio capital across countries. Finally, we compare how the hard-friction channel (ω) and the informational-friction channel (γ) differ in their aggregate and country-level model-implied output effects.

2.4 Approach and Scope

The empirical strategy combines four blocks. First, a gravity model is estimated on bilateral portfolio positions to recover friction coefficients for financial-development distance, informational distance, linguistic distance, and geographic distance. Second, these coefficients are transformed into bilateral wedge matrices. Third, CMU is simulated as proportional reductions in selected intra-EU wedges, and portfolio allocations are recomputed in general equilibrium. Fourth, simulated capital reallocation is mapped into output through a calibrated Cobb–Douglas production block.

The scope is intentionally focused. The analysis covers portfolio investment (not all capital flows), uses

2019 as the base year, and is implemented on a sample of EU countries plus three non-EU comparators. Results are interpreted as model-implied counterfactuals under stated assumptions, not as direct forecasts of realised macroeconomic outcomes. Portfolio allocation is simulated directly, while output effects are derived from the calibrated Cobb–Douglas production block.

2.5 Contributions and Main Findings

This thesis contributes by turning the CMU debate into a quantified capital-allocation problem with explicit friction channels. The framework separates policy-sensitive hard frictions from informational frictions and traces how each channel affects simulated allocation and model-implied output.

The main findings are fourfold. First, all four bilateral friction variables are negatively associated with portfolio holdings in the gravity model, supporting the friction-based interpretation of cross-border allocation. Second, CMU-style friction reductions lower home bias across EU countries and increase cross-border portfolio integration within the model. Third, aggregate model-implied EU output effects are positive in most scenarios (with a central outcome around 0.75 percent and a mean of 1.692 percent across the simulation grid), but the informational channel (γ) is the stronger model-implied output-generating lever relative to the hard channel (ω). Fourth, gains are unevenly distributed across countries: several smaller and less capital-rich economies gain strongly in percentage terms, while some larger economies experience weaker or negative model-implied effects.

2.6 Thesis Structure

Section 3 reviews the literature on home bias, EU financial fragmentation, CMU policy design, and financial gravity modelling. Section 4 presents the data sources, sample construction, variable definitions, and diagnostic checks. Section 5 develops the methodological framework, including gravity estimation, wedge construction, counterfactual design, and production mapping. Section 6 reports the simulation results on aggregate effects, channel decomposition, capital reallocation, country-level outcomes, and robustness. Section 7 interprets the findings in light of theory and policy, and discusses limitations. Section 8 concludes.

3 Literature Review

3.1 The Home Bias Puzzle: Origins and Persistence

In the existing body of literature in international finance, one of the most persistent anomalies is the home bias puzzle: the systematic tendency of investors to hold a far greater share of domestic equities in their portfolios than standard portfolio theory would predict. Solnik (1974) showed empirically that international equity portfolios could reduce risk to roughly half that of a well-diversified domestic portfolio of equivalent size. This logic follows directly from the mean-variance framework of Markowitz (1952) and its international extension by Grubel (1968): given that national business cycles and equity returns are imperfectly correlated across countries, investors should hold broadly diversified global portfolios roughly in proportion to each country's share of world market capitalisation. Yet the data reveal a striking departure from this benchmark. French and Poterba (1991) documented that most investors hold nearly all of their wealth in domestic assets, and showed that rationalising this behaviour required implying that investors in each nation expected returns in their domestic equity market to be several hundred basis points higher than returns in other markets. The puzzle, as they observed, appeared to be the result of investor choices rather than institutional constraints, a framing that has shaped the subsequent literature and that motivates further research.

Furthermore, French and Poterba (1991) documented that in the late 1980s American investors held approximately 94 percent of their equity wealth in domestic stocks, Japanese investors held around 98 percent domestically, and British investors around 82 percent. These weights are far larger than what mean-variance diversification fundamentals would imply, given each country's share of the world equity index. This finding was not an artifact of a single country or time period. Tesar and Werner (1995) extend the analysis to the G5 economies and show that home bias was pervasive. They also document an important paradox: although international equity holdings increased during the early 1990s, turnover rates on foreign holdings were substantially higher than those on domestic holdings. This suggests that the preference for domestic assets cannot be explained simply by passive inertia.

What makes home bias a genuine puzzle rather than a mere empirical regularity is its persistence despite the sharp decline in explicit barriers to international investment over the subsequent decades. The liber-

alisation of capital accounts, the growth of international mutual funds, the reduction of transaction costs through electronic trading, and the progressive dismantling of regulatory restrictions might all have been expected to close the gap between actual and optimal international diversification. Yet surveys of the evidence through the 2000s and 2010s consistently found that, while home bias has moderated somewhat, it remains quantitatively large. Lewis (1999), in an influential survey of the literature, demonstrated that the standard explanations rooted in observable barriers to trade, such as explicit transaction costs, withholding taxes, and foreign ownership restrictions, are quantitatively too small to account for the observed magnitude of domestic bias. Even generous estimates of these frictions could rationalise only a modest tilt towards home assets, not the near-complete concentration in domestic markets observed empirically.

The search for deeper explanations has generated a substantial body of theoretical and empirical work, which can be broadly organised around three competing channels. The first channel emphasises information asymmetry and familiarity. Investors possess better information about domestic firms than about comparable foreign firms, and this informational advantage may induce a rational preference for home assets (Gehrig, 1993; Brennan and Cao, 1997). This could include information about governance structures, competitive positions, and macroeconomic exposures. Related work by Huberman (2001) on investors' tendency to hold shares in their local electricity provider, and by Grinblatt and Keloharju (2001) on Finnish investors' preference for companies communicating in their native language, lends support to the notion that familiarity itself acts as a soft investment barrier. However, as Lewis (1999) and Coeurdacier and Rey (2013) point out, pure information-based models struggle to explain why familiarity biases are not competed away in equilibrium by informed intermediaries, and why they persist even among sophisticated institutional investors with extensive analytical resources.

The second channel invokes the hedging of non-traded income risk. Baxter and Jermann (1997) argued that since human capital returns are highly correlated with domestic equity returns, investors rationally bias their financial portfolios away from domestic assets to hedge their non-tradeable labour income. This argument, however, predicts the opposite of the observed pattern. Subsequent theoretical work by Pesenti and Wincoop (2002) and Obstfeld and Rogoff (2000) showed that non-tradeable goods and the hedging of real exchange rate risk can provide a partial rationale for domestic bias when goods markets are also segmented, but the quantitative magnitude of the effect remains insufficient to explain the full extent of

home bias, particularly in equity as opposed to bond markets.

The most comprehensive treatment of these competing explanations is provided by Coeurdacier and Rey (2013) in their survey of the home bias literature. They document that home bias in equities, while declining from its late-1980s peak as financial globalisation accelerated, remained substantial across developed economies well into the 2000s and showed signs of persistence even after controlling for improved data quality. Crucially, they argue that no single mechanism provides an adequate quantitative explanation on its own; the residual bias, once explicit frictions are accounted for, must be attributed to a combination of soft barriers, including information asymmetries, regulatory segmentation, investor familiarity, and institutional constraints. These are all harder to measure directly but can be proxied by bilateral distance, shared language, legal heritage, and cultural affinity. This insight is critical for the present analysis: if home bias were driven purely by transaction costs that financial integration has already reduced, we would expect bilateral portfolio allocations within the EU to converge to market-cap weights as integration proceeded. Instead, the evidence of persistent fragmentation points to structural impediments that resist convergence, motivating the use of a gravity framework to decompose their bilateral incidence.

Coeurdacier and Rey (2013) further document that home bias has evolved over distinct phases corresponding to major liberalisation episodes, including the wave of capital account liberalisations in the 1980s, the European Financial Services Action Plan period, the introduction of the euro, and the post-2008 financial crisis adjustment. This evolution implies that gravity estimates of bilateral friction are most reliable when applied to a sample period in which the underlying integration regime is stable, since pooling across structurally distinct phases would conflate friction levels with regime change. The empirical implementation in the thesis takes this implication seriously by anchoring the analysis on a base year that postdates the major regime breaks of the prior two decades and predates the macroeconomic dislocations of the post-pandemic period.

3.2 EU-Specific Fragmentation: Evidence and Patterns

The home bias documented in the preceding section is a global phenomenon, but the European Union presents a particularly instructive case. The Single Market was explicitly designed to eliminate barriers to the free movement of capital, and successive legislative programmes have removed most formal legal

impediments to cross-border investment within the bloc. If hard barriers were the primary driver of home bias, the EU should by now approximate an integrated capital market in which bilateral portfolio allocations reflect relative market size rather than national borders. The evidence does not support this reading. Bilateral equity positions within the EU remain well below the levels predicted by a frictionless benchmark, and the pattern of under-allocation is systematic rather than idiosyncratic, pointing to soft barriers that single market legislation has not reached.

Lane and Milesi-Ferretti (2008), drawing on IMF External Wealth of Nations data, document that while intra-EU portfolio investment expanded substantially through the early 2000s, a persistent bilateral gap remains between observed holdings and the CAPM-weighted benchmark. The expansion was partly mechanical and does not in itself constitute evidence of integration in the sense of barrier elimination. More telling is the cross-sectional dispersion: bilateral flows within the EU vary systematically with country-pair characteristics in ways that mirror the gravity patterns observed in goods trade, suggesting that the residual friction is bilateral in nature and not simply a property of individual markets.

Eurozone membership constitutes the most significant institutional step toward integration short of full political union, eliminating exchange rate risk, reducing transaction costs, and harmonising monetary conditions across member states. Coeurdacier and Martin (2009) show that euro adoption is associated with a meaningful increase in bilateral equity investment among member countries, consistent with the currency union removing one identifiable barrier. However, the effect is partial: even controlling for common currency, bilateral equity positions within the Eurozone fall substantially short of what frictionless portfolio theory would predict, and the euro effect does not extrapolate to non-Eurozone EU members. The implication is that monetary integration facilitates but does not complete financial integration; the residual gap reflects barriers that a common currency cannot dissolve.

A further dimension of the EU fragmentation pattern is its asymmetry across market size. Large economies, Germany and France in particular, are systematically over-represented in intra-EU equity portfolios relative to their market capitalisation shares, while peripheral economies such as Greece and Portugal are persistently underweighted. This asymmetry is not fully explained by differences in market liquidity, institutional quality, or macroeconomic volatility, though these variables correlate with it.

The Coeurdacier and Martin (2009) euro premium was estimated on pre-crisis data covering roughly 1997 to 2003. A subsequent literature documents that the within-Eurozone integration premium attenuated substantially after the 2008 financial crisis and the 2010 to 2012 sovereign debt episode. Lane (2013) documents the sharp reversal of intra-Eurozone capital flows during this period, with peripheral economies experiencing sustained outflows. Lane and Milesi-Ferretti (2018) extend this evidence using the updated External Wealth of Nations data and confirm that cross-border positions within the Eurozone fell substantially below their pre-crisis peak. The European Central Bank’s annual Financial Integration in Europe reports document a partial recovery from 2013 onward without a return to pre-crisis levels (European Central Bank, 2020). Read alongside Coeurdacier and Rey’s (2013) regime-stability argument cited in section 3.1, this evidence implies that the original Coeurdacier-Martin estimate captured a specific integration regime that intervening events partially dismantled, so post-2013 cross-sections recover an attenuated and structurally different parameter.

3.3 The Capital Markets Union Agenda

The Capital Markets Union (CMU) initiative, launched by the European Commission in 2015, has generated a substantial body of policy and academic literature examining both the rationale for deeper EU capital market integration and the realistic prospects for achieving it. This section reviews that literature along two axes: the policy design and objectives of CMU, and the empirical evidence on prior integration episodes that informs expectations about integration depth.

3.3.1 Policy design and scope

The foundational policy rationale for CMU is set out in the European Commission’s Green Paper (2015) and subsequent Action Plans (2015, 2020), which identify fragmentation in insolvency law, withholding tax regimes, post-trade infrastructure, and supervisory practice as the principal obstacles to cross-border capital allocation within the single market. Véron and Wolff (2016) offer an early and influential independent assessment, arguing that CMU should be understood not as a programme of hard legal unification but as an effort to achieve convergence across regulatory and supervisory practices that remain nationally fragmented despite a common legislative framework. A consistent feature of their treatment, reinforced

in the subsequent assessments by Sapir, Véron, and Wolff (2018) and Letta (2024), is that CMU operates through regulatory and institutional channels that are distinct from monetary union: the supervisory fragmentation it targets persists within the Eurozone, where currency union has already removed one bilateral friction. Their distinction between formal legal harmonisation and effective regulatory convergence has shaped much of the subsequent academic treatment. Lannoo and Thomadakis (2019), in a CEPS-ECMI Task Force report, develop this point further, arguing that CMU should be reframed as a “market finance action plan” and documenting persistent gaps in insolvency harmonisation, withholding tax procedures, and post-trade integration despite legislative progress on securitisation, covered bonds, and prospectus regulation. Constâncio (2017), in a speech as ECB Vice-President, makes a similar point from a supervisory perspective, arguing that without a single EU-level supervisor for capital markets, analogous to the Single Supervisory Mechanism in banking, divergent national implementation of formally common rules will continue to generate material cross-border frictions. The implication across this literature is that the distance between legislative output and effective integration is itself a central research problem.

A parallel strand of the policy literature examines CMU in comparison with historical precedents. The most directly relevant is the Financial Services Action Plan (FSAP), implemented between 1999 and 2005. Kalemli-Ozcan, Papaioannou, and Peydró (2010) show that the legislative harmonisation achieved under the FSAP had a statistically significant positive effect on cross-border banking flows, though with considerable heterogeneity across country pairs. For equity markets specifically, the evidence is more modest. Coeurdacier and Martin (2009) estimate that euro adoption, which simultaneously removed exchange rate risk and reduced transaction costs, increased bilateral equity positions among eurozone members, but that substantial unexplained bilateral gaps persisted. Their gravity-based decomposition suggests that the euro accounted for a meaningful but incomplete share of the friction reduction, leaving institutional distance and informational barriers largely unresolved. Balta and Delgado (2009) corroborate this pattern, documenting large cross-country differences in the degree of equity home bias across EU member states and finding that traditional integration policies had done little to reduce these disparities. Lane and Milesi-Ferretti (2008) provide broader evidence consistent with this pattern, showing that geographic and cultural proximity remain powerful predictors of bilateral equity positions even within highly integrated economic blocs.

3.3.2 Empirical expectations for integration depth

The question of what level of friction reduction CMU can realistically achieve has received direct attention in several policy assessments. Sapir, Véron, and Wolff (2018), in a Bruegel policy contribution, characterise CMU as a long-term structural endeavour and argue that effective integration requires not only legislative harmonisation but institutional reform, particularly of the European Securities and Markets Authority (ESMA) and the supervisory architecture, that has repeatedly been blocked by political constraints. Constâncio (2017) frames the issue in terms of supervisory convergence specifically, arguing that the scope for regulatory arbitrage and national divergence will remain a binding constraint on integration depth as long as capital markets supervision stays fragmented across national authorities. Bhatia et al. (2019) provide the most comprehensive recent assessment and conclude that while legislative momentum has been sustained, the gap between formal harmonisation and effective integration remains large, with supervisory convergence lagging particularly far behind. Their analysis supports treating full convergence as a long-run theoretical ceiling rather than a plausible near-term outcome. The Letta report (Letta, 2024) reinforces this conclusion from a political economy perspective, arguing that the fragmentation of national capital markets is sustained by deeply embedded institutional structures, including pension system design, tax treatment of savings, and corporate governance conventions, that formal regulation can influence only slowly.

A consistent implication of this literature is that regulatory and institutional barriers, on the one hand, and informational and cultural barriers, on the other, are unlikely to respond to policy intervention at the same rate or to the same degree. This motivates a two-dimensional scenario structure. Taken together, the literature establishes two points that are directly consequential for the thesis. First, CMU targets precisely the class of bilateral frictions, institutional distance, regulatory heterogeneity, and informational opacity, that a gravity decomposition is designed to quantify. Second, the empirical record of prior integration episodes supports intermediate rather than full convergence as the policy-relevant benchmark.

3.4 Gravity Models of Bilateral Capital Flows

The gravity equation is the most empirically successful framework in the study of bilateral economic flows. Its application to international finance, and to bilateral equity positions specifically, provides the

methodological foundation for the thesis. This section reviews the development of the gravity model from its origins in trade theory to its current status as the standard framework for modelling cross-border portfolio investment.

The gravity equation entered economics as an empirical regularity. Tinbergen (1962) observed that bilateral trade flows between two countries are well described by a function increasing in both countries' economic size and decreasing in the geographic distance between them, in direct analogy to Newton's law of gravitation. For several decades the equation was regarded as an empirical workhorse without clear theoretical foundations, a status that limited its use in counterfactual and welfare analysis. J. E. Anderson (1979) provided the first formal derivation from a general equilibrium model of national product differentiation with CES preferences, though the contribution was largely overlooked at the time. Bergstrand (1985) offered an alternative microfoundation through a monopolistic competition framework, and Deardorff (1998) showed that gravity-like predictions emerge from a range of standard trade models, establishing that the equation's empirical success was not an accident of functional form but a robust implication of theory.

The theoretical standing of the gravity equation was transformed by J. E. Anderson and Wincoop (2003), whose structural gravity model introduced the concept of multilateral resistance. Their key result is that bilateral trade depends not on the absolute level of bilateral frictions but on those frictions relative to each country's average friction with all trading partners. A country pair with high bilateral costs may nonetheless trade intensively if both countries face even higher barriers with the rest of the world. The implication for empirical work is fundamental: omitting multilateral resistance terms from a gravity regression biases the estimated coefficients on bilateral variables, because part of the observed bilateral variation reflects differences in average openness rather than pair-specific barriers. J. E. Anderson and Wincoop (2003) demonstrated that controlling for multilateral resistance, either structurally or through origin and destination fixed effects, substantially changes the estimated magnitude of border effects. Head and Mayer (2014), in a comprehensive handbook chapter consolidating two decades of development, establish the structural gravity equation with appropriate fixed effects as the consensus specification in both trade and financial applications. The multilateral resistance insight is equally consequential for financial gravity: a country that is generally open to foreign investment will, all else equal, allocate a

smaller share of its portfolio to any single bilateral partner than a country that is generally closed, and failing to account for this will misattribute country-level openness to bilateral-level friction.

The extension of gravity to financial flows was pioneered by Portes and Rey (2005), who estimated a gravity equation for gross bilateral equity flows across fourteen countries over the period 1989 to 1996. Their central finding is that geographic distance is a strong and robust predictor of cross-border equity transactions, even after controlling for market size, transaction costs, and institutional variables. This result was initially surprising: unlike goods, financial assets are weightless, and the cost of executing a trade in Tokyo from London is essentially the same as executing one in Paris. Portes and Rey (2005) argued that distance in the financial context proxies not for shipping costs but for informational frictions: the cost of acquiring, processing, and monitoring information about foreign assets declines with proximity, and investors systematically underweight distant markets whose firms they find harder to evaluate. They supported this interpretation by showing that bilateral information flow proxies, specifically telephone call traffic, had significant explanatory power for equity flows and partially displaced the distance coefficient when included. Portes, Rey, and Oh (2001), in the working paper that preceded the 2005 publication, had already documented that information variables dominate traditional transaction cost measures in explaining bilateral equity flows, and that the information channel operates through both the extensive margin (whether two countries exchange equity at all) and the intensive margin (the volume conditional on positive flows). The two papers together established that the gravity framework captures a substantive economic mechanism in finance: bilateral information asymmetry as the primary friction governing cross-border equity allocation.

This finding has been replicated and extended across different samples, time periods, and asset classes. Aviat and Coeurdacier (2007) show that bilateral goods trade and bilateral financial flows are jointly determined, with trade in goods creating informational spillovers that facilitate cross-border investment, and financial linkages in turn reducing the cost of trade finance. Their results imply that treating goods gravity and financial gravity as independent systems understates the role of information in both and that the two forms of bilateral integration are complements rather than substitutes. Lane and Milesi-Ferretti (2008), using the IMF's Portfolio Investment Positions (PIP), estimate a gravity model for bilateral equity holdings rather than flows and confirm that the same variables, distance, common language, colonial ties,

and bilateral trade intensity, are significant determinants of portfolio positions. Their use of holdings data rather than flow data is methodologically important: holdings capture the stock of bilateral investment at a point in time and are less subject to the timing noise and round-tripping that can contaminate flow data. They also document that the gravity structure holds not only across the full global sample but within subsets of highly integrated economies, including EU member states, establishing that financial gravity is not simply a developing-versus-developed-country phenomenon but operates within the most institutionally advanced economic blocs.

A number of features distinguish financial gravity from its trade counterpart and carry implications for how the framework should be applied to portfolio allocation. The first distinction concerns the nature of the assets being modelled. Unlike traded goods, financial assets are not consumed at their destination. An equity claim on a German firm held by a French investor does not physically move; what moves is the claim to a stream of future returns, along with associated governance rights and information requirements. The frictions captured by a financial gravity model are therefore informational and institutional rather than logistical: the cost of evaluating foreign corporate disclosures, monitoring foreign management, navigating unfamiliar legal systems for shareholder protection, and bearing the residual uncertainty of operating outside one's home institutional environment. Okawa and Wincoop (2012) formalise this point in a theoretical model of bilateral asset trade in which information acquisition costs generate gravity-like predictions where distance proxies for the cost of learning about foreign assets rather than transporting them. Their framework provides a structural interpretation for the empirical finding, consistent across Portes and Rey (2005), Lane and Milesi-Ferretti (2008), and Coeurdacier and Martin (2009), that information proxies tend to dominate traditional cost measures in financial gravity regressions.

A second distinction concerns the measurement of country economic size. In trade gravity, GDP is the standard size variable on both the origin and destination sides. In financial gravity, where the underlying allocation problem concerns financial wealth and equity market capitalisation, market capitalisation might in principle be the more direct measure of destination size. In practice, however, the empirical literature has consistently used GDP as the dominant measure of country economic size in financial gravity applications (Portes and Rey, 2005; Lane and Milesi-Ferretti, 2008; Coeurdacier and Martin, 2009). This

choice reflects two considerations. First, GDP has broader and more consistent country coverage than market capitalisation, particularly for smaller economies and for time periods in which equity market data are sparse. Second, GDP and market capitalisation are highly correlated across the advanced-economy samples for which portfolio gravity work is typically conducted, so that the choice of size measure has limited practical effect on the estimated bilateral friction coefficients.

Taken together, the trade-to-finance gravity literature establishes two results that are foundational for the present analysis. First, bilateral equity positions are well described by a gravity structure in which economic size attracts investment and bilateral frictions repel it, with informational distance playing the role that transport costs play in goods trade (Portes and Rey, 2005; Lane and Milesi-Ferretti, 2008). Second, the structural gravity framework requires controlling for multilateral resistance to obtain unbiased friction estimates (J. E. Anderson and Wincoop, 2003; Head and Mayer, 2014), with the appropriate strategy depending on the research question. Third, within-EU equity allocation exhibits the same gravity regularities as global allocation, with informational and institutional barriers persisting even after currency union, which means a gravity-based decomposition can credibly identify the channels through which further integration would operate (Coeurdacier and Martin, 2009).

3.5 Estimation: Poisson Pseudo Maximum Likelihood and Sample Design

The gravity equation as derived by J. E. Anderson and Wincoop (2003) is a multiplicative model: bilateral flows are an exponential function of bilateral frictions and multilateral resistance terms. Early empirical implementations, following Tinbergen (1962), estimated the model by taking logarithms and applying OLS. Santos Silva and Tenreyro (2006) demonstrate that this log-linear approach introduces two distinct and serious biases when applied to bilateral financial data.

The first problem concerns zero bilateral flows. Portfolio investment data, whether from the IMF’s Coordinated Portfolio Investment Survey or national sources, contain a substantial number of country pairs for which no bilateral holdings are reported. Log-linearisation drops these observations entirely, since the logarithm of zero is undefined. This is not an innocuous data-cleaning step. If the zeros are not randomly distributed but instead concentrate among country pairs with high bilateral frictions, small markets, or weak institutional linkages, precisely the pairs that carry the most information about the fric-

tion structure, their exclusion introduces a systematic selection bias that attenuates the estimated friction coefficients. In a European sample, this is particularly relevant for bilateral positions involving smaller peripheral economies, which are exactly the countries for which the CMU question is most consequential.

The second problem is a consequence of Jensen’s inequality. Under heteroskedasticity, which is pervasive in bilateral financial data where a few large country pairs dominate the distribution, the expected value of the log-transformed dependent variable does not equal the log of the expected value: $E[\ln y] \neq \ln E[y]$. Santos Silva and Tenreyro (2006) show that this inequality means the log-linear OLS estimator is inconsistent whenever the variance of the error term is correlated with the regressors, a condition that is virtually guaranteed in gravity data where larger and closer country pairs exhibit both higher mean flows and different variance structures than smaller and more distant pairs.

The solution proposed by Santos Silva and Tenreyro (2006) is the Poisson Pseudo Maximum Likelihood (PPML) estimator, which estimates the multiplicative gravity model directly in levels rather than logs. The estimator requires only that the conditional mean is correctly specified, that is, $E[y | X] = \exp(X\beta)$, and does not require that the data follow a Poisson distribution. It retains zero observations naturally, since the exponential function maps any linear index to a positive predicted value, and remains consistent under heteroskedasticity of arbitrary form. Santos Silva and Tenreyro (2011) provide Monte Carlo evidence confirming that PPML performs well across a range of data-generating processes, including those with large shares of zeros and heavy-tailed distributions, both of which characterise bilateral portfolio data. Head and Mayer (2014) consolidate the case for PPML in their handbook treatment and identify it as the current standard for gravity estimation in both trade and finance.

A related estimation issue concerns the treatment of multilateral resistance. As J. E. Anderson and Wincoop (2003) showed, bilateral frictions must be measured relative to each country’s average friction with all partners. In cross-sectional gravity regressions, the standard approach is to include origin and destination fixed effects, which absorb all country-specific variation, including multilateral resistance, market size, and institutional quality, and isolate the bilateral friction component. In panel settings, the analogous approach uses origin-time and destination-time fixed effects. However, the inclusion of comprehensive fixed effects comes at a cost: any bilateral friction variable that is collinear with the fixed

effects structure cannot be separately identified. This creates a tension in applications where the research objective is to estimate the effect of country-level characteristics, such as financial market development or regulatory quality, on bilateral flows. If these variables enter as origin or destination attributes, they are absorbed by the fixed effects; if they enter as bilateral differences, they can be identified but only to the extent that they vary across country pairs independently of the fixed effects. The empirical literature has handled this trade-off in different ways depending on the research question. Portes and Rey (2005) use origin and destination fixed effects and identify friction from purely bilateral variables such as distance and information proxies. Coeurdacier and Martin (2009) adopt a similar approach for their European sample. Studies that aim to identify the role of country-level institutions in shaping bilateral flows, such as Papaioannou (2009) sacrifice full fixed effects in order to preserve identification of the country-level variables of interest, accepting the potential for multilateral resistance bias in exchange.

The choice between these two approaches is fundamentally tied to the research question. When the empirical objective is to estimate clean bilateral friction coefficients in isolation, full origin and destination fixed effects are appropriate. When the objective is to quantify the role of country-level institutional variables, including measures of financial market development that vary at the country rather than the bilateral level, the approach of substituting GDP and other country-level controls for the structural fixed effects is the standard alternative, as exemplified by Papaioannou (2009) in the bilateral financial-flows context. The present analysis adopts this latter specification because the financial development measure that enters the wedge construction is a country-level characteristic expressed bilaterally as an absolute difference between the financial development indices of the two countries, and absorbing all country-level variation into fixed effects would prevent its identification. The methodology section discusses the implications of this trade-off explicitly and presents the resulting specification as a deliberate choice along the lines of the Papaioannou (2009) tradition rather than as an oversight relative to the Portes and Rey (2005) or Coeurdacier and Martin (2009) specifications.

The choice of fixed effects structure also interacts with sample design. In a sample of n countries, origin and destination fixed effects consume $2n - 1$ degrees of freedom, and the bilateral friction coefficients are identified from the within-country-pair variation that remains. For a sample of 29 countries, the panel yields 812 directed bilateral pairs (excluding the diagonal), which provides ample variation for identifying a

parsimonious set of friction variables. Restricting the estimation sample to a relatively homogeneous set of advanced European economies, augmented with the United Kingdom, Switzerland, and Norway as outside options, has the cost of narrowing the variation in the friction measures relative to a global sample but has the analytical advantage of identifying friction coefficients from the precise distribution of bilateral characteristics most relevant for the CMU counterfactual. Coefficients estimated on a global sample including emerging economies would absorb friction structure governing flows to and from countries with very different institutional and informational environments, and may not provide the most appropriate calibration for a within-EU policy experiment. The inclusion of the three non-EU comparators expands the variation in linguistic and financial development distance beyond what is observed within the EU alone, and provides a benchmark against which intra-EU frictions can be assessed without diluting the policy-relevant identification with structurally distinct emerging-market observations.

3.6 Bilateral Frictions and Their Empirical Measurement

The gravity framework establishes that bilateral equity positions are shaped by frictions beyond explicit transaction costs. A large empirical literature has investigated which specific channels drive this bilateral variation, and a related strand has developed standardised measurement approaches and data sources for each channel. This section reviews the evidence and the measurement standards for the four bilateral friction channels operationalised in the methodology, and concludes by formalising the hard-versus-soft taxonomy that organises the policy counterfactual.

3.6.1 Geographic distance

Geographic distance is the most consistently significant predictor of bilateral equity positions across samples and specifications. Portes and Rey (2005) estimate a distance elasticity of bilateral equity flows of approximately -0.5 to -0.8 , comparable to goods trade despite the near-zero marginal cost of executing cross-border financial transactions. The standard interpretation is that distance proxies for information acquisition and monitoring costs rather than transport (Okawa and Wincoop, 2012). Lane and Milesi-Ferretti (2008) confirm that the coefficient remains significant even within Europe, where physical distances are modest, implying that informational decay with distance is a material friction even within

a highly integrated bloc.

The standard data source for bilateral geographic distance in gravity applications is the CEPII GeoDist database (Mayer and Zignago, 2011), which provides multiple distance measures including population-weighted bilateral distances between primary urban centres and simple great-circle distances between capital cities. The CEPII measures have been adopted as the de facto standard in both trade and financial gravity work because of their consistent country coverage and methodological transparency.

3.6.2 Linguistic proximity

Linguistic proximity is closely related to both the information and cultural channels but empirically distinct. Melitz and Toubal (2014) construct a multidimensional measure encompassing direct communication (shared native language), linguistic similarity between distinct languages, and translation capacity, and show that all three dimensions significantly affect bilateral flows. In the financial context, common language lowers both the cost of processing corporate disclosures and the risk of legal misinterpretation. Lane and Milesi-Ferretti (2008) and Coeurdacier and Martin (2009) confirm the result within Europe.

Two main approaches to measuring linguistic distance have emerged in the empirical literature. The first uses dummy variables for shared official languages or shared language families, which has the virtue of simplicity but treats all language pairs in the same family as equivalent and assigns zero proximity to all pairs without a shared language regardless of their actual linguistic similarity. The second, which has gained ground in more recent gravity work, uses lexicostatistical distance measures derived from comparative linguistics. The Automated Similarity Judgment Program (ASJP), developed by Bakker, Müller, Velupillai, Wichmann and colleagues (Bakker et al., 2009), provides bilateral lexical distance scores derived from systematic comparison of standardised word lists across thousands of languages. ASJP-based linguistic distance offers a continuous measure that can distinguish, for example, Dutch and German (closely related within the West Germanic family) from German and French (related but more distant Indo-European languages) and from German and Finnish (unrelated, separated by the Indo-European-Uralic boundary). For a European sample with substantial linguistic heterogeneity that is poorly captured by official-language indicators, the continuous ASJP measure has stronger discriminating power than dummy variables. Linguistic proximity captures a dimension of soft friction that regulation

cannot eliminate, since it is a property of the country pair rather than of the regulatory environment.

3.6.3 Information asymmetry and direct information proxies

Information asymmetry, measured directly through bilateral information flow indicators, has been examined since the earliest financial gravity work. Portes, Rey, and Oh (2001) show that telephone call traffic is a strong predictor of bilateral equity flows, with an elasticity comparable to the distance effect, and that it partially displaces the distance coefficient when included. Their interpretation is that the predictive power of geographic distance reflects, at least in part, the underlying decay of information flow with distance, and that direct measures of bilateral information transmission can substitute for distance as proxies for the informational channel.

Subsequent empirical work has extended this approach using digital information flow proxies. Mondria, Wu, and Zhang (2010), using bilateral internet search query data, find that attention-based information measures predict bilateral equity holdings even after controlling for distance and standard gravity covariates, with effects that are quantitatively significant and theoretically consistent with rational inattention models of portfolio choice. The general finding across this work is that direct information proxies support the interpretation that distance matters for financial flows primarily through information rather than any direct cost channel.

The methodological implication is that bilateral measures of information flow, including telephone traffic, search queries, news coverage intensity, and analyst coverage, can be used in place of, or alongside, geographic distance to capture the informational dimension of bilateral friction. Bilateral news coverage in particular has the appealing property that it varies over time at the country-pair level and can therefore be exploited in panel specifications, in contrast to time-invariant measures such as geographic distance.

However, news-based information distance carries one methodological disadvantage: bilateral news coverage is partly endogenous to cross-border investment activity, since greater investment volumes generate more financial reporting. This bidirectional relationship complicates causal interpretation of any estimated news-distance coefficient. It does not, however, disqualify news distance as the operational proxy for the informational dimension of bilateral friction. The complementary measures that the CMU agenda

directs at this channel, including the European Single Access Point for issuer disclosures, consolidated tape provision, and harmonised reporting standards, operate on exactly the informational environment that bilateral news coverage indexes. A reduction in news distance in the policy counterfactual therefore approximates the effect of these complementary policies, distinct from changes in hard regulatory frictions.

3.6.4 Financial development distance

Regulatory barriers and financial market development constitute the hard friction channel most directly targeted by policy. Bekaert, Harvey, and Lundblad (2005) find that equity market liberalisation episodes are associated with significant increases in portfolio inflows and GDP growth, though the magnitude varies with institutional quality. Within the EU, where formal capital account restrictions have been eliminated, the relevant friction is not the presence of legal barriers but the degree of regulatory divergence across member states in insolvency regimes, withholding tax procedures, supervisory practice, and the depth and accessibility of capital markets. Bekaert, Harvey, and Lundblad (2007) show that the elasticity of capital flows to financial openness is heterogeneous, with stronger effects where investor protection and disclosure are better, suggesting that regulatory harmonisation and institutional quality are complements rather than substitutes.

Operationalising the hard friction channel for empirical work requires a measure of financial development that captures the depth, accessibility, and stability of national financial systems beyond the binary liberalised-versus-restricted classification used in earlier work. Sahay et al. (2015) develop a comprehensive multi-dimensional framework for measuring financial development that distinguishes between financial institutions and financial markets, and within each category captures depth, access, and efficiency on a consistent basis across countries. Svirydzenka (2016a) extends this framework to construct the IMF Financial Development Index, which provides a single summary measure aggregated from the underlying sub-indices and covers a broad cross-section of countries on a consistent annual basis. The bilateral analogue, the absolute difference in financial development index scores between two countries, is a natural measure of institutional distance and operates as the hard friction component of the bilateral wedge in gravity applications. Differences in financial development reflect precisely the kind of institu-

tional gaps that regulatory harmonisation under CMU is designed to narrow, making the index a credible operationalisation of the policy-amenable component of bilateral friction.

3.6.5 The hard, policy-sensitive soft, and immutable structural taxonomy

Across these channels, the literature points to a consistent hierarchy for EU samples. Informational distance, captured by geographic distance, language, and direct information proxies, tends to have robust and quantitatively significant effects on bilateral equity positions. Regulatory and institutional divergence, captured by financial development distance, is also material but harder to measure precisely within the EU, where it takes the form of implementation differences rather than legal barriers. Linguistic and geographic frictions overlap with the informational channel but are conceptually and empirically distinct.

This hierarchy supports a tripartite classification of bilateral frictions for policy counterfactual analysis. The first category comprises hard frictions, defined as institutional and regulatory differences that policy intervention can target directly. Financial development distance is the canonical example: differences in regulatory frameworks, supervisory practice, and the depth and accessibility of financial systems are precisely the gaps that the CMU agenda aims to narrow through harmonised legislation and supervisory convergence (Véron and Wolff, 2016). The second category comprises policy-sensitive soft frictions, defined as informational barriers that are not directly targeted by policy but respond endogenously to the level of bilateral economic integration. Bilateral information flows, including news coverage and analyst attention, fall into this category: they are not policy levers in themselves, but greater financial integration generates greater bilateral attention and information production as a secondary effect, a complementarity that Véron and Wolff (2016) emphasise in their assessment of CMU. The third category comprises immutable structural frictions, defined as bilateral characteristics that are largely invariant to policy intervention. Geographic distance and linguistic proximity belong here: physical distance is fixed, and language differences evolve over horizons far longer than any plausible policy intervention.

The total welfare effect of CMU therefore depends on the composition of bilateral friction across these three categories. Policy intervention can reduce the hard component directly and the policy-sensitive soft component endogenously, but cannot affect the immutable structural component.

3.7 Constructing Friction Wedges from Regression Coefficients

A separate question, addressed by a distinct strand of the literature, is how to aggregate these estimates into a single summary measure of bilateral friction suitable for counterfactual analysis. The concept of a bilateral friction wedge, a scalar Δ_{ij} that captures the total cost of investing across a given country pair, has its roots in the trade cost literature and has been progressively adapted to financial flows.

J. E. Anderson and Wincoop (2004), in their survey of trade costs published in the *Journal of Economic Literature*, formalise the iceberg cost framework in which bilateral trade is discounted by a multiplicative friction factor. They estimate that total trade costs for rich countries carry an ad valorem equivalent of approximately 170 percent and document substantial heterogeneity across country pairs. The key analytical property of their formulation is separability: because friction variables enter the gravity equation additively inside a log-linear specification, total bilateral cost decomposes into the product of its constituent components. A tariff reduction can therefore be simulated by scaling the tariff component of the wedge while holding transport and information components fixed. This multiplicative structure is what makes the iceberg framework useful for policy counterfactuals, and it carries directly into financial applications where the relevant frictions are informational and regulatory rather than logistical.

This decomposability has a direct analytical consequence for the parameterisation of policy counterfactuals. If a friction component enters the bilateral wedge multiplicatively as Δ_{ij}^c for component c , then a proportional reduction in that component can be represented as a power transformation of the form $\Delta_{ij}^{c,\text{policy}} = \left(\Delta_{ij}^c\right)^{1-\phi}$, where $\phi \in [0, 1]$ governs the share of the friction that is removed. Setting $\phi = 0$ leaves the wedge unchanged, $\phi = 1$ removes the friction entirely by setting the relevant component to unity, and intermediate values parameterise partial elimination. This functional form preserves positivity, maps cleanly into the iceberg cost structure, and allows separate policy parameters to govern reductions in different friction components without altering the multiplicative structure of the overall wedge.

Novy (2013) takes a different approach. Rather than constructing the wedge from estimated coefficients on observable distance measures, he inverts the structural gravity equation of J. E. Anderson and Wincoop (2003) to back out bilateral trade costs as a function of observed bilateral and domestic trade flows. The resulting measure is a comprehensive “gravity residual” that captures all bilateral frictions, observed

and unobserved, as a single scalar. Novy (2013) shows that the measure is consistent with Ricardian, monopolistic competition, and heterogeneous-firm models, and applies it to demonstrate that U.S. bilateral trade costs declined by roughly 40 percent between 1970 and 2000. The strength of this approach is its comprehensiveness; its limitation is that, precisely because it is a residual, it cannot decompose total friction into constituent channels. For applications where the research question concerns the effect of reducing a specific barrier type, as in the CMU context, the coefficient-based construction is more informative.

The extension to financial flows is provided by Okawa and Wincoop (2012), who derive a structural gravity equation for bilateral asset trade in which information acquisition costs play the role of transport costs. In their model, bilateral portfolio positions are discounted by a friction that enters multiplicatively, exactly as iceberg costs enter the trade gravity equation, generating a wedge that reduces the attractiveness of each foreign destination relative to the frictionless benchmark.

A challenge specific to financial applications concerns the diagonal element of the wedge matrix. In trade, domestic transactions are directly observable and anchor the cost measure, as in Novy (2013). In portfolio investment, domestic equity holdings are not reported in the PIP and must be inferred residually from market capitalisation and reported foreign liabilities (Lane and Milesi-Ferretti, 2008). There is no bilateral friction between a country and itself, yet the home bias literature reviewed in Section 3.1 establishes that observed domestic portfolio shares far exceed what any configuration of off-diagonal frictions would predict. Coeurdacier and Rey (2013) confirm that this residual remains quantitatively large even in the most comprehensive gravity specifications. Any gravity-based portfolio model must therefore include a country-specific home bias residual on the diagonal, written here as Δ_{ii} to maintain consistent notation with the off-diagonal wedges, and calibrated so that the model exactly reproduces observed domestic shares. This residual is the financial analogue of the border effect in trade gravity: it absorbs familiarity effects, domestic legal protections, currency preferences, and all other forces not captured by observable bilateral variables.

A further empirical concern in financial gravity applications concerns the treatment of countries that function as cross-border financial centres. Coppola et al. (2021) document that a substantial fraction

of reported bilateral portfolio positions involves intermediary jurisdictions such as Ireland, Luxembourg, and the Cayman Islands, where the legal counterparty recorded in International Monetary Fund Portfolio Investment Positions (IMF PIP) data differs from the ultimate issuer or ultimate investor. Once positions are restated on an ultimate-ownership basis, the apparent bilateral structure of global portfolio holdings shifts substantially, with implications for how home bias and bilateral friction should be measured. For policy counterfactual analysis, the diagonal home bias residual is particularly susceptible to this pass-through distortion in financial centres, where reported domestic holdings may not map cleanly to the genuine domestic-foreign decision faced by ultimate investors. The standard practical responses are either to reassign holdings to ultimate ownership where data permit, or to constrain the diagonal calibration for affected countries to prevent the residual from absorbing pass-through structure that has no economic interpretation as home bias.

A final implementation point concerns the construction of policy-relevant wedge components from estimated coefficients. The structural gravity framework predicts negative coefficients on all bilateral distance variables, on the theoretical grounds that greater geographic distance, lower linguistic similarity, larger financial development gaps, and lower bilateral information flow should each suppress bilateral investment. Empirical estimates that contradict these theoretical signs are typically interpreted as evidence of specification error, multicollinearity among friction variables, or sample-specific anomalies, rather than as economic findings to be incorporated into policy simulations. Wedge components built from coefficients with the opposite sign would imply, counter to theory, that increased distance raises bilateral investment, and would therefore not provide a meaningful basis for counterfactual analysis. Standard practice in the structural gravity literature is to treat such coefficients as evidence to be addressed through specification refinement and sensitivity analysis, rather than to embed them in a structural simulation.

The decomposability of the wedge into separately identified components, established in the trade literature by J. E. Anderson and Wincoop (2004) and extended to financial flows by Okawa and Wincoop (2012), is what makes the gravity framework suitable for evaluating policies that target specific friction channels. Without this property, a policy simulation could quantify the aggregate effect of a generic friction reduction but could not distinguish between reducing regulatory divergence and reducing informational or cultural barriers, a distinction that is central to the CMU debate reviewed in Section 3.3.

3.8 Productivity Spillovers from Cross-Border Capital

The gravity-based portfolio reallocation framework developed in the preceding sections quantifies the static allocative effects of bilateral friction reduction. A complementary literature considers whether cross-border capital flows generate dynamic productivity gains beyond their effect on the level of capital, through technology transfer, managerial spillovers, and improvements in corporate governance. This section reviews that literature with particular attention to the distinction between foreign direct investment (FDI) and portfolio equity, which has direct implications for how productivity spillovers should be modelled in a portfolio-focused application.

The empirical foundation of the productivity-spillover literature was established in the FDI context. Borensztein, De Gregorio, and Lee (1998), using cross-country panel data, find that FDI is positively associated with productivity growth in recipient economies, but only above a threshold level of human capital. Below that threshold, the spillover effect is statistically and economically negligible. Their interpretation is that absorptive capacity is required for foreign-supplied technology and managerial practices to be assimilated by domestic firms. Alfaro et al. (2004) extend this finding by showing that the strength of FDI-driven productivity growth depends on the development of local financial markets, which mediate the diffusion of new technologies through credit allocation to domestic firms. The conditional nature of FDI spillovers, contingent on absorptive capacity and institutional quality, has been a recurring finding across subsequent empirical work, and implies that a single linear elasticity is at best an approximation of the underlying mechanism.

A parallel strand of literature examines productivity spillovers through technology and R&D channels. Coe and Helpman (1995) document that aggregate R&D capital stocks in trading partners are significant determinants of total factor productivity in the recipient country, with spillover effects increasing in the bilateral intensity of trade linkages. Coe, Helpman, and Hoffmaister (1997) extend the framework to North-South spillovers and confirm that developing-country productivity benefits from the R&D stocks of advanced trading partners through similar transmission channels. The R&D spillover literature provides a structural basis for modelling productivity as a function of foreign-economy characteristics weighted by bilateral linkage measures, although the original applications focus on trade rather than capital linkages.

A critical caveat for the present analysis concerns the type of capital flow through which spillovers are most plausibly transmitted. Kose et al. (2009), in their reappraisal of the financial globalisation literature, argue that the empirical evidence for productivity spillovers is substantially stronger for FDI than for portfolio equity. The mechanism is intuitive: FDI typically involves managerial and operational involvement by the foreign investor, generating direct opportunities for technology transfer, governance improvements, and labour-market spillovers. Portfolio equity, by contrast, is largely passive: minority shareholders typically lack the operational influence required to transmit firm-specific technology or managerial practices, and any spillovers must operate through indirect channels such as the disciplining effect of foreign capital on domestic governance, demand for higher-quality corporate disclosure, and market-driven competition for capital allocation. Kose et al. (2009) emphasise that financial globalisation as a whole, including portfolio flows, can deliver indirect benefits by improving institutional quality and discipline, but the direct productivity-transmission case for portfolio equity is weaker than for FDI.

The implication for modelling is twofold. First, productivity spillover elasticities estimated on FDI data should be treated as upper bounds when applied to portfolio equity, since the underlying mechanism is theoretically weaker for passive minority investment than for controlling stakes. Second, the productivity channel in a portfolio-focused application should be presented as a structured robustness extension rather than a primary mechanism, with explicit acknowledgement that the empirical literature does not provide direct estimates of portfolio-equity-specific spillover elasticities.

4 Data

This section describes the data used to estimate the gravity model and construct the simulation baseline. The final dataset combines bilateral portfolio investment positions, aggregate portfolio totals, macroeconomic fundamentals, and bilateral friction variables. The raw merged dataset contains 1,535,274 observations over the period 2000–2025. After restricting the data to the simulation country set, the final country-filtered dataset contains 21,866 observations. The simulation baseline is constructed for 2019.

4.1 Data Sources

Table 1 summarises the datasets used in the final model. The core investment variable is bilateral portfolio holdings. Aggregate foreign holdings and total portfolio wealth are used to reconstruct domestic holdings. Penn World Tables (PWT) variables provide the macroeconomic fundamentals. Bilateral frictions are measured using geographic distance, linguistic distance, news-based informational distance, and financial-development distance.

Table 1: Data sources and model use

Dataset	Variables used	Frequency / years
IMF bilateral portfolio holdings (PIP)	a_{ij} , $\text{ForeignPortfolio}_i$, $\text{ForeignPortfolio}_j$	Annual, 1997–2024
Eurostat total portfolio wealth	TotalPortfolio_i , TotalPortfolio_j	Annual, 2016–2025
Penn World Tables	Y_i , Y_j , k_i , k_j , L_i , L_j , A_i , A_j , α_i , α_j	Annual, 1950–2024
IMF Financial Development Index	FD_i , FD_j , d_{ij}^{fin}	Annual, 2000–2020
CEPII language	d_{ij}^{ling}	Time-invariant
Country coordinates (GitHub)	d_{ij}^{geo}	Time-invariant
IntenSE / news coverage	d_{ij}^{news}	Annual, 2015–2025

All data sources are standard academic or institutional datasets with the exception of the news coverage variable. The IntenSE (Information Intensity Spatial Explorer) dataset provides bilateral counts of cross-country news article coverage derived from large-scale automated media scraping across national news outlets (CEPII, 2025). For each directed country pair (i, j) , the raw variable captures the volume of articles published in country i ’s media that reference country j . Coverage is available annually from 2015 onwards. Because this source is less established than the other datasets used here, its properties and limitations are discussed in detail in Section 4.6.1.

4.2 Sample Construction

The sample is restricted to the countries used in the simulation model and consists of 29 countries: 26 EU member states (Croatia excluded due to missing data) and three non-EU comparators (United Kingdom, Switzerland, Norway). The non-EU countries provide outside-sample price anchors and are not directly subjected to the CMU shock. The 2019 base-year cross-section therefore contains 841 possible bilateral country pairs, of which 29 are domestic pairs and 812 are foreign pairs.

Table 2: Sample construction

Metric	Value
Raw observations	1.535×10^6
Final observations after country filtering	21,866
Country count	29
Year range	2016–2020
Base year	2019
Bilateral pairs, N^2	841
Domestic pairs	29
Foreign pairs	812
PPML observations	2,858
Base-year simulation observations	841

The base year is 2019. This is the most recent pre-pandemic year for which all required data sources are jointly available: bilateral portfolio investment positions from IMF PIP, total portfolio wealth from Eurostat, macroeconomic fundamentals from Penn World Tables, financial development indices from the IMF, and bilateral news coverage from IntenSE. Portfolio allocations and capital stocks from 2019 are more likely to reflect medium-run equilibrium patterns than observations from the COVID-19 pandemic period, the post-pandemic recovery, or the energy price shocks of 2021 and 2022. The methodological implications of anchoring the simulation to a stable pre-pandemic integration regime are discussed in Section 5.3.

Domestic holdings are reconstructed from aggregate portfolio totals rather than taken directly from the

bilateral portfolio matrix. Foreign bilateral pairs are used in the PPML gravity estimation.

Negative bilateral holdings are treated as data artefacts and floored at zero. Zero bilateral holdings are retained because the PPML estimator can use zero-flow observations directly.

4.3 Variable Construction

The bilateral portfolio variable a_{ij} measures the portfolio position held by investor country i in destination country j . For foreign pairs, this variable is observed directly in the bilateral portfolio data. Domestic holdings are reconstructed as

$$a_{ii} = \text{TotalPortfolio}_i - \text{ForeignPortfolio}_i. \quad (1)$$

Observed portfolio shares are constructed by row-normalising the bilateral holdings matrix:

$$\pi_{ij}^{data} = \frac{a_{ij}}{\sum_{l=1}^n a_{il}}. \quad (2)$$

The observed home share is the diagonal element of the portfolio-share matrix:

$$\pi_{ii}^{data}. \quad (3)$$

The return proxy is constructed from the marginal product of capital and normalised to have mean one:

$$R_j = \frac{\alpha_j Y_j / k_j}{\frac{1}{n} \sum_{l=1}^n \alpha_l Y_l / k_l}. \quad (4)$$

The marginal product of capital is used as the return proxy because it is directly computable from Penn World Tables variables for all 29 countries in the sample and provides a consistent, forward-looking measure of capital scarcity. A higher MPK indicates that capital is relatively scarce and productive, making the destination more attractive to foreign investors. Market-based return measures such as historical

equity returns are volatile, backward-looking, and unavailable at the required coverage and frequency for several smaller EU economies in the sample.

Financial-development distance is constructed as the absolute bilateral difference in the IMF Financial Development Index:

$$d_{ij}^{fin} = |FD_i - FD_j|. \quad (5)$$

The IMF Financial Development Index (Sviryzdenka, 2016a) is used rather than a binary capital-account liberalisation indicator because the relevant friction within the EU is not the presence of capital controls, which have been formally eliminated across member states, but the residual institutional and regulatory divergence in the depth, accessibility, and stability of national financial systems. The composite index captures this multi-dimensional institutional distance on a consistent annual basis across all countries in the sample. The choice of this measure over alternatives is discussed in Section 3.6.

Country economic size is measured by GDP throughout the variable construction and model. Market capitalisation would in principle be a more direct measure of equity market size, but consistent market capitalisation data are unavailable at the required frequency and coverage for several smaller EU economies in the sample. GDP and market capitalisation are highly correlated across the advanced-economy sample used here, and GDP is the standard size variable in the financial gravity literature (Portes and Rey, 2005; Lane and Milesi-Ferretti, 2008; Coeurdacier and Martin, 2009). The implications of this choice for the portfolio allocation rule are discussed in Section 5.6.

Geographic distance is constructed from country coordinates using the haversine formula:

$$\begin{aligned} d_{ij}^{geo} &= R \cdot 2 \arctan 2 \left(\sqrt{h_{ij}}, \sqrt{1 - h_{ij}} \right), \\ h_{ij} &= \sin^2 \left(\frac{\Delta \phi_{ij}}{2} \right) + \cos(\phi_i) \cos(\phi_j) \sin^2 \left(\frac{\Delta \lambda_{ij}}{2} \right). \end{aligned} \quad (6)$$

with $R = 6371$ km, $\Delta \phi_{ij} = \phi_j - \phi_i$, and $\Delta \lambda_{ij} = \lambda_j - \lambda_i$.

The linguistic and news variables are min–max normalised to the unit interval before conversion to distance measures:

$$x^{\text{scaled}} = \frac{x - \min(x)}{\max(x) - \min(x)} \in [0, 1]. \quad (7)$$

Normalisation is applied to d_{ij}^{ling} and d_{ij}^{news} because their underlying indicators, ASJP lexical proximity scores and bilateral news coverage intensity, have no natural cardinal unit scale, so the conversion to a distance via $1 - x$ requires them to lie on $[0, 1]$ to remain non-negative and interpretable. Geographic distance is already in kilometres and requires no rescaling. Financial development distance is an absolute difference of IMF FD Index scores, which are bounded in $[0, 1]$ by construction, so the bilateral difference inherits a natural $[0, 1]$ range without further transformation.

The PPML gravity model uses a_{ij} as the dependent variable and the regressors d_{ij}^{ling} , d_{ij}^{geo} , d_{ij}^{news} , d_{ij}^{fin} , Y_i , and Y_j .

Table 3 defines the final variables used in the model.

Table 3: Variable dictionary

Variable	Type	Formula or derivation	Source
a_{ij}	Raw	Observed bilateral holdings from investor i to destination j	IMF Portfolio Investment Positions (IMF PIP)
a_{ii}	Constructed	$\text{TotalPortfolio}_i - \text{ForeignPortfolio}_i$	Eurostat and IMF PIP
π_{ij}	Constructed	$a_{ij} / \sum_j a_{ij}$	Bilateral matrix
π_{ii}	Constructed	Diagonal of π_{ij}	Bilateral matrix and totals
Y_i	Raw	GDP of investor country i	PWT
Y_j	Raw	GDP of destination country j	PWT
k_i	Raw	Physical capital stock	PWT

Variable	Type	Formula or derivation	Source
L_i	Raw	Labour input	PWT
A_i	Raw	TFP level	PWT
α_i	Raw	Capital share	PWT
R_j	Constructed	$(\alpha_j Y_j / k_j) / \text{mean}(\alpha Y / k)$	PWT variables
TotalPortfolio $_i$	Raw	Country total portfolio wealth	Eurostat totals
ForeignPortfolio $_i$	Raw	Country total foreign portfolio holdings	IMF PIP totals
d_{ij}^{ling}	Raw	Continuous lexical distance derived from ASJP scores via CEPIL; preferred over binary language dummies for within-Europe heterogeneity (see Section 3.6)	CEPIL / ASJP
d_{ij}^{geo}	Constructed	Bilateral distance in km computed with the haversine formula using country latitude/longitude coordinates	GitHub country dataset
d_{ij}^{news}	Raw	News/information distance from IntenSE coverage	IntenSE
d_{ij}^{fin}	Constructed	Absolute difference in financial development index	IMF Financial Development Index

4.4 Descriptive Statistics

Table 4 reports the descriptive statistics for the main variables used in the gravity model and simulation baseline. Bilateral portfolio holdings are highly right-skewed in levels. The log-transformed holdings variable is included as a diagnostic transformation, although the gravity model itself is estimated using PPML rather than log-linear OLS.

Table 4: Main descriptive statistics

Variable	Mean	Std. dev.	p25	p50	p75	Skewness	Kurtosis
a_{ij}	20,828.1	57,930.7	161.3	974.5	12,690.9	5.513	43.234
$\log(1 + a_{ij})$	7.069	2.867	5.089	6.883	9.449	-0.095	-0.593
Y_i	1.165×10^6	1.372×10^6	2.843×10^5	5.164×10^5	1.879×10^6	1.411	0.842
Y_j	1.183×10^6	1.369×10^6	2.916×10^5	5.269×10^5	1.909×10^6	1.389	0.802
d_{ij}^{ling}	0.614	0.227	0.517	0.679	0.776	-0.894	-0.164
d_{ij}^{geo}	1,402.0	737.6	821.2	1,303.4	1,851.1	0.611	-0.055
d_{ij}^{news}	0.996	0.012	0.998	0.999	1.000	-8.927	106.242
d_{ij}^{fin}	0.245	0.172	0.101	0.213	0.368	0.622	-0.470
R_j	1.000	0.418	0.757	0.860	1.171	1.190	0.337
π_{ii}	0.629	0.230	0.510	0.646	0.781	-0.732	0.598
TotalPortfolio _{<i>i</i>}	1.690×10^6	2.293×10^6	1.023×10^5	6.222×10^5	2.556×10^6	1.720	2.329
f_i	0.450	1.622	0.030	0.077	0.157	5.176	27.349

4.5 Missingness and Cleaning

Table 5 reports coverage and cleaning rules for the variables used in the gravity model. Observations with missing gravity regressors are dropped from the PPML sample. Zero bilateral holdings are retained.

Table 5: Missingness and cleaning rules

Variable	Expected obs.	Available obs.	Missing share	Zero share	Treatment
a_{ij}	21,866	17,931	0.180	0.022	Negative values floored at zero; zeros retained for PPML.
Y_i	21,866	20,184	0.077	0.000	Dropped from PPML sample if missing.
Y_j	21,866	20,184	0.077	0.000	Dropped from PPML sample if missing.
d_{ij}^{ling}	21,866	19,006	0.131	0.035	Dropped from PPML sample if missing.
d_{ij}^{geo}	21,866	21,866	0.000	0.035	Dropped from PPML sample if missing.
d_{ij}^{news}	21,866	5,574	0.745	0.000	Dropped from PPML sample if missing.
d_{ij}^{fin}	21,866	17,661	0.192	0.028	Constructed as absolute difference; dropped if still missing.

4.6 Distributional Diagnostics

The portfolio holdings variable is strongly skewed in levels, so Figure 1 shows the distribution of $\log(1 + a_{ij})$ for foreign bilateral pairs. The log-transformed distribution has skewness -0.095 and kurtosis -0.593 .

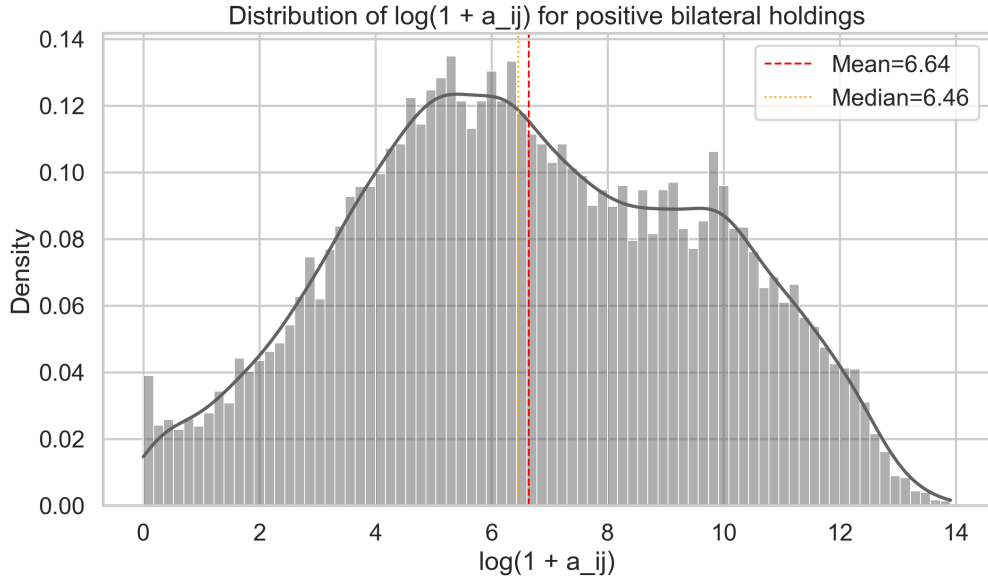


Figure 1: Distribution of $\log(1 + a_{ij})$ for foreign bilateral portfolio positions.

Observed home shares differ substantially across countries. Figure 2 reports the observed diagonal portfolio share π_{ii}^{data} by country.

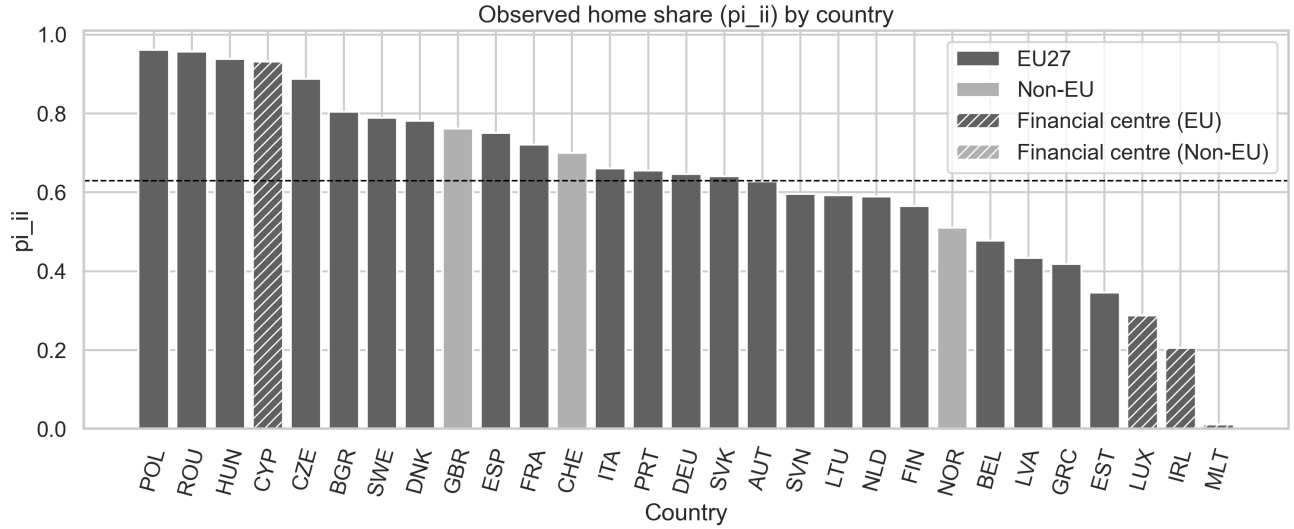


Figure 2: Observed home share π_{ii}^{data} by country.

The four wedge variables are shown in Figure 3. The histograms are density-normalised, so the distribu-

tions are comparable despite the variables having different units.

Wedge-variable distributions (normalised proportion, PPML sample)

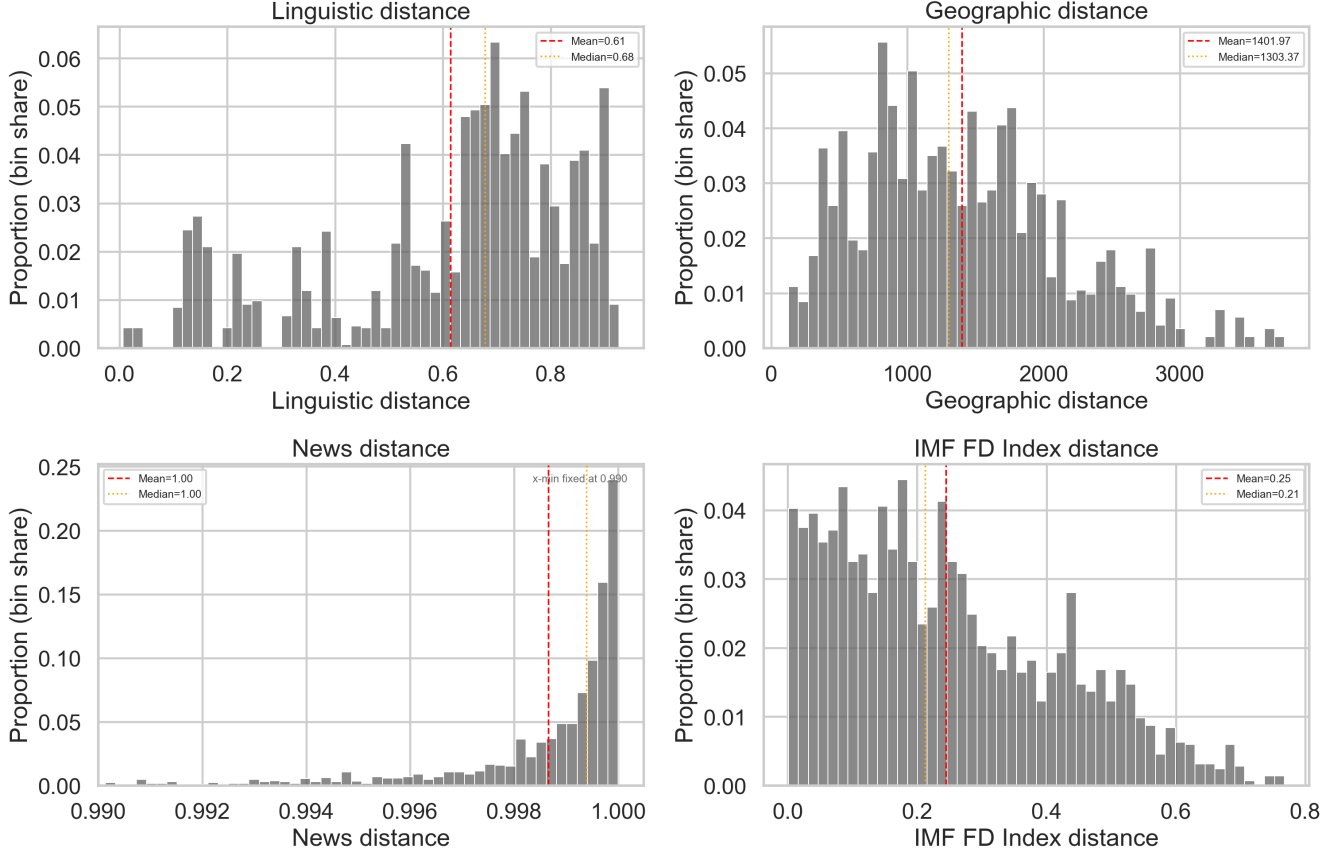


Figure 3: Density-normalised distributions of the four bilateral wedge variables.

4.6.1 News coverage and informational distance

The news-based informational-distance variable proxies how visible destination country j is to investors in origin country i . The motivation is that investors are less likely to allocate capital to countries that receive little attention in their domestic media environment, and that CMU may reduce these informational frictions through complementary measures such as the European Single Access Point and harmonised disclosure standards, as discussed in Section 3.6.

The variable has three material limitations. First, news coverage is highly skewed: large, economically central countries dominate media attention, so bilateral coverage is much thinner for peripheral country

pairs than for pairs involving Germany, France, Italy, or the United Kingdom. Second, the variable has 74.5 per cent missingness in the PPML sample, the highest of any regressor (Table 5). Third, bilateral coverage is partly endogenous to investment activity, since greater cross-border investment generates more financial reporting, which complicates causal interpretation of the estimated coefficient.

The parameter γ should therefore be read as a broad reduction in informational frictions rather than a literal increase in news articles, and the results involving γ are interpreted cautiously throughout.

4.6.2 Domestic holdings and within-sample portfolio coverage

Domestic portfolio holdings are not directly reported in the IMF PIP data and are constructed residually as total portfolio wealth (Eurostat) minus total foreign portfolio investment (IMF PIP), as defined in Section 4.3.

This construction overstates the domestic component because total portfolio wealth is a worldwide figure while the deducted foreign holdings cover only bilateral positions within the 29-country sample. Positions held outside the sample are therefore classified as domestic. The resulting home share is a sample-consistent residual rather than a direct measure of domestic investment preference; calibrated diagonal wedges Δ_{ii} absorb this overstatement by matching model-implied home shares to the observed residual. The within-sample counterfactual remains valid because the same portfolio universe is applied consistently to baseline and CMU scenarios, but the level of home bias is less precisely measured than the within-sample changes in allocation.

4.7 Correlation and Multicollinearity Diagnostics

Table 6 reports the correlation matrix for the four wedge variables and log bilateral holdings. The wedge variables are positively correlated with one another, but the correlations are not close to unity. This supports treating them as distinct bilateral frictions. The correlations between the wedge variables and log holdings are negative, consistent with the interpretation that larger bilateral distances are associated with lower bilateral investment positions.

Table 6: Correlation matrix for wedge variables and bilateral holdings

	d_{ij}^{ling}	d_{ij}^{geo}	d_{ij}^{news}	d_{ij}^{fin}	$\log(1 + a_{ij})$
d_{ij}^{ling}	1.000	0.217	0.156	0.275	-0.457
d_{ij}^{geo}	0.217	1.000	0.023	0.184	-0.207
d_{ij}^{news}	0.156	0.023	1.000	0.135	-0.178
d_{ij}^{fin}	0.275	0.184	0.135	1.000	-0.378
$\log(1 + a_{ij})$	-0.457	-0.207	-0.178	-0.378	1.000

The variance inflation factors in Table 7 provide an additional multicollinearity diagnostic. The news-distance variable has a VIF above 10, while language distance is in the moderate range. Geographic distance and financial-development distance have low VIF values.

Table 7: Variance inflation factors for wedge variables

Variable	VIF	Interpretation
d_{ij}^{ling}	9.336	Moderate, 5–10
d_{ij}^{geo}	4.922	Low, below 5
d_{ij}^{news}	10.518	High, above 10
d_{ij}^{fin}	3.330	Low, below 5

5 Methodology

The methodology is presented in chronological order to match the structure of the empirical analysis and the simulation model.

5.1 Declaration of Artificial Intelligence Use

AI-based tools were used as supporting tools during the thesis process. They were used to generate search ideas, identify potentially relevant literature, and obtain suggestions for improving clarity, structure, and academic wording. The tools were not used to produce the thesis analysis, empirical results, model output, or final written arguments. All substantive choices, data processing, econometric modelling, interpretation of results, and final text are the authors' own responsibility.

5.2 Modelling Overview

The analysis proceeds in five sequential stages. First, we estimate a gravity model on bilateral portfolio investment data to recover the structural coefficients on four bilateral frictions: linguistic distance, financial development distance, informational distance, and geographic distance. These coefficients quantify how strongly each friction is associated with the amount of bilateral portfolio investment between countries. Second, we use the estimated coefficients to construct bilateral wedge matrices that summarise the frictions between all country pairs in the sample. The model distinguishes between three wedge categories: hard wedges (institutionally determined frictions directly targeted by CMU policy), policy-sensitive soft wedges (informational barriers that respond to integration), and immutable structural wedges (linguistic and geographic frictions that policy cannot alter). The diagonal elements of the wedge matrix are the intrinsic wedges between a country and itself. Since there is no reliable data on domestic investment positions, we calibrate the diagonal wedges to match the observed domestic portfolio shares in the base year. This ensures that the model can reproduce the observed pattern of home bias in the baseline scenario. Third, we combine the calibrated wedges with country level returns and GDP weights to construct a gravity style portfolio allocation model that matches the observed pattern of portfolio holdings. Fourth, we simulate Capital Markets Union by proportionally reducing the hard wedge and the policy sensitive soft wedge for intra EU country pairs, using the policy parameters ω and γ , and then recompute portfolio

allocations and capital stocks under each scenario. Fifth, we augment the production side of the model with an endogenous TFP channel in which countries that receive more foreign equity capital experience higher productivity, governed by the spillover elasticity θ . This structure allows each part of the model to be evaluated separately. The gravity coefficients are estimated once and then held fixed. The diagonal wedge calibration is performed for the baseline scenario and then held fixed in CMU counterfactuals. The CMU shock is then applied as a clean counterfactual relative to the calibrated baseline. Within each CMU scenario, expected returns are updated iteratively in general equilibrium until convergence before final portfolio shares are computed. Any difference between baseline and counterfactual outcomes is therefore attributable to the change in bilateral frictions and the induced return adjustment.

5.3 Base Year

The base year for the simulation is 2019, selected on the data-availability and stability grounds described in Section 4.2. The methodological implication of this choice is that the gravity coefficients and calibrated wedges reflect a stable post-crisis integration regime, as the regime-stability argument of Coeurdacier and Rey (2013) requires: pooling across the pre-crisis integration peak, the 2008–2012 reversal, and the post-pandemic period would conflate friction levels with regime change, producing unreliable coefficient estimates for a single-scenario counterfactual.

5.4 Sample and Country Coverage

The model is built on a sample of 29 countries. The core of the sample consists of 26 EU member states. Croatia is excluded due to missing data. To this EU group, we add three non-EU countries: the United Kingdom, Switzerland, and Norway. This allows us to observe how the CMU shock affects portfolio allocation and output within the EU, while the non-EU countries serve as comparison economies that are not directly shocked and only adjust through general-equilibrium return changes.

The EU countries form the policy-relevant core of the analysis because the CMU shock is applied only to bilateral pairs where both countries belong to the EU sample. This allows the simulation to capture how lower intra-EU frictions affect portfolio allocation, capital reallocation, and output across the member states included in the sample.

The three outside countries are included because the EU is not a closed financial system. They are not directly shocked, and their outcomes move only through general equilibrium return adjustments. They therefore serve as comparison economies, while the main policy interpretation remains centered on intra EU reallocation.

5.4.1 Limitations of the Sample Restriction

Restricting the model to 29 countries creates several limitations. The most direct issue is that the observed portfolio matrix only includes bilateral positions among the countries in the sample. As a result, the total portfolio wealth of each investor country, defined as the row sum of the observed bilateral matrix, excludes holdings in countries outside the sample. This means that the within sample portfolio shares can overstate the importance of domestic holdings and of the included foreign destinations relative to the true global portfolio. This is one reason why the diagonal wedge calibration is necessary. By calibrating the diagonal wedge to match the observed domestic portfolio share, we ensure that the model can reproduce the observed home bias in the baseline scenario, even if the bilateral matrix is incomplete.

A related issue concerns the domestic portfolio share. In the code, domestic holdings are constructed from aggregate portfolio totals as total portfolio wealth minus total foreign portfolio investment. This is internally consistent with the available data, but the bilateral matrix used in the simulation still omits out of sample destinations. Consequently, the portfolio share denominator remains sample specific even when the diagonal is constructed from aggregate data.

There are also implications for the gravity regression. Although the raw dataset contains a much wider set of countries, the estimated PPML regression is run only on the same 29-country sample used in the simulation. The estimated coefficients therefore reflect variation within a relatively homogeneous set of advanced economies rather than the full world distribution of financial and informational distance. This is an acceptable restriction for our research question, but it should be kept in mind when interpreting the estimated friction coefficients. Additionally, there are data gaps for Eastern European countries, which may affect the precision of the estimated coefficients on financial and informational distance. However, PPML is designed to be robust to zero observations, so the estimation should still be valid even with these gaps.

The model should therefore be understood as a focused representation of the portfolio allocation problem most relevant for the CMU question, rather than as a full model of the global financial system.

5.5 Gravity Model

We use a gravity model to estimate how bilateral frictions affect portfolio investment positions. In this setting, hard frictions refer to differences in financial systems and institutions, while soft frictions refer to barriers related to language and information. The gravity framework is useful because it links bilateral investment to country size and bilateral distance measures in a transparent way.

The model is estimated with a Poisson Pseudo Maximum Likelihood (PPML) estimator. PPML is appropriate here because bilateral portfolio positions are non negative and include many zero observations. It is also robust to heteroskedasticity and allows the conditional mean to be modelled in levels through the exponential link function.

Our empirical specification is:

$$E[a_{i,j,t} | X_{i,j,t}] = \exp\left(\beta_0 + \beta_1 d_{i,j}^{\text{ling}} + \beta_2 d_{i,j,t}^{\text{news}} + \beta_3 d_{i,j,t}^{\text{fin}} + \beta_4 d_{i,j}^{\text{geo}} + \gamma_1 Y_{i,t} + \gamma_2 Y_{j,t} + \delta_t\right) \quad (8)$$

where $a_{i,j,t}$ is the bilateral portfolio investment position from country i to country j in year t ; $d_{i,j}^{\text{ling}}$ is linguistic distance based on ASJP lexical distance (CEPII, 2026; ASJP Consortium, 2026; Bakker et al., 2009); $d_{i,j,t}^{\text{news}}$ is the informational distance measure based on the number of bilateral news articles (CEPII, 2025); $d_{i,j,t}^{\text{fin}}$ is the absolute difference in the IMF Financial Development Index between country i and country j (Svirydzenka, 2016b; International Monetary Fund, 2016); $d_{i,j}^{\text{geo}}$ is bilateral great-circle distance in kilometres, computed using the haversine formula applied to country coordinates from a publicly available dataset covering all sovereign states (dr5hn, 2026). The haversine method is identical to the computational approach used by CEPII GeoDist (Mayer and Zignago, 2011); the choice of coordinate source has negligible effect on bilateral distances within the European sample, where all country pairs fall within a well-mapped geographic area. $Y_{j,t}$ are GDP in levels (Groningen Growth and Development Centre, 2026; Feenstra, Inklaar, and Timmer, 2015); and δ_t denotes year fixed effects.

The geographic distance is computed as

$$d_{i,j}^{\text{geo}} = R \cdot 2 \arctan 2 \left(\sqrt{h_{i,j}}, \sqrt{1 - h_{i,j}} \right), \quad (9)$$

where

$$h_{i,j} = \sin^2 \left(\frac{\Delta \phi_{i,j}}{2} \right) + \cos(\phi_i) \cos(\phi_j) \sin^2 \left(\frac{\Delta \lambda_{i,j}}{2} \right), \quad (10)$$

with $\Delta \phi_{i,j} = \phi_j - \phi_i$, $\Delta \lambda_{i,j} = \lambda_j - \lambda_i$, and Earth radius $R = 6371$ km.

This specification follows Papaioannou (2009) in substituting country-level economic size variables for full origin–destination fixed effects, in order to preserve identification of $d_{i,j,t}^{\text{fin}}$, which is constructed as a bilateral difference in a country-level characteristic. The trade-off this entails is discussed in Section 3.5. Furthermore, bilateral goods trade is not included as a covariate despite its presence in standard financial gravity specifications (Lane and Milesi-Ferretti 2008). The reason is that bilateral goods trade and bilateral equity holdings are jointly determined (Aviat and Coeurdacier 2007), so trade enters the bilateral system as an outcome rather than as an exogenous control. Including it as a regressor would absorb part of the underlying bilateral friction structure that the gravity decomposition aims to recover, and would also weaken the mapping from estimated coefficients to the CMU policy lever, since the same informational and institutional frictions that CMU targets also drive bilateral trade.

The PPML estimation is run on the available panel years in the final dataset for the 29-country simulation sample. Domestic pairs where $i = j$ are excluded since we do not have reliable data on domestic investment portfolio, only an approximation. Zero bilateral positions are retained, since PPML is well suited to non negative data with many zeros. Negative portfolio positions are treated as data artefacts and are floored to zero before estimation. Observations are dropped only when one or more regressors are missing. Standard errors are clustered at the bilateral pair level.

For numerical stability, the bilateral portfolio position variable and the GDP variables are rescaled before estimation. This rescaling does not affect the estimated coefficients on the friction variables and therefore

does not affect the wedge construction used in the simulation. We do this by dividing by one million. This ensures that we do not experience overflow issues in the Python code which can arise when exponentiating large numbers in the PPML link function and then also impact the robustness of the model.

The expected sign of all distance coefficients is negative, so that a higher distance is associated with a lower bilateral portfolio position. This is consistent with the intuition that greater soft and hard distances should all be associated with lower levels of bilateral portfolio investment. In implementation, only friction coefficients with the expected negative sign are activated in wedge construction; if a coefficient is estimated with the opposite sign, that wedge component is left neutral.

5.6 Portfolio Weights

The portfolio weight $\pi_{i,j}$ denotes the share of investor country i 's observed portfolio allocated to destination country j . Using the bilateral portfolio matrix, the observed portfolio share is defined as

$$\pi_{i,j}^{\text{data}} = \frac{a_{i,j}}{\sum_{\ell=1}^n a_{i,\ell}} \quad (11)$$

Where $a_{i,j}$ is the observed bilateral portfolio investment position and the denominator is the sum of all bilateral investment positions from country i to all destination countries in the sample.

The model implied portfolio allocation follows a gravity style rule in which destination attractiveness depends on expected returns, destination size, and bilateral frictions. This is a reduced form representation of a more general portfolio choice problem in which investors allocate their wealth across a set of destinations to maximise expected utility. The model is static and does not include inter temporal dynamics, but it captures the key cross sectional forces that shape portfolio allocation in the baseline and counterfactual scenarios. For investor country i , the portfolio share allocated to destination country j is

$$\pi_{i,j} = \frac{R_j^\eta \cdot Y_j / \Delta_{i,j}}{\sum_{\ell=1}^n R_\ell^\eta \cdot Y_\ell / \Delta_{i,\ell}} \quad (12)$$

Where R_j is the expected return for destination j , Y_j is GDP, η is the return elasticity, and $\Delta_{i,j}$ is the

bilateral wedge. GDP is used as the destination size variable throughout; the rationale for this choice over market capitalisation is set out in Section 4.3. The return elasticity η governs how strongly investors respond to return differentials across destinations. The bilateral wedge $\Delta_{i,j}$ captures the frictions between investor country i and destination country j .

5.6.1 Domestic Portfolio Weights

Domestic holdings are not directly reported in the bilateral IMF data and are constructed residually as described in Section 4.3. The residual construction means the diagonal portfolio share captures domestic holdings plus any foreign positions held outside the sample, as discussed in Section 4.6.2. This is one reason the diagonal wedge calibration is necessary: by calibrating $\Delta_{i,i}$ to match the observed domestic share, the model reproduces the baseline home bias pattern even though the bilateral matrix is incomplete. A floor of 1% of total portfolio wealth is applied if the residual is very small or negative.

5.7 Bilateral Wedge Matrices

After estimating the gravity model, we use the friction coefficients to construct four bilateral wedge matrices. The first is a hard wedge based on financial development distance, measured as the absolute difference in the IMF Financial Development Index between countries i and j (Sviryzdenka, 2016b; International Monetary Fund, 2016). The second is a fixed soft wedge based on linguistic distance (CEPII, 2026; ASJP Consortium, 2026; Bakker et al., 2009). The third is a fixed soft wedge based on bilateral geographic distance in kilometres, computed with the haversine formula using country latitude and longitude coordinates from a GitHub dataset of all countries rather than from CEPII (dr5hn, 2026). The fourth is a policy-sensitive soft wedge based on informational distance, which we measure based on the number of news articles about country j in country i 's media. The informational wedge is the policy-sensitive soft lever because the CMU complementary measures targeting the informational dimension, the European Single Access Point for issuer disclosures, consolidated tape provision, and harmonised reporting standards, operate on exactly the informational environment that bilateral news coverage indexes. A reduction in γ therefore captures the effect of these complementary policies, distinct from changes in hard regulatory frictions. The financial development wedge is the primary hard policy lever because differences

in the depth, accessibility, and stability of financial systems are directly targeted by the regulatory and institutional harmonisation that CMU policies aim to deliver. The linguistic and geographic wedges are treated as fixed because neither language differences nor physical distance are directly affected by CMU policies and are more likely to reflect long-run structural factors rather than short-run policy changes. By including both linguistic and geographic distance as fixed wedges, we can isolate the informational distance component of news coverage from the cultural and physical proximity channels that independently shape bilateral communication and investment flows.

These wedges are defined as

$$\Delta_{i,j}^{\text{fin}} = \exp\left(-\hat{\beta}_{\text{fin}} \cdot d_{i,j}^{\text{fin}}\right) \quad (13)$$

$$\Delta_{i,j}^{\text{ling}} = \exp\left(-\hat{\beta}_{\text{ling}} \cdot d_{i,j}^{\text{ling}}\right) \quad (14)$$

$$\Delta_{i,j}^{\text{news}} = \exp\left(-\hat{\beta}_{\text{news}} \cdot d_{i,j}^{\text{news}}\right) \quad (15)$$

$$\Delta_{i,j}^{\text{geo}} = \exp\left(-\hat{\beta}_{\text{geo}} \cdot d_{i,j}^{\text{geo}}\right) \quad (16)$$

5.7.1 Classification of Wedge Components

The four wedge components play distinct roles in the model, and it is important to be precise about which are treated as policy levers and which are treated as structural constants.

The financial development wedge $\Delta_{i,j}^{\text{fin}}$, constructed from the absolute difference in the IMF Financial Development Index, is the primary hard policy lever governed by ω . Financial development gaps reflect differences in the depth, stability, and accessibility of financial systems across countries. These are institutional characteristics that CMU policies directly target through harmonisation of regulatory standards,

supervisory convergence, and the removal of cross-border barriers to capital market access. A reduction in ω therefore represents progress towards a more integrated and uniformly developed European financial system.

The informational wedge $\Delta_{i,j}^{\text{news}}$, constructed from bilateral news article distance, is the policy-sensitive soft lever governed by γ . CMU does not directly produce news articles, but the complementary measures it directs at the informational channel, the European Single Access Point, consolidated tape provision, and harmonised reporting standards, improve the informational environment in ways that bilateral news coverage indexes. A reduction in γ therefore captures the effect of these complementary policies on the soft-friction component, distinct from the changes to hard regulatory frictions parameterised by ω .

The linguistic wedge $\Delta_{i,j}^{\text{ling}}$ and the geographic wedge $\Delta_{i,j}^{\text{geo}}$ are both treated as immutable structural constants. Language similarity and physical distance are deep bilateral characteristics that are not altered by financial regulation or capital market policy. They are retained in the total wedge matrix because they are empirically important determinants of bilateral portfolio investment - omitting them would cause the gravity regression to over-attribute variation in investment flows to the policy-sensitive variables - but they are held fixed in every CMU scenario and are not subject to any policy shock.

This classification can be summarised as follows:

Component	Variable	Type	Policy lever
$\Delta_{i,j}^{\text{fin}}$	Financial development distance	Hard	ω
$\Delta_{i,j}^{\text{news}}$	News article distance	Soft	γ
$\Delta_{i,j}^{\text{ling}}$	Linguistic distance	Immutable	—
$\Delta_{i,j}^{\text{geo}}$	Geographic distance	Immutable	—

The baseline bilateral wedge is therefore the product of all four components:

$$\Delta_{i,j}^{\text{baseline}} = \Delta_{i,j}^{\text{fin}} \cdot \Delta_{i,j}^{\text{ling}} \cdot \Delta_{i,j}^{\text{news}} \cdot \Delta_{i,j}^{\text{geo}} \quad (17)$$

The off diagonal entries are pinned down directly by the gravity coefficients and the observed bilateral distances. The diagonal entries are not identified in the gravity regression because there is no bilateral

distance from a country to itself. These diagonal terms are therefore calibrated to reproduce the observed domestic portfolio shares.

For each country i , the diagonal wedge must satisfy

$$\pi_{i,i} = \frac{R_i^\eta \cdot Y_i / \Delta_{i,i}}{\sum_{\ell=1}^n R_\ell^\eta \cdot Y_\ell / \Delta_{i,\ell}} \quad (18)$$

Since $\Delta_{i,i}$ appears in both the numerator and the denominator, this equation is solved iteratively. Convergence is defined by

$$\max_i \left| \hat{\pi}_{i,i} - \pi_{i,i}^{\text{data}} \right| < 10^{-5} \quad (19)$$

5.7.2 Calibration of the Domestic Wedge

The diagonal wedge $\Delta_{i,i}$ captures residual home bias that is not explained by cross country return differentials, destination size, and bilateral off diagonal frictions. Intuitively, a lower $\Delta_{i,i}$ makes the domestic option more attractive and raises the model implied home share, while a higher $\Delta_{i,i}$ has the opposite effect.

The calibration is performed by fixed point iteration. In each iteration, we compute the model implied portfolio shares from equation (12), compare the implied home share $\hat{\pi}_{i,i}$ to the observed home share $\pi_{i,i}^{\text{data}}$, and then update the diagonal wedge until the convergence criterion in equation (19) is satisfied.

The calibrated diagonal wedge can be interpreted as a residual domestic preference term. It absorbs all country specific forces that increase or decrease domestic portfolio allocation beyond what is predicted by returns, GDP size, and observed bilateral frictions. In economic terms, the magnitude of $\Delta_{i,i}$ reflects how much more or less attractive a country's investors perceive domestic investment opportunities relative to what the model would predict from fundamentals alone. A country with a low calibrated diagonal wedge is one where investors exhibit stronger home bias than returns, size, and bilateral frictions can explain, while a high diagonal wedge indicates that investors allocate less to domestic assets than the frictionless

benchmark would suggest.

Countries are excluded from the diagonal calibration if the observed home share is missing, equal to zero, or arbitrarily close to one. For these cases, the diagonal wedge is fixed at unity rather than calibrated. If a wedge is at unity, it means it is set to 1, which neither biases home nor foreign investment. This is a practical solution to the issue of missing or extreme home shares, but it also means that the model may not fully capture the true level of home bias in these cases. For example, if a country has an observed home share of zero, this could reflect a very high level of home bias that cannot be directly calibrated. By setting the diagonal wedge to unity, we are effectively assuming that there is no additional domestic preference beyond what is captured by returns, size, and bilateral frictions, which may not be accurate for all countries. If a wedge is set to unity, it is equal to 1. This corresponds to a neutral friction: it neither increases nor decreases the perceived attractiveness of that country. Mathematically, perceived attractiveness is inversely proportional to the wedge, since the mass term (GDP) is divided by the wedge. A lower wedge therefore increases perceived attractiveness, while a higher wedge reduces it. See equation (12).

5.7.3 Treatment of Financial Centres

Following Coppola, Maggiori, Neiman, and Schreger (2021), as discussed in Section 3.7, the simulation treats a subset of countries as financial centres whose domestic holdings may be distorted by pass-through investment structures, where the legal counterparty in PIP data does not correspond to the ultimate investor or issuer. These countries are Ireland, Luxembourg, Cyprus, and Malta. For these countries, the diagonal wedge is not calibrated to observed home bias and is instead fixed at unity.

The motivation is practical. In countries with unusually large intermediary sectors, the observed domestic share may reflect reporting structures or conduit activity rather than underlying portfolio preferences. Fixing the diagonal at unity prevents these distortions from driving the residual home bias calibration.

The off diagonal wedges for these countries are still constructed in the same way as for all other countries, using the estimated gravity coefficients and observed bilateral distance measures.

5.8 Returns and Return Elasticity

Expected returns are proxied by the normalised marginal product of physical capital, constructed from Penn World Tables variables as defined in Section 4.3. In the portfolio allocation rule, R_j serves as the attractiveness weight for destination j : a higher MPK indicates capital is relatively scarce, generating higher expected returns and drawing portfolio inflows.

Higher returns make a destination more attractive in the portfolio allocation rule given in equation (12). The strength of this response is governed by the return elasticity η . In the baseline calibration, we set $\eta = 1.0$. Robustness checks consider $\eta \in \{0.5, 1.0, 2.0\}$ to ensure that there is linearity in expectations and allocations. A lower value of η implies a weaker response to return differentials, while a higher value implies a stronger response. If the qualitative results hold across these values, then the model can be viewed as robust to the return elasticity assumption.

5.9 General Equilibrium: Return Updates After Capital Reallocation

The CMU shock reduces bilateral frictions, which reallocates capital across destinations. However, capital reallocation simultaneously changes the capital stock in each country, which affects the marginal product of capital and hence expected returns. Without accounting for this feedback, the model would systematically over-predict capital flows to initially high-return destinations. To capture the endogenous adjustment of returns after capital reallocation, we employ an iterative general equilibrium procedure.

For each CMU scenario (ω, γ, θ) , returns are updated through fixed-point iteration. Beginning with baseline returns R_j^0 , we compute portfolio shares from equation (12), aggregate portfolio flows to obtain capital stocks k_j^{fin} for each destination, and then update returns using the marginal product of capital:

$$R_j^{t+1} = \alpha_j \cdot \frac{Y_j}{k_j^{\text{fin},t}} \quad (20)$$

Returns are renormalised to have a cross-country mean of one before each iteration. We repeat this cycle until convergence, defined as

$$\max_j \left| R_j^{t+1} - R_j^t \right| < 10^{-7} \quad (21)$$

The resulting equilibrium returns R_j^∞ satisfying equation (21) are then used in all downstream calculations of capital and output effects.

This general equilibrium loop is essential for two reasons. First, it prevents the model from treating returns as exogenous. A destination with high baseline returns would otherwise receive implausibly large capital inflows, leading to unrealistic capital concentration. Second, it captures the economic adjustment mechanism: the CMU shock changes relative returns, inducing capital reallocation, which then changes returns again, until a new equilibrium is reached. By iterating to convergence, we ensure that any difference in capital and output between baseline and counterfactual scenarios is attributable to the change in bilateral frictions and the market’s endogenous adjustment to it.

5.10 Total Factor Productivity Spillovers

A key extension of the baseline capital reallocation model is an endogenous productivity channel. The idea is that countries that receive more foreign capital may also benefit from knowledge transfer, managerial learning, or technology diffusion. However, the literature reviewed in Section 3.8 shows that the strongest evidence for such spillovers comes from FDI rather than portfolio investment. In this thesis, the TFP channel is therefore treated as a structured sensitivity extension rather than as a core estimated mechanism.

5.10.1 Portfolio-Capital Mapping to Productive Capital

An important modelling choice is that the bilateral and aggregate portfolio data (IMF PIP and Eurostat financial balance-sheet data) capture portfolio equity and debt positions, not total economy-wide productive capital. Therefore, the simulation does not replace national capital stocks with portfolio claims. Instead, we use portfolio reallocation to generate a proportional shock to the productive capital base from Penn World Tables.

For any portfolio allocation matrix Π , portfolio-linked financial capital received by country i is

$$k_i^{\text{fin}} = \sum_{j=1}^n \pi_{j,i} \cdot s_j, \quad s_j = \sum_{\iota=1}^n a_{j,\iota} \quad (22)$$

with baseline and counterfactual values denoted $k_i^{\text{fin,base}}$ and $k_i^{\text{fin,CMU}}$.

To translate this into production-relevant capital, we define a bounded exposure weight

$$\phi_i = \frac{k_i^{\text{fin,base}}}{k_i^{\text{fin,base}} + k_i^{\text{PWT}}}, \quad \phi_i \in [0, 1] \quad (23)$$

and financial-capital growth

$$g_i^{\text{fin}} = \frac{k_i^{\text{fin,CMU}} - k_i^{\text{fin,base}}}{k_i^{\text{fin,base}}}. \quad (24)$$

Effective productive capital under CMU is then

$$k_i^{\text{eff,CMU}} = k_i^{\text{PWT}} \left(1 + \phi_i g_i^{\text{fin}} \right) \quad (25)$$

or equivalently

$$k_i^{\text{eff,CMU}} = k_i^{\text{PWT}} \left[(1 - \phi_i) + \phi_i \frac{k_i^{\text{fin,CMU}}}{k_i^{\text{fin,base}}} \right]. \quad (26)$$

This mapping keeps baseline capital anchored to PWT, avoids arbitrary capping rules, and prevents implausible full replacement of productive capital by financial claims. It also makes the scaling role of ϕ_i explicit. Since ϕ_i measures the share of total capital exposure accounted for by portfolio equity and debt relative to the broader capital base, any portfolio-induced effect is attenuated when foreign finance is small compared with total capital. Countries with lower financial exposure therefore experience smaller capital and productivity responses even when portfolio positions change.

A limitation is that portfolio equity and debt positions are financial claims and may overlap conceptually with the productive capital stock already measured in PWT. For this reason, k_i^{fin} is not interpreted as an additional physical capital stock and is not added directly to k_i^{PWT} . Instead, it is used only to construct a bounded exposure weight and a proportional reallocation shock. The mapping should therefore be interpreted as a modelling bridge from portfolio reallocation to productive capital, not as a national-accounting identity.

5.10.2 Endogenous TFP

We model productivity as a linear function of finance exposure intensity:

$$A_i(\theta) = \bar{A}_i \cdot (1 + \theta \cdot \phi_i) \quad (27)$$

where $\bar{A}_i$ is baseline TFP from the Penn World Tables, ϕ_i is the bounded exposure weight from equation (23), and $\theta \geq 0$ is the spillover elasticity. When $\theta = 0$, the model collapses to the fixed-TFP baseline. When $\theta > 0$, countries with higher finance exposure receive a larger productivity level adjustment. Because this adjustment is scaled by ϕ_i , the model does not assume that all foreign portfolio claims translate one for one into productive spillovers or into FDI-style effects.

This scaling is important for interpretation. The TFP extension is designed to show the likely direction and relative amplification of the CMU effect under stronger financial integration, not to forecast the true quantitative size of productivity gains. The absolute magnitudes should therefore be interpreted cautiously.

5.10.3 Output with Endogenous TFP

Country level output is given by a Cobb Douglas production function:

$$y_i^{\text{base}} = A_i(\theta) \cdot (k_i^{\text{PWT}})^{\alpha_i} \cdot L_i^{1-\alpha_i} \quad (28)$$

$$y_i^{\text{CMU}} = A_i(\theta) \cdot (k_i^{\text{eff,CMU}})^{\alpha_i} \cdot L_i^{1-\alpha_i} \quad (29)$$

where L_i is labour input and α_i is the capital share of income. Hence, CMU affects GDP through the capital channel in equation (25): changes in portfolio allocations first change k_i^{fin} , then $k_i^{\text{eff,CMU}}$, and then output through the Cobb–Douglas production function.

The total output effect of a CMU scenario relative to baseline can be decomposed into a capital deepening component and a TFP component:

$$\frac{\Delta y_i}{y_i} \approx \alpha_i \cdot \frac{\Delta k_i^{\text{eff}}}{k_i^{\text{PWT}}} + \frac{\Delta A_i}{A_i} \quad (30)$$

A limitation of this extension is that the spillover literature typically studies foreign direct investment rather than portfolio equity investment. Our model applies this mechanism to portfolio equity and debt exposure, which are likely to transmit weaker spillovers than direct investment with control rights and managerial involvement. For this reason, the productivity channel should be interpreted cautiously and primarily as a structured robustness extension.

5.10.4 Calibrated Baseline Productivity

Rather than using raw Penn World Tables TFP directly as the productivity input, we calibrate a baseline productivity residual A_i^0 so that the model-implied output at $\theta = 0$ exactly reproduces observed GDP for each country. This ensures internal consistency: the production block is anchored to true national income data rather than to the stated TFP from the Penn World Tables.

The calibrated productivity is defined as

$$A_i^0 = \frac{Y_i}{(k_i^{\text{PWT}})^{\alpha_i} \cdot L_i^{1-\alpha_i}} \quad (31)$$

so that with $\theta = 0$ and $k_i = k_i^{\text{PWT}}$, baseline output satisfies $y_i^{\text{base}} = Y_i$ exactly. The raw PWT TFP series

is retained in the data export as a diagnostic variable but is not used in the production block. A_i^0 is not a new structural estimate; it is a calibration residual, analogous in spirit to the diagonal wedge calibration, that forces one model object to match one data moment exactly. The PWT TFP is used as a diagnostic to verify that A_i^0 is broadly consistent with the measured productivity series across countries.

When $\theta > 0$, baseline output rises proportionally with finance exposure:

$$y_i^{\text{base}}(\theta) = A_i^0 \cdot (1 + \theta \cdot \phi_i) \cdot (k_i^{\text{PWT}})^{\alpha_i} \cdot L_i^{1-\alpha_i} = Y_i \cdot (1 + \theta \cdot \phi_i) \quad (32)$$

and CMU changes output by shifting $k_i^{\text{eff,CMU}}$ through the (ω, γ) shock. All output effects reported in the results section are therefore measured relative to this GDP-anchored baseline.

5.10.5 Calibration of θ

The spillover elasticity θ is calibrated rather than estimated from our own data. Following Kose et al. (2009), we consider $\theta \in [0, 0.10]$ as a sensitivity range informed by evidence from the broader international capital-flows literature. In a portfolio setting this range should be interpreted as an upper bound, since the underlying spillover mechanism is theoretically weaker for passive minority investment than for FDI with managerial involvement, as discussed in Section 3.8. The lower bound $\theta = 0$ corresponds to no productivity spillovers, while higher values provide illustrative benchmarks for how much CMU gains could be amplified under stronger spillover assumptions.

In the simulation code, θ is evaluated on a grid of 20 equally spaced values between 0 and 0.10. This allows us to assess whether the sign and relative size of the CMU effect are robust to alternative assumptions about productivity spillovers. It does not imply that the resulting GDP changes should be read as point forecasts of the true productivity effect of portfolio integration.

5.11 CMU Shock

The central policy experiment in the thesis is a reduction in bilateral frictions within the EU sample. The model distinguishes between two policy parameters. The parameter ω governs the reduction in

hard frictions, which are represented by the financial development wedge. The parameter γ governs the reduction in the policy-sensitive soft frictions, which are represented by the informational wedge. The linguistic and geographic wedges are treated as fixed and are not directly altered by policy.

Let $\mathcal{E}$ denote the set of EU countries in the sample. The CMU hard wedge is defined as

$$\Delta_{ij,\text{fin}}^{\text{CMU}} = \begin{cases} \left(\Delta_{ij}^{\text{fin}}\right)^{1-\omega} & \text{if } i \neq j \text{ and } i, j \in \mathcal{E} \\ \Delta_{ij}^{\text{fin}} & \text{otherwise} \end{cases} \quad (33)$$

The CMU informational wedge is defined as

$$\Delta_{ij,\text{news}}^{\text{CMU}} = \begin{cases} \left(\Delta_{ij}^{\text{news}}\right)^{1-\gamma} & \text{if } i \neq j \text{ and } i, j \in \mathcal{E} \\ \Delta_{ij}^{\text{news}} & \text{otherwise} \end{cases} \quad (34)$$

The linguistic and geographic wedges remain unchanged:

$$\Delta_{ij}^{\text{ling,CMU}} = \Delta_{ij}^{\text{ling}} \quad (35)$$

$$\Delta_{ij}^{\text{geo,CMU}} = \Delta_{ij}^{\text{geo}} \quad (36)$$

The full CMU wedge is therefore

$$\Delta_{ij}^{\text{CMU}} = \Delta_{ij,\text{fin}}^{\text{CMU}} \cdot \Delta_{ij}^{\text{ling}} \cdot \Delta_{ij}^{\text{geo}} \cdot \Delta_{ij,\text{news}}^{\text{CMU}} \quad (37)$$

This functional form ensures that the wedges remain positive and that a full reduction, such as $\omega = 1$ or $\gamma = 1$, sets the relevant friction component equal to unity for intra EU pairs. In implementation, the diagonal wedges are calibrated once in the baseline and then held fixed across all (ω, γ) scenarios,

while only off diagonal wedges are shocked. For each scenario, expected returns are updated iteratively in general equilibrium until convergence, and portfolio shares are recomputed using the converged return vector.

In the simulation code, both ω and γ are evaluated on grids of 20 equally spaced values on the interval $[0, 1]$. Portfolio allocations, portfolio-linked financial capital, effective productive capital, TFP scenarios, and output are recomputed for every (ω, γ) pair.

The hard-friction value $\omega = 0.30$ is anchored to Coeurdacier and Martin (2009), who estimate that euro adoption reduced bilateral equity transaction costs by approximately 27% through the combined currency-union and preferential financial-liberalisation channels. This is treated as the closest available integration analogue for CMU. As reviewed in Section 3.2, however, the Coeurdacier-Martin premium was estimated on pre-crisis data covering roughly 1997 to 2003, and subsequent evidence from Lane (2013) and Lane and Milesi-Ferretti (2018) documents that the within-Eurozone integration premium attenuated substantially after the 2008 financial crisis and the 2010 to 2012 sovereign debt episode. The central $\omega = 0.30$ value therefore reflects the upper end of what a post-crisis CMU programme might plausibly achieve, and the simulation grid evaluates lower values of ω to bound the attenuated-regime case identified in Section 3.2.

The soft-friction value $\gamma = 0.10$ is set below ω to reflect two considerations established in the literature review. First, Véron and Wolff (2016) characterise CMU as operating directly on hard institutional frictions, with bilateral information flows responding more indirectly. Second, the CMU complementary measures targeting the informational dimension reviewed in Section 3.6, the European Single Access Point for issuer disclosures, consolidated tape provision, and harmonised reporting standards, are partial rather than complete reductions in the underlying informational environment. The $\gamma = 0.10$ value therefore represents a modest but non-zero informational channel consistent with these complementary policy instruments, and the simulation grid evaluates the full range of γ values to characterise the model's sensitivity to this calibration.

The spillover value $\theta = 0.03$ reflects Kose et al. (2009), who find that portfolio equity liabilities raise TFP growth but with effects substantially weaker than for FDI. The central value is therefore set toward the lower end of the FDI-based literature surveyed in Section 3.8, consistent with the modelling implication

that FDI-based spillover elasticities should be treated as upper bounds when applied to portfolio equity. All three values are reference points; sensitivity is evaluated across the full grid.

5.11.1 Return Elasticity

The return elasticity η governs how strongly investors respond to return differentials across destination countries. In the portfolio allocation rule, returns enter as R_j^η , so a higher value of η makes investors more sensitive to differences in expected returns, while a lower value dampens this response. In the baseline, we set $\eta = 1.0$ and compare this with alternative values $\eta \in \{0.5, 2.0\}$.

It is important to distinguish between the direct and indirect role of η . The parameter does not directly change country productivity, capital shares, or the return formula itself. Instead, it changes how strongly expected returns affect portfolio allocation. However, because the model updates returns after capital has been reallocated, different values of η can indirectly affect equilibrium returns. A higher return elasticity may shift more capital toward initially high-return destinations, which then changes their capital stock and therefore their marginal product of capital in the next iteration. Thus, η should be interpreted as an investor sensitivity parameter that can influence equilibrium returns through the capital-allocation feedback mechanism.

The robustness checks therefore examine whether the main conclusions depend on the baseline assumption $\eta = 1.0$. If the qualitative results remain unchanged across $\eta \in \{0.5, 1.0, 2.0\}$, such as lower home bias, higher EU output, and lower dispersion in the marginal product of capital, then the model can be viewed as robust to this parameter choice.

5.11.2 TFP Spillover Elasticity

The baseline case for the productivity extension is $\theta = 0$, which shuts down the endogenous TFP channel completely. We then evaluate higher values of θ up to 0.10. This allows us to test whether the gains from CMU are amplified when foreign capital also raises productivity.

If EU GDP gains increase monotonically in θ for a given policy experiment, then the no-spillover case serves as a benchmark against which the sensitivity of the results can be evaluated.

5.11.3 Policy Parameters

The most important robustness exercise concerns the policy parameters themselves. The model evaluates the full grid of ω and γ values on $[0, 1]$, which allows us to compare partial and full implementation of CMU.

We study cases where only hard frictions are reduced, by varying ω while setting $\gamma = 0$, and cases where informational frictions are reduced, by varying γ while holding ω fixed. We also study joint changes in both parameters. Since the linguistic and geographic wedges remain fixed throughout, the parameter γ should be interpreted specifically as a reduction in informational barriers rather than in all soft barriers more generally.

Taken together, these exercises show whether the main conclusions of the model depend on a narrow parameterisation or whether they hold across a broader set of plausible assumptions.

6 Results

This section presents the results from the gravity estimation and the counterfactual Capital Markets Union simulation. The interpretation follows the sequence of the model. First, the gravity model estimates how bilateral frictions affect observed portfolio investment. Second, these estimated frictions are used to construct bilateral wedges. Third, the CMU counterfactual reduces selected wedges and generates new portfolio allocations. Finally, the resulting capital reallocation is translated into model-implied output effects through the calibrated Cobb–Douglas production block.

The results should therefore be interpreted as model-implied effects of financial integration, not as direct forecasts of realised GDP. The model directly simulates portfolio reallocations. Output changes are secondary effects obtained by mapping changes in effective capital into production. To make aggregate output comparisons meaningful, the production block is calibrated so that baseline country-level model output matches observed GDP in the base year when $\theta = 0$.

6.1 Gravity Model and Friction Estimates

The first stage of the analysis estimates how bilateral frictions affect observed portfolio investment positions. The gravity regression provides the empirical link between bilateral investment flows and the four friction variables used in the simulation: linguistic distance, geographic distance, informational distance, and financial-development distance. These estimated coefficients determine the strength of the corresponding wedge components in the counterfactual model.

The estimated coefficients support the interpretation of these variables as frictions to cross-border portfolio investment. All four distance variables enter with negative signs. Linguistic distance has a coefficient of -2.041 ($p < 0.001$) and is statistically significant at the 1 percent level. News/information distance has a coefficient of -7.190 ($p = 0.044$) and is statistically significant at the 5 percent level. Financial-development distance has a coefficient of -4.215 ($p < 0.001$) and is statistically significant at the 1 percent level. Geographic distance is also negative, with a coefficient of -3.130×10^{-4} ($p = 0.012$), and is statistically significant at the 5 percent level. The signs therefore indicate that countries invest less in destinations that are linguistically, informationally, financially, or geographically more distant.

The two GDP controls enter positively and significantly. Origin GDP has a coefficient of 0.452 ($p < 0.001$), while destination GDP has a coefficient of 0.435 ($p < 0.001$), and both are statistically significant at the 1 percent level. This is consistent with the gravity structure of the model, where larger origin economies have more investable portfolio wealth and larger destination economies attract more investment. The regression is estimated on 2,858 bilateral observations, includes year fixed effects, and has a pseudo R^2 of 0.249.

Table 8 reports the PPML coefficient estimates. The coefficients are not interpreted as structural elasticities in isolation. Their main role is to discipline the wedge matrices used in the simulation. A negative coefficient means that the corresponding distance variable raises the bilateral wedge and therefore lowers the attractiveness of the destination country in the portfolio allocation rule. The fact that all four friction variables enter negatively supports the use of these variables as barriers inhibiting free-market portfolio allocation across countries.

Table 8: PPML gravity model estimates

Regressor	Dependent variable: a_{ij}
	PPML
Linguistic distance, d_{ij}^{ling}	-2.041^{***} (0.447)
News/information distance, d_{ij}^{news}	-7.190^{**} (3.575)
Financial-development distance, d_{ij}^{fin}	-4.215^{***} (0.921)
Geographic distance, d_{ij}^{geo}	$-3.130 \times 10^{-4**}$ (1.240×10^{-4})
Origin GDP, Y_i	0.452^{***} (0.047)
Destination GDP, Y_j	0.435^{***} (0.039)
Year fixed effects	Yes
Observations	2,858
Pseudo R^2	0.249
Clustered standard errors	Bilateral pair

Notes: The table reports Poisson pseudo-maximum likelihood estimates of the bilateral portfolio investment gravity model. Standard errors clustered at the bilateral-pair level are reported in parentheses. Year fixed effects are included but not reported. Monetary variables are scaled for numerical stability. Significance levels are denoted by $***p < 0.01$, $**p < 0.05$, and $*p < 0.10$.

To quantify the economic magnitude of each friction, the PPML coefficients can be translated into percentage changes in bilateral portfolio holdings via $(\exp(\hat{\beta}) - 1) \times 100$. A move from zero to the maximum linguistic distance is associated with a change of $(\exp(-2.041) - 1) \times 100 \approx -87.0$ percent in bilateral holdings, implying that a country pair at opposite ends of the linguistic distance scale holds roughly seven times less

cross-border portfolio investment than an otherwise identical pair sharing a common linguistic heritage. The news/informational distance coefficient implies an even larger effect: $(\exp(-7.190) - 1) \times 100 \approx -99.9$ percent across the full $[0, 1]$ range of the variable, reflecting the extreme concentration of news coverage in this sample. Most observations are clustered above 0.99, so a full one-unit decrease is unlikely in practice. Financial-development distance implies $(\exp(-4.215) - 1) \times 100 \approx -98.5$ percent over its full range, indicating that a large gap in financial development is associated with near-zero bilateral investment. Geographic distance, which is measured in kilometres, implies $(\exp(-3.130 \times 10^{-4} \times 1000) - 1) \times 100 \approx -26.9$ percent per additional 1,000 km, a more moderate but still economically meaningful friction. These magnitudes confirm that all four distance variables constitute substantial barriers to cross-border portfolio allocation and justify their inclusion as wedge components in the counterfactual simulation.

The gravity model is not expected to predict every bilateral position exactly. Some country pairs involve financial-centre effects, tax structures, or institutional ownership patterns that are difficult to capture with a parsimonious friction specification. The relevant validation exercise is whether the model produces plausible signs and enough cross-sectional variation to support the simulation. The predicted-versus-actual diagnostic plots are therefore reported in the appendix.

6.2 Aggregate EU Output Effects and Policy Channel Decomposition

The central result is that the simulated CMU shock raises model-implied EU output over most of the policy space. Figures 4 and 5 report the percentage change in aggregate EU output across the policy parameters ω and γ , and the productivity amplification parameter θ . The parameter ω reduces hard financial-development wedges, γ reduces informational wedges, and θ changes how strongly finance exposure enters the productivity term in the production function. Unlike ω and γ , θ does not change portfolio allocation directly.

Across the full simulation grid of 8,000 scenarios, the distribution of model-implied EU output effects is centred around a positive value. The mean change in EU output is 1.692 percent, the median is the same at 1.692 percent, and the central outcome is an increase of 0.75 percent. The standard deviation across scenarios is 0.909 percentage points. The minimum simulated effect is -1.053 percent and the maximum is 3.937 percent, giving a total range of approximately 4.99 percentage points. The largest output gain

occurs when both policy channels are fully activated, $\omega = 1$ and $\gamma = 1$, where model-implied EU output increases by approximately 3.94 percent.

Figure 4 shows that the largest gains occur in the upper-right part of the policy space, where both ω and γ are high. However, the slope of the heatmap is not symmetric across the two policy dimensions. Moving upward along the γ dimension produces a clear increase in EU output. Moving along the ω dimension produces a weaker and more ambiguous effect. In the hard-only case, where $\omega = 1$ and $\gamma = 0$, model-implied EU output falls by approximately 1.05 percent. This shows that reducing hard financial-development wedges alone does not necessarily raise aggregate output in the model.

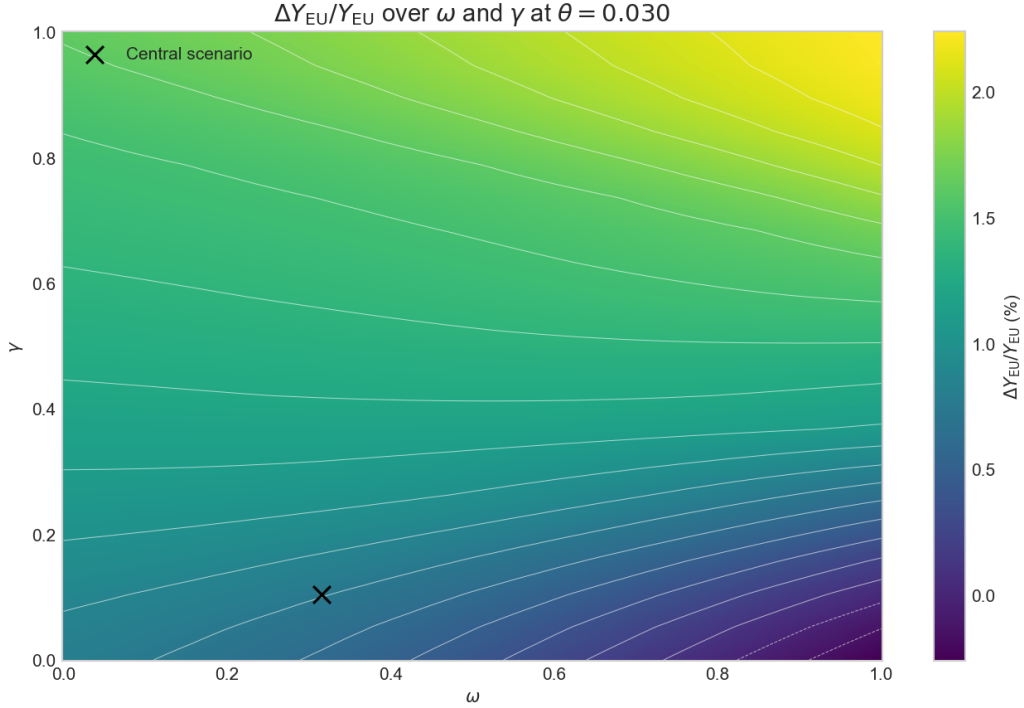


Figure 4: Model-implied EU output change over ω and γ at the main value of θ .

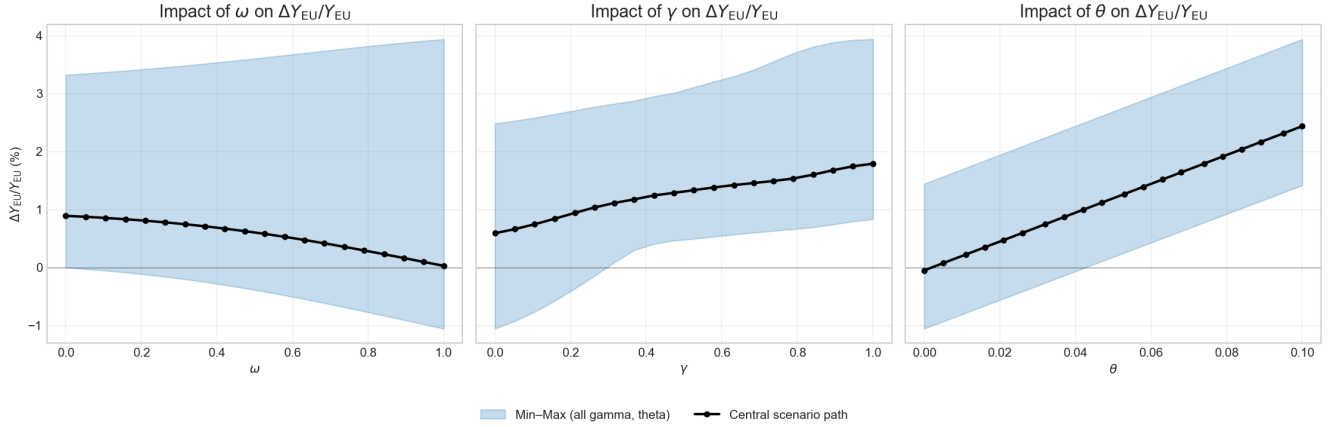


Figure 5: Marginal effects of ω , γ , and θ on model-implied EU output.

Figure 5 decomposes the same result into one-dimensional relationships. The shaded areas show the full range of outcomes across the remaining parameter combinations, while the black line reports the central scenario path which shows the average across all other variables for that specific value of e.g. γ . The three parameters have very different relationships with $\Delta Y_{EU}/Y_{EU}$. The correlation between ω and EU output changes is almost zero, at -0.011. This confirms that the hard-friction channel has little linear relationship with aggregate output across the full scenario grid. Its effect can be positive or negative depending on the values of γ and θ .

By contrast, the informational channel has a clear positive relationship with EU output. The correlation between γ and $\Delta Y_{EU}/Y_{EU}$ is 0.515. As γ increases, the distribution of output effects shifts upward, which is visible both in the black path and in the shaded min-max band. This means that scenarios with larger reductions in informational frictions are generally associated with higher model-implied EU output. Among the two policy parameters, γ therefore has the stronger positive association with aggregate output.

The productivity parameter θ has the strongest correlation with EU output changes, at 0.832. This reflects its role as an amplification parameter. Higher values of θ raise the measured output effect because productivity scales the output impact of the capital reallocation. Since θ does not enter the portfolio allocation rule, it does not determine where capital flows. Instead, it changes how strongly a given capital reallocation is translated into output.

Taken together, the aggregate results show that model-implied EU output gains are positive on average, but the policy channels differ substantially. The informational channel is consistently associated with higher EU output. The TFP channel amplifies the output effect. The hard-friction channel is much more ambiguous and, in the hard-only case, can generate a negative aggregate output effect.

The aggregate result should be read as the model-implied change in EU GDP conditional on the production mapping. Since baseline output is calibrated to observed GDP, the aggregate effect can be interpreted as the percentage change in EU output implied by the simulated capital reallocation. However, it remains a model result rather than a direct macroeconomic forecast.

6.3 Capital Reallocation Across Europe

The main mechanism behind the aggregate output effect is capital reallocation. CMU lowers intra-European wedges and changes the relative attractiveness of destination countries. As a result, model-implied portfolio capital is reallocated across countries even though total financial capital is conserved.

Figure 6 shows the percentage change in model-implied portfolio capital received under the main CMU scenario. The reallocation is highly uneven across countries. The largest proportional increase is observed in Estonia, where portfolio capital received rises by 38.5 percent. Lithuania receives the second-largest proportional increase, with a gain of 27.1 percent. Slovakia follows with an increase of 22.8 percent. Latvia and Romania also receive large proportional gains, with increases of 22.2 percent and 21.5 percent, respectively.

The largest proportional declines are also significant. Italy records the largest fall, with model-implied portfolio capital received declining by 12.0 percent. France follows with a decline of 9.5 percent. Germany loses 5.7 percent, Spain loses 5.2 percent, and the United Kingdom records a decline of 2.4 percent. These numbers show that the CMU shock does not increase portfolio capital received uniformly across all countries. Instead, it changes the cross-country distribution of capital within the model.

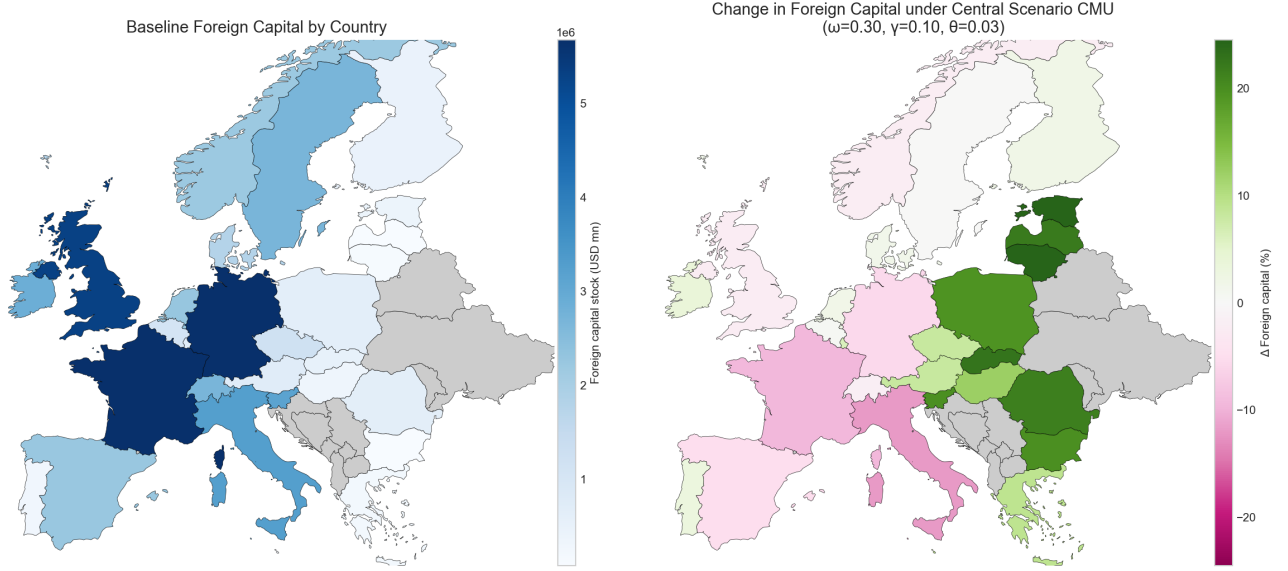


Figure 6: Percentage change in model-implied portfolio capital received under the main CMU scenario.

The figure therefore shows that CMU reallocates capital across the European portfolio network rather than increasing inflows everywhere. Some economies, such as Estonia, Lithuania, Slovakia, Latvia, and Romania, receive large proportional gains. Other countries, including Italy, France, Germany, Spain, and the United Kingdom, receive less portfolio capital in the main scenario. It is best described as a reallocation of model-implied portfolio capital across countries according to changes in relative attractiveness after wedge reductions.

This result is central for the economic interpretation of CMU. The model does not suggest that CMU merely deepens existing financial centres or increases capital everywhere. Instead, it predicts that lower frictions redirect existing portfolio capital toward a new cross-country allocation. These portfolio-capital changes are not interpreted as direct changes in physical capital. They are later translated into effective productive capital through the bounded exposure weight ϕ_i . In the thesis discussion, this should be interpreted as a capital-market reallocation and convergence mechanism rather than as redistribution in a fiscal sense.

6.4 Country-Level GDP Winners and Losers

The country-level output effects are reported in Figures 7 and 8. These results translate the simulated portfolio-capital reallocation into model-implied GDP changes through the calibrated production block. The largest percentage gains are concentrated in smaller Eastern European and Baltic economies. Estonia records the largest model-implied GDP gain, at 8.79 percent. Slovenia follows with a gain of 5.59 percent, while Slovakia gains 3.87 percent. Lithuania and Romania also record sizeable gains, at 2.63 percent and 2.60 percent, respectively.

The weakest country-level effects are concentrated among larger economies and some non-EU countries. Italy records the largest percentage decline, at -0.81 percent, followed closely by France at -0.81 percent. Norway, Germany, and Spain also record negative model-implied GDP effects, with changes of -0.54 percent, -0.49 percent, and -0.31 percent, respectively.

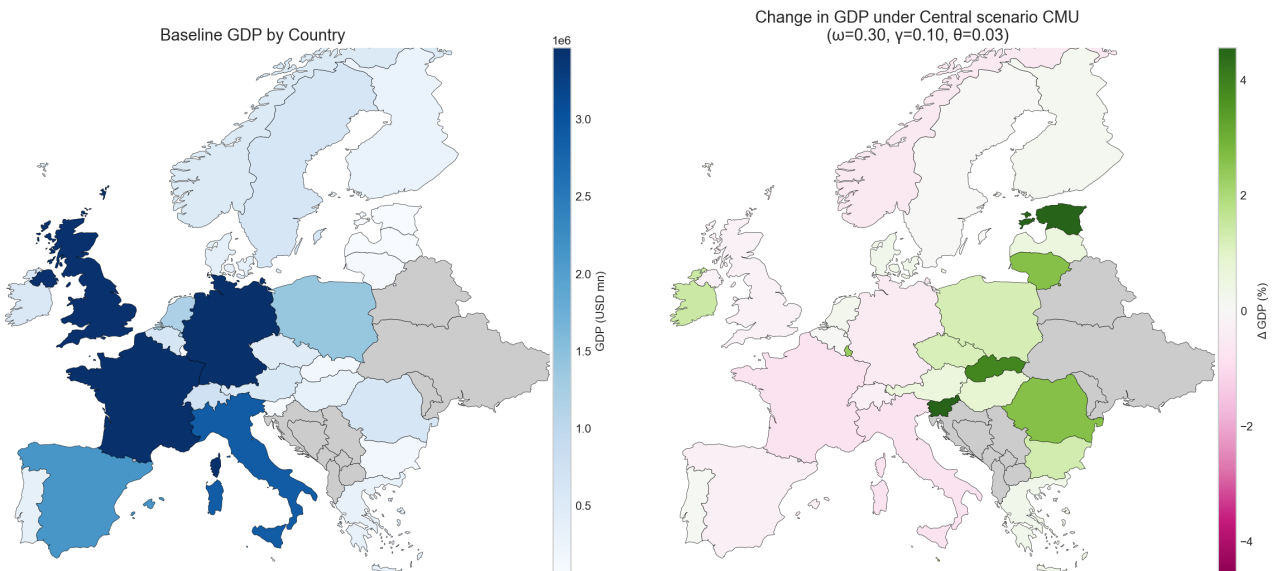


Figure 7: Model-implied GDP change under the main CMU scenario.

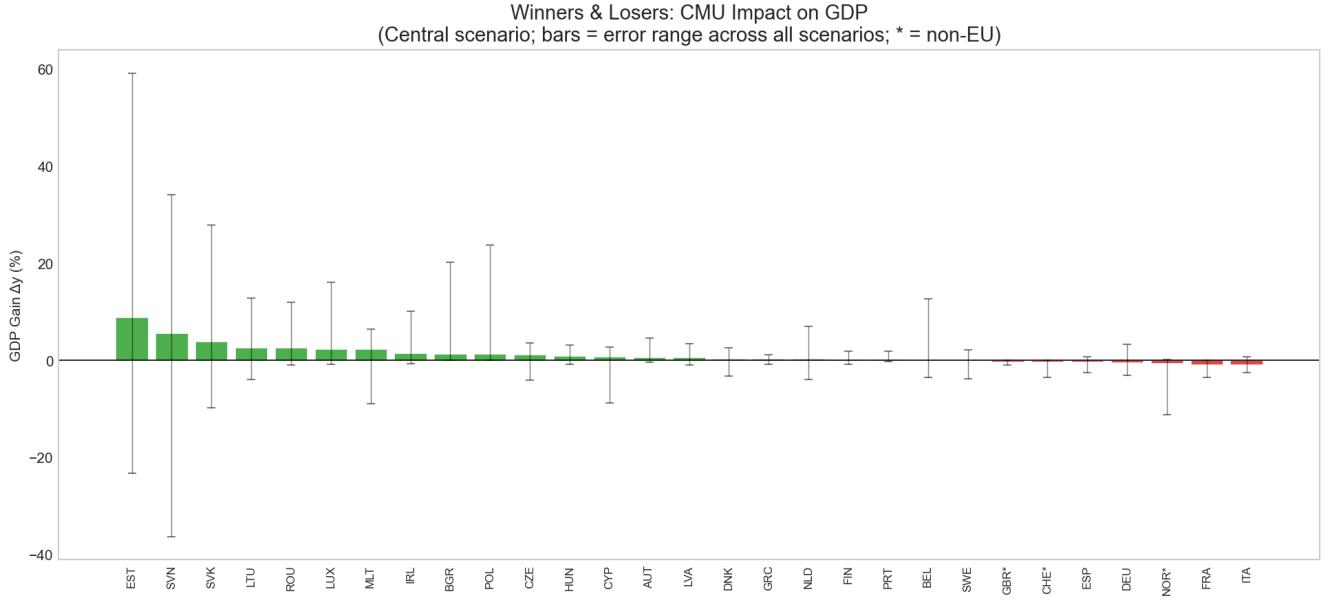


Figure 8: Country-level winners and losers under the main CMU scenario.

The GDP winners and losers map closely onto the portfolio-inflow results. Estonia stands out as a clear example: it receives the largest proportional increase in portfolio capital at 38.5 percent, and it also records the largest GDP gain at 8.79 percent. This tight alignment between capital gains and output gains illustrates the direct transmission mechanism: increased portfolio capital translates into higher effective productive capital and stronger output effects. This reflects the fact that the GDP effect is calculated through the bounded exposure weight ϕ_i , where portfolio reallocation is first translated into effective productive capital, and output is then computed through the Cobb–Douglas production block.

The capital and GDP results are consistently aligned for the large losers as well. Italy and France both appear among the largest proportional declines in portfolio capital received and also record the weakest GDP outcomes. Italy loses 12.0 percent of portfolio capital and records a GDP change of -0.81 percent, while France loses 9.5 percent of portfolio capital and records a GDP change of -0.81 percent. Germany, Spain, and the United Kingdom also experience negative portfolio capital flows and negative GDP effects, reinforcing the pattern that weaker capital inflows translate into lower model-implied output. This indicates a consistent transmission mechanism: the capital reallocation induced by CMU frictions operates systematically through the production block to generate the observed GDP outcomes.

The main conclusion is that CMU is not distributionally neutral. It raises aggregate EU output in the main scenario, but the gains are unevenly distributed across countries. Smaller economies such as Estonia, Slovenia, Slovakia, Lithuania, and Romania experience the largest percentage GDP gains, while some larger economies such as Italy, France, Germany, and Spain experience negative percentage effects. These results provide the basis for the discussion of convergence, political economy, and the uneven distribution of gains from financial integration.

6.5 Domestic Share and Home-Bias Reduction

A direct implication of lower cross-border frictions is a reduction in home bias. Figure 9 shows the change in domestic portfolio shares under the main CMU scenario. The result is clear: domestic shares fall for all EU countries in the sample, while the only positive changes occur among the non-EU countries.

The largest increases in domestic share are observed outside the EU. Norway's domestic share rises from 0.510 to 0.526, an increase of 1.61 percentage points, or 3.16 percent relative to baseline. Switzerland's domestic share rises from 0.700 to 0.713, an increase of 1.28 percentage points, or 1.82 percent. The United Kingdom's domestic share rises from 0.762 to 0.766, an increase of 0.43 percentage points, or 0.57 percent. These are the only countries with positive changes in domestic share.

Among EU countries, domestic shares decline throughout. The largest fall is in Italy, where the domestic share falls from 0.661 to 0.503, a decline of 15.76 percentage points, or 23.84 percent relative to baseline. Belgium follows with a fall from 0.478 to 0.323, a decline of 15.50 percentage points, or 32.41 percent. Germany's domestic share falls from 0.646 to 0.519, a decline of 12.69 percentage points, while France falls from 0.721 to 0.594, also a decline of 12.69 percentage points. Spain's domestic share falls from 0.751 to 0.629, a decline of 12.16 percentage points.

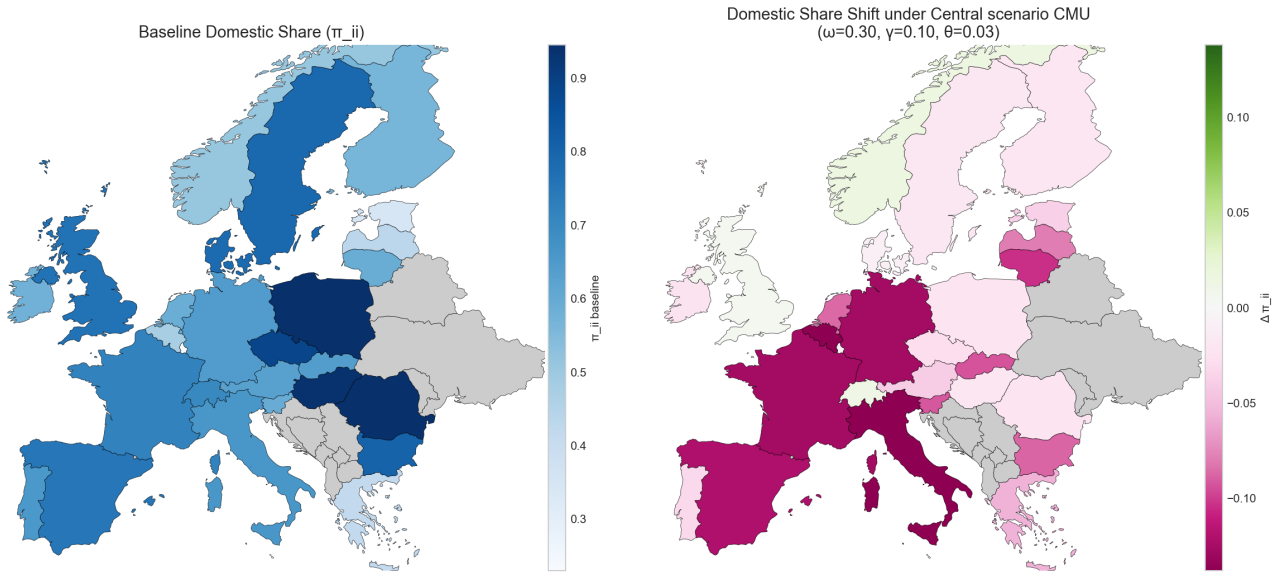


Figure 9: Change in domestic portfolio share under the main CMU scenario.

These domestic-share results complement the capital-flow and GDP results. Italy and France are among the largest domestic-share decliners, and they also appear among the countries with weaker outcomes in both portfolio capital received and model-implied GDP. Italy's model-implied GDP falls by 0.81 percent and France's by 0.81 percent. Germany and Spain also experience large domestic-share reductions and negative GDP effects of 0.49 percent and 0.31 percent, respectively.

The pattern confirms that the CMU shock operates through the intended financial-integration channel. The model does not generate output gains through an exogenous increase in total capital; it reallocates existing portfolio wealth across destinations. Lower domestic shares across EU countries therefore indicate lower home bias and greater cross-border portfolio integration within the model. The increase in domestic shares for Norway, Switzerland, and the United Kingdom reflects portfolio reweighting in the counterfactual: because only EU–EU frictions are directly reduced, outside countries adjust to changed relative returns and allocations, which can alter their domestic shares without implying a new capital injection into the system.

6.6 Financing Conditions and the Marginal Product of Capital

The model-consistent measure of financing scarcity is the marginal product of capital. If CMU improves allocative efficiency, capital should move toward countries where baseline MPK is high, since these are countries where additional capital is most productive. Figure 10 plots baseline MPK against the percentage change in foreign portfolio capital received under the main CMU scenario.

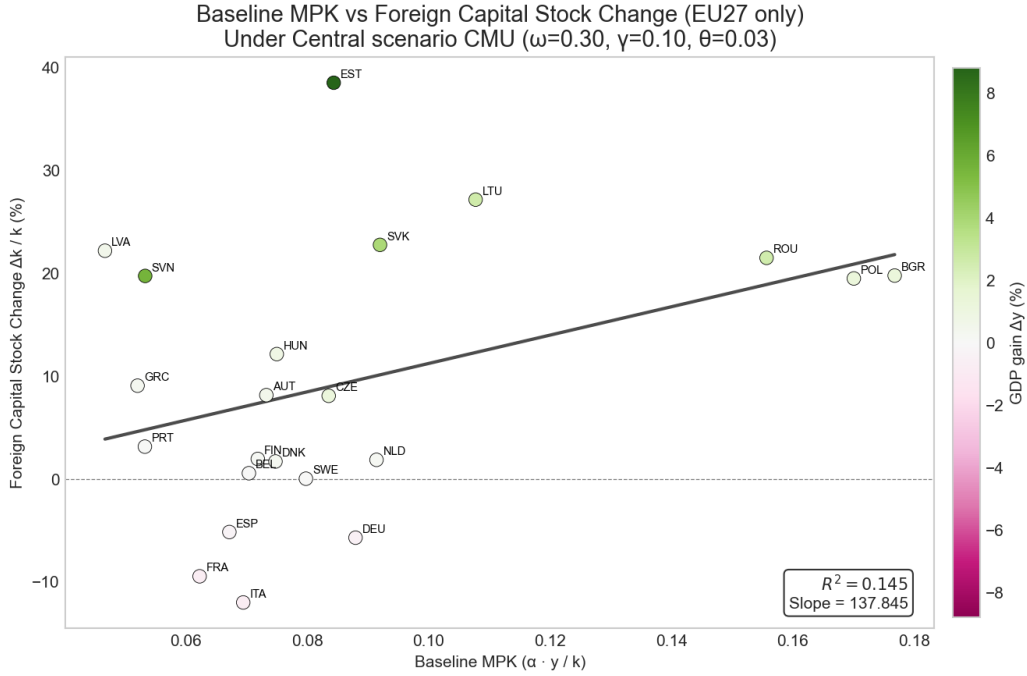


Figure 10: Baseline MPK and percentage change in foreign portfolio capital received under the main CMU scenario.

The scatterplot shows a positive relationship between baseline MPK and the change in foreign portfolio capital received. The fitted line has a positive slope of 137.845, and the regression fit has an R^2 of 0.145. Countries with higher baseline MPK therefore tend to receive larger proportional increases in foreign portfolio capital under the main scenario. Poland, Bulgaria, and Romania are positioned on the right part of the figure, with both high baseline MPK and large positive changes in foreign portfolio capital received. By contrast, countries such as France, Italy, and Spain record negative changes in foreign portfolio capital received.

Figure 11 further shows that the hard-friction and informational-friction channels have different relation-

ships with average MPK and MPK dispersion. Along the ω path, holding γ and θ fixed, average MPK falls almost monotonically. The average and median value along this path are both 0.0900 with and a standard deviation of 0.0014. The minimum is 0.0878 and the maximum is 0.0922. The correlation between ω and average MPK is -0.9999 , indicating a near-perfect negative relationship along the plotted path. Importantly, MPK dispersion along this path remains relatively constant, with a mean standard deviation of 0.0390, and a small positive correlation with ω of 0.7151, indicating that MPK convergence is not the primary mechanism along the hard-friction dimension.

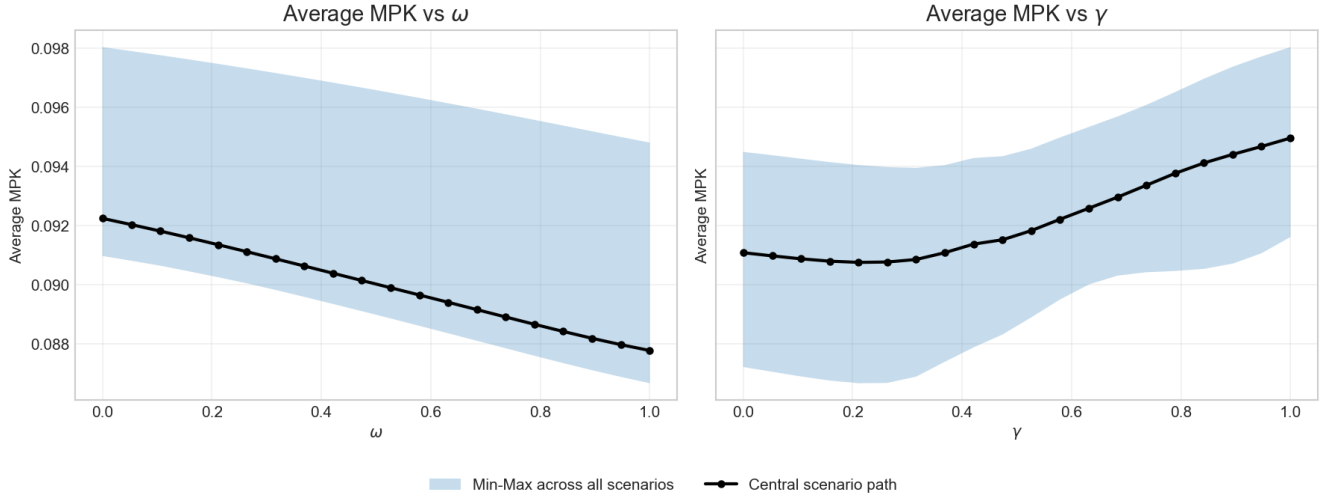


Figure 11: Average MPK along the ω and γ dimensions.

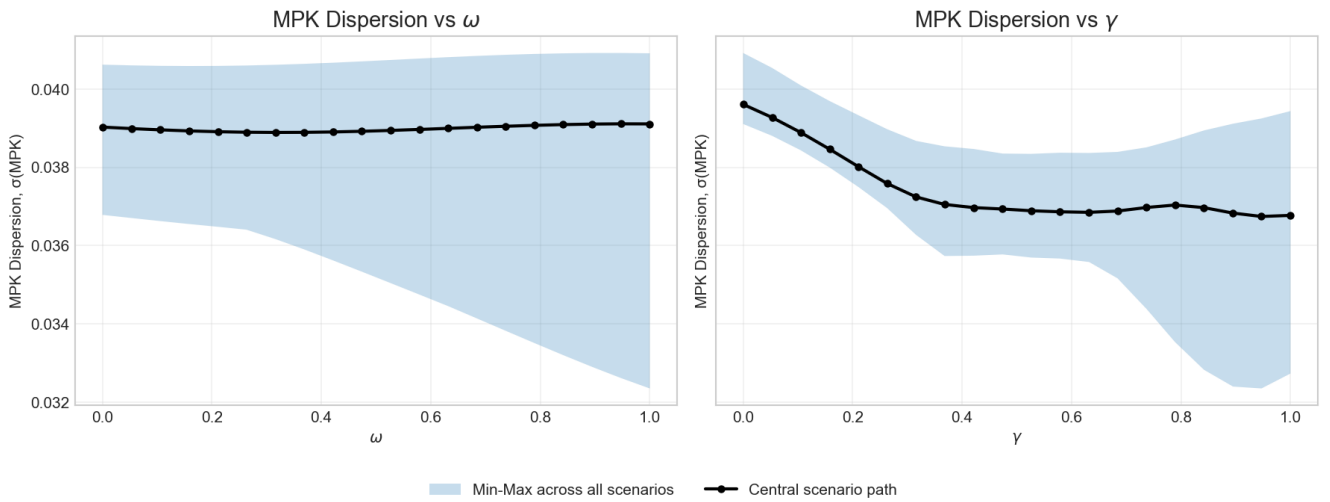


Figure 12: Standard Deviation of MPK along the ω and γ dimensions.

Along the γ path, holding ω and θ fixed, average MPK rises. The average value along this path is 0.0922, with a median of 0.0917 and a standard deviation of 0.0015. The minimum is 0.0907 and the maximum is 0.0950. The correlation between γ and average MPK is 0.9432. In contrast to the ω path, MPK dispersion falls significantly as informational frictions are reduced. The standard deviation of MPK dispersion along this path has a mean of 0.0374, ranging from a minimum of 0.0367 to a maximum of 0.0396. The correlation between γ and MPK dispersion is -0.8262 , indicating that informational-friction reductions are strongly associated with reduced cross-country variation in marginal products of capital. This pattern demonstrates that the informational-integration channel works partly through an efficiency mechanism: by reducing information asymmetries, CMU aligns marginal products of capital across countries, enabling capital to flow toward equalized productivity levels rather than being trapped by informational barriers. This shows that average MPK increases as informational frictions are reduced, while MPK dispersion declines, while by contrast, the hard-friction parameter ω leaves dispersion relatively unchanged.

The interpretation in terms of cost of capital should be cautious. The model does not estimate a market interest rate directly. However, MPK is the production-side object that corresponds most closely to capital scarcity. A lower MPK after receiving capital can be interpreted as an easing of financing scarcity, while movements in average MPK describe how the simulated allocation changes the productivity of marginal capital across the model economy.

6.7 Portfolio Reallocation Between EU and Non-EU Destinations

To examine how the CMU scenario changes the geographical allocation of portfolio investment, Figure 13 reports percentage changes in four aggregate flow categories under the central scenario. The four categories are: total EU-to-EU investment including domestic holdings, cross-border EU-to-EU investment only, EU investment in non-EU destinations, and non-EU investment in EU destinations.

The results show a clear reallocation of EU portfolios toward other EU assets. Total EU-to-EU investment, including domestic holdings, increases by 1.75 percent. The effect is much stronger when focusing only on cross-border intra-EU investment: cross-border EU-to-EU investment increases by 86.47 percent.

At the same time, EU investment in non-EU destinations declines by 20.52 percent. This shows that

EU investors substitute away from non-EU assets as European assets become relatively more attractive within the model.

By contrast, non-EU investment into the EU falls by 4.43 percent. In the current modelling setup, this means that the CMU shock mainly reallocates EU investors' own portfolios toward the EU, rather than generating higher external inflows from the non-EU countries included in the sample.

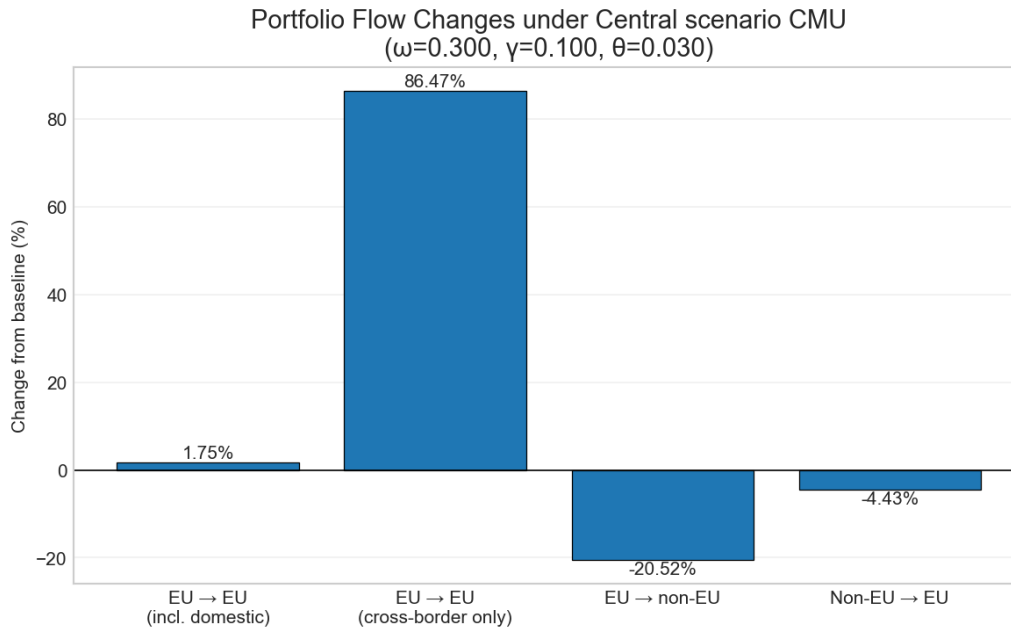


Figure 13: Percentage changes in portfolio flow categories under the Central CMU scenario.

An important limitation should be noted here. The model only reduces frictions on EU–EU links. It does not model a secondary reduction in frictions between non-EU investors and EU destinations that could arise if CMU made EU capital markets more transparent, harmonised, or financially developed from the perspective of outside investors. As a result, non-EU countries in the model do not benefit from lower effective frictions when investing in the EU; they only react to equilibrium changes in relative returns and allocations. The negative non-EU-to-EU result should therefore not be interpreted as a full estimate of how global investors would respond to a real-world CMU. Instead, the result should be interpreted more narrowly as showing that, within the present model, the dominant effect of CMU is a reallocation of European portfolios toward other European assets.

6.8 TFP Amplification

The productivity channel is designed as an amplification mechanism. The model does not assume that CMU directly changes TFP. Instead, each country has a baseline finance-exposure weight ϕ_i , and the parameter θ determines how strongly that exposure enters the production function. The TFP-adjusted productivity term is

$$A_i(\theta) = A_i^0(1 + \theta\phi_i), \quad (38)$$

where A_i^0 is calibrated so that baseline model output equals observed GDP when $\theta = 0$. For a given value of θ , the same $A_i(\theta)$ is used in both the baseline and CMU scenario. The TFP effect is therefore measured as an amplification of the output gap relative to the $\theta = 0$ case, not as a direct difference between baseline and CMU productivity.

Figure 14 shows the country-level TFP amplification in the main scenario. The measure is defined as $\Delta y_i(\theta) - \Delta y_i(\theta = 0)$, where $\Delta y_i(\theta) = y_i^{CMU}(\theta) - y_i^{baseline}(\theta)$. Across the 29 matched countries, the mean amplification is small, at 0.019 percent of baseline output at $\theta = 0$. The range is also modest, from a minimum of -0.009 percent to a maximum of 0.170 percent.

The largest positive amplification is observed in Estonia, where the TFP channel adds 0.170 percent relative to the $\theta = 0$ output effect. Slovenia follows with an amplification of 0.152 percent. Luxembourg, Slovakia, and Malta also have positive amplification effects of 0.053 percent, 0.049 percent, and 0.039 percent, respectively. At the lower end, Norway records the most negative amplification effect, at -0.009 percent, followed by France at -0.006 percent, Switzerland at -0.004 percent, Germany at -0.004 percent, and Italy at -0.004 percent.

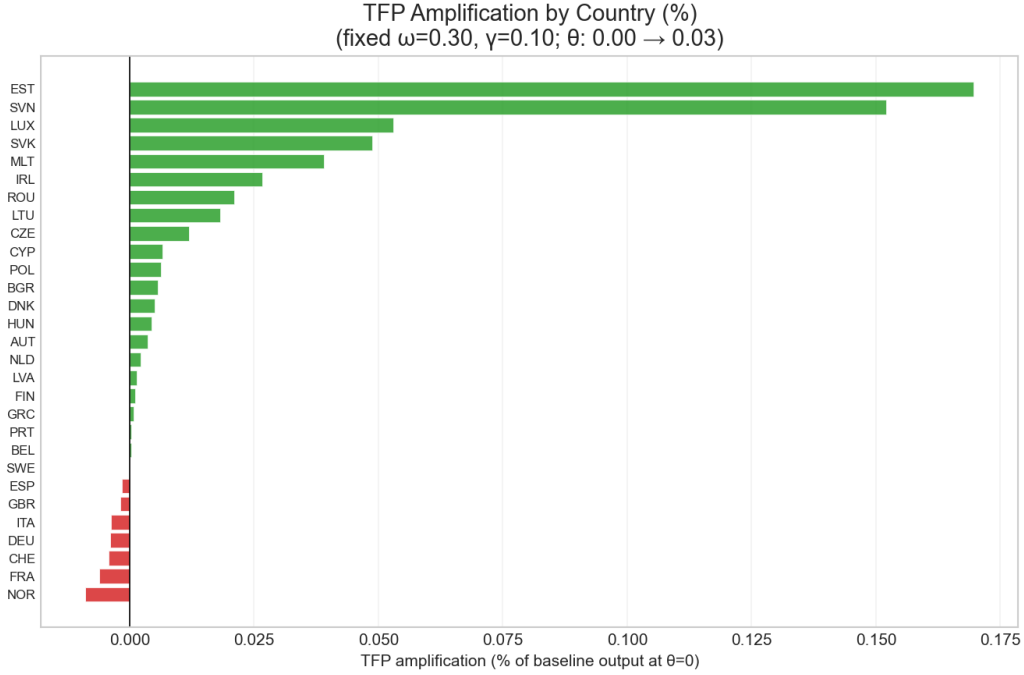


Figure 14: Country-level TFP amplification under the main CMU scenario.

The TFP result is therefore secondary to the main allocation result. The amplification effects are much smaller than the main country-level GDP effects reported above. For example, Estonia’s total model-implied GDP gain is 8.79 percent, while its TFP amplification is 0.170 percent. Slovenia’s total GDP gain is 5.59 percent, while its TFP amplification is 0.152 percent. The main driver of the simulation is therefore still the reallocation of portfolio capital caused by changes in ω and γ . The role of θ is to show how sensitive the output results are to the assumption that finance exposure raises productivity. Higher θ increases the output effect, but the underlying capital-flow pattern is determined by the wedge reductions.

6.9 Robustness and Model Validation

The final set of results evaluates whether the main conclusions depend on narrow parameter choices or on a particular calibration of the model. Overall, the qualitative findings remain stable across the main scenario grid. The model continues to predict positive EU-wide output gains across a broad range of policy combinations, the informational channel remains the strongest source of aggregate gains, and

the country-level results continue to show a reallocation of capital toward smaller and less capital-rich economies.

A central robustness concern is the assumed return elasticity, η . In the model, η governs how strongly investors respond to return differentials in the portfolio allocation rule. It should therefore be interpreted as an allocation-sensitivity parameter rather than as a direct change in country productivity or in the return formula itself. However, because the model updates returns after capital has been reallocated, different values of η can indirectly affect equilibrium returns through their effect on country-level capital stocks. The robustness exercise therefore tests whether the main results depend on the assumed strength of this return-sensitive portfolio response.

Figure 15 shows the sensitivity of the model-implied EU output effect to alternative values of $\eta \in \{0.5, 1.0, 2.0\}$. The figure indicates that the main qualitative pattern is preserved. The informational-friction parameter γ remains clearly positive across all three values of η , which supports the conclusion that informational integration is the strongest growth channel in the model. By contrast, the hard-friction parameter ω remains weaker and more ambiguous. Its effect is not uniformly positive and becomes less favourable in some parts of the parameter space. The TFP parameter θ remains an amplification channel rather than a primary driver of portfolio reallocation.

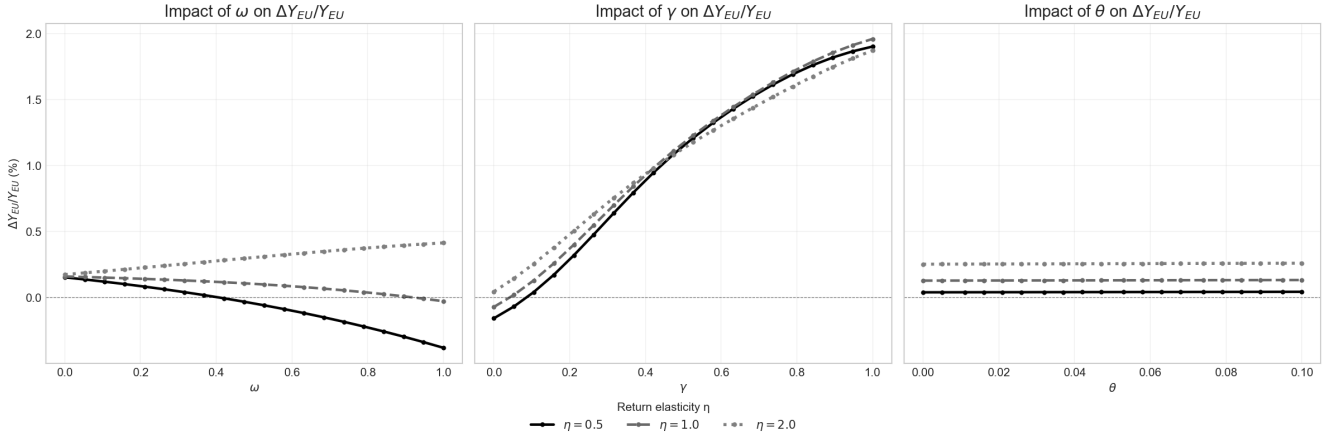


Figure 15: Robustness of the EU output effect to alternative values of the return elasticity η .

Taken together, these results suggest that the central conclusions are not an artefact of the baseline choice

$\eta = 1.0$. Lower and higher return sensitivity change the magnitude of the simulated output effects, but they do not overturn the main result: reducing informational frictions generates the clearest and most robust EU output gains, while the hard-friction channel is comparatively weaker.

Several additional validation exercises are consistent with the intended interpretation of the model, while not constituting independent causal tests. First, the domestic-share results show that the CMU shock lowers domestic portfolio shares within the EU, which is consistent with a reduction in model-implied home bias and with the intended financial-integration mechanism. Second, the capital-flow maps show that the policy shock reallocates existing portfolio capital across countries rather than mechanically increasing the total quantity of capital in the system. Third, the marginal-product-of-capital results are consistent with an allocative-efficiency interpretation: countries with higher baseline MPK tend to receive larger inflows under the main scenario, particularly when informational frictions are reduced. This evidence is suggestive rather than definitive, since a full efficiency test would require showing that MPK dispersion falls after reallocation. Fourth, the Lorenz-curve and Gini diagnostics point to a modest reduction in the concentration of gains, which is consistent with, but does not by itself prove, a convergence interpretation. To keep the main text focused, these supporting diagnostics are reported in the appendix.

The robustness exercises also clarify the limits of the framework. The output effects are model-implied and depend on the Cobb–Douglas mapping from capital reallocation to GDP. The model does not incorporate all macroeconomic adjustment channels, such as endogenous savings, labour-market responses, fiscal policy, banking-sector balance-sheet constraints, or firm entry and exit. In addition, percentage gains can appear large in small countries even when their absolute contribution to total EU output remains modest. These limitations do not overturn the capital-reallocation result, but they mean that the output effects should be interpreted as conditional model-implied changes rather than unconditional forecasts.

The overall conclusion from the robustness analysis is therefore that CMU works in the model primarily as a reallocation policy. It reduces home bias, redirects capital toward smaller and less integrated economies, and raises model-implied EU output mainly through the informational-integration channel. These conclusions remain qualitatively stable across alternative assumptions about return sensitivity and across a broad range of policy and productivity parameters.

7 Discussion and Implications

7.1 Interpretation of Key Findings

The results reported in Section 6 admit three levels of interpretation that must be distinguished. The first concerns what the gravity estimates tell us about the structure of bilateral barriers to EU portfolio investment. The second concerns what the counterfactual simulation tells us about the aggregate and distributional consequences of removing those barriers under the CMU programme. The third concerns a specific and policy-consequential asymmetry between barrier types that the simulation is uniquely positioned to quantify. This subsection addresses each in turn.

7.1.1 Bilateral Frictions in EU Portfolio Investment

The PPML coefficients in Table 8 should be interpreted as multiplicative effects on expected bilateral portfolio holdings. A coefficient β implies that a change Δx in the corresponding regressor changes expected holdings by the factor $\exp(\beta\Delta x)$. This interpretation is important because the regressors are measured on different scales: linguistic distance, news distance, and financial-development distance lie on a $[0, 1]$ interval in the data, while geographic distance is measured in kilometres.

Linguistic distance has a large and precisely estimated negative coefficient ($\hat{\beta}_{\text{ling}} = -2.041$, $p < 0.001$). Since the variable is scaled to $[0, 1]$, moving from the least to the most linguistically distant country pair is associated with expected bilateral holdings of approximately $\exp(-2.041) \approx 0.13$ of their previous level, or about 87 per cent lower. This indicates that linguistic proximity remains an important correlate of bilateral portfolio allocation within the sample.

News distance has the largest coefficient in absolute value ($\hat{\beta}_{\text{news}} = -7.190$, $p = 0.044$), but its interpretation requires particular caution. Since news distance is also scaled to $[0, 1]$, the point estimate implies that moving from the best to the worst information connection in the sample is associated with expected bilateral holdings of approximately $\exp(-7.190) \approx 0.00075$ of their previous level. This is an economically very large association. However, the coefficient is less precisely estimated than the linguistic, financial-development, or geographic coefficients, with a clustered standard error of 3.575, and the identification concerns discussed in Section 7.3.6 imply that it should not be interpreted as a clean causal information

effect.

Financial-development distance is large, negative, and precisely estimated ($\hat{\beta}_{\text{fin}} = -4.215$, $p < 0.001$). Over its empirical range, the coefficient implies that expected holdings are approximately $\exp(-4.215) \approx 0.015$ of their level at zero distance, or about 98.5 per cent lower. Unlike the news coefficient, this estimate is both large and precisely estimated, which makes it central to the CMU policy experiment.

Geographic distance is measured in kilometres, so its coefficient is naturally small in absolute magnitude ($\hat{\beta}_{\text{geo}} = -3.130 \times 10^{-4}$, $p = 0.012$). Per kilometre the effect is modest, but over larger distances it is economically meaningful: an additional 1,000 kilometres is associated with expected holdings of approximately $\exp(-0.313) \approx 0.73$ of their previous level, or about 27 per cent lower. The result is consistent with the financial-gravity literature’s finding that geographic distance reduces bilateral positions, while reflecting that the EU sample is geographically more compact than global samples such as Portes and Rey (2005) and Lane and Milesi-Ferretti (2008).

The GDP coefficients are positive, consistent with the standard gravity prediction that larger origin and destination economies are associated with larger bilateral portfolio positions. Because the monetary variables are scaled before estimation, these coefficients should not be interpreted as literal effects of a one-dollar change but as effects on the scaled GDP variable. The substantive point is that economic size attracts portfolio investment on both sides of the bilateral relationship.

This pattern has not been established in prior within-EU gravity work. Coeurdacier and Martin (2009) estimate a gravity model for bilateral equity positions among European countries but do not include bilateral financial-development distance as a regressor; their specification identifies frictions through geographic, linguistic, and institutional-dummy variables. The present paper’s inclusion of the IMF Financial Development Index as a bilateral distance measure, following the country-level tradition of Bekaert, Harvey, and Lundblad (2005), adds a hard-friction channel that is directly relevant to the CMU policy question. The large and precisely estimated negative coefficient on financial-development distance suggests that institutional and regulatory divergence is an important measurable barrier to bilateral portfolio investment within the sample.

The policy implication is therefore suggestive rather than mechanical. The CMU programme targets the hard institutional channel, and the gravity estimates indicate that this channel is empirically relevant for bilateral allocation. However, the coefficient ranking should not be read as a direct ranking of policy payoffs. The simulation results below show that the friction with the largest or most precisely estimated gravity coefficient is not necessarily the friction whose reduction generates the largest aggregate output gain. In particular, the hard financial-development channel is important in gravity identification, but the informational channel plays a larger role in the counterfactual output effects.

The identification concerns surrounding the news-distance coefficient, including the concentration of variation among a thin tail of country pairs and the reverse-causality channel running from bilateral investment to bilateral media coverage, are treated in detail in Section 7.3. Throughout the remainder of this discussion, the news-distance coefficient is treated as a statistically significant but cautiously interpreted channel whose policy relevance depends on the modelling assumptions used in the counterfactual simulation.

7.1.2 Domestic Share Reduction and the Confirmation of the Financial Integration Channel

As reported in Section 6.5, domestic portfolio shares fall across EU countries under the central CMU scenario. EU investors allocate a smaller share of their portfolios to domestic assets and a larger share to foreign assets within the integrated market. Non-EU countries are affected only indirectly, since the CMU shock is applied to EU–EU wedges. Their domestic shares rise in the simulation because equilibrium return changes alter the relative attractiveness of foreign and domestic destinations, not because the model directly reduces non-EU investors’ frictions when investing in Europe.

This pattern is important for establishing what the CMU shock actually does in the model. The simulation is not generating output gains through an exogenous increase in capital. It is changing where investors allocate their existing portfolio wealth. The fall in domestic shares therefore indicates that the counterfactual operates through the intended financial-integration channel: lower bilateral frictions reduce home bias and increase cross-border portfolio integration. This is consistent with the interpretation of Coeurdacier and Rey (2013): even in comprehensive gravity specifications, a quantitatively large and patterned residual home bias remains across EU countries, driven by factors including institutional

investor structures and regulatory constraints that are not directly observable at the bilateral level.

The implication connects directly to the country-level output distribution discussed in Section 6.4. As domestic shares fall across the EU, capital is redirected toward a new cross-country allocation. The countries that receive larger inflows are those whose attractiveness increases most under the wedge reductions. As shown in Section 6.3, the largest proportional inflow gains accrue to smaller economies such as Poland, Belgium, and Bulgaria, while the largest losses are concentrated in Italy, France, and Germany. The capital reallocation is therefore not a uniform increase but a redistribution of existing portfolio wealth according to the new relative attractiveness of each destination after friction reduction. This asymmetry matters for the CMU simulation and generates the asymmetric output effects reported in Section 6.4.

7.1.3 Capital Reallocation and the Centre-periphery Output Gradient

As reported in Section 6.2, the simulation generates a positive mean change in EU output across the scenario grid. The most probable scenario outcome corresponds to an increase of 0.75 per cent, while the mean across all scenarios is 1.692 per cent. As shown in Figures 7 and 8, the largest percentage gains accrue to smaller Eastern European and Baltic economies, while several large Western and Southern European economies experience output losses under the main scenario.

The mechanism is straightforward. Bilateral friction reduction makes peripheral destinations accessible at lower cost to EU investors. Capital flows toward peripheral destinations with above-average returns. The capital-deepening channel is central to the simulation because output changes are generated through the model's mapping from portfolio reallocation to effective capital and then to Cobb-Douglas output. The centre-periphery gradient in output effects is therefore a robust model outcome across the simulated parameter grid, rather than a feature of the central calibration alone.

Section 6.9 indicates that this broad sign structure is preserved across the simulated parameter space: large core economies tend to exhibit negative output effects, while peripheral high-return economies tend to exhibit positive effects. Only the magnitude of the gains and losses shifts across the parameter space; the distributional pattern is preserved.

These results are consistent with the asymmetry in EU bilateral portfolio positions documented by Lane

and Milesi-Ferretti (2008): large economies dominate intra-EU equity holdings, so friction reduction naturally redirects capital outward from the dominant players. The analogy to the post-crisis renationalisation of capital flows documented by Lane (2013) runs in reverse: just as stress episodes renationalised EU capital flows disproportionately affecting the periphery, integration-driven friction removal de-renationalises them, disproportionately benefiting peripheral destinations.

The aggregate output effect across the scenario grid is partial rather than complete, reflecting friction reductions that represent a plausible near-term CMU integration scenario rather than full harmonisation. The gain is therefore modest relative to estimates from earlier EU integration episodes, consistent with the partial and gradual nature of the CMU programme.

7.1.4 Distributional and Allocative Efficiency Implications

Two results suggest that the aggregate output effect understates the broader allocative significance of CMU in the model. First, as reported in Section 6.9, the Gini diagnostics across EU27 countries point to a modest reduction in the concentration of output gains under the main scenario, with the Lorenz-curve diagnostic indicating that gains are concentrated among countries in the lower portion of the output distribution. Although the compression is modest under partial friction reduction, the direction suggests that capital market integration can act as a market-driven convergence mechanism within the model, channelling capital toward peripheral economies through reallocation rather than through fiscal transfers.

This finding connects to two distinct literatures. In the EU convergence tradition, cohesion funds have been the primary instrument for reducing regional output disparities, operating through direct fiscal transfers. The present result suggests that capital market integration can generate convergence through an entirely different channel: by allowing capital to flow from capital-abundant, low-return core economies toward capital-scarce, high-return peripheral economies, friction removal compresses the output distribution through market allocation. This mechanism is distinct from fiscal redistribution and in principle complementary to it. The implication for the CMU agenda, as framed by Letta (2024), is that CMU is not only a supply-side efficiency measure but also a distributional instrument, a framing that has received less attention in the policy literature than the aggregate efficiency argument.

Second, as reported in Section 6.6, countries with higher baseline MPK tend to receive larger foreign capital inflows under the main scenario, and average EU MPK rises as the informational channel (γ) increases. This provides evidence that the model is not only reallocating capital mechanically but directing it toward countries where it generates higher output. Persistent MPK dispersion across countries is a measure of misallocation: if capital were allocated to its highest-value use, marginal returns would equalise in equilibrium. The positive relationship between baseline MPK and simulated inflows, visible in Figure 10, indicates that the CMU-scale friction reduction moves the cross-country capital allocation toward greater efficiency.

The improvement falls short of full equalisation because the friction reduction is partial and because the model abstracts from labour market adjustment, goods trade, and other channels through which marginal returns equalise in the long run. But the direction is consistent with the view that EU capital market fragmentation generates meaningful allocative inefficiency, not merely missed portfolio diversification gains.

7.1.5 Hard Versus Soft Frictions and the CMU Design Problem

The most policy-consequential finding in this paper is not the aggregate output gain under the central scenario but the differential output contribution of the hard and informational channels revealed by Figures 4, 5, and 11. These three figures together establish a result that is counterintuitive given the gravity estimates: although financial-development distance is one of the largest and most precisely estimated hard-friction coefficients in the gravity model, the simulation indicates that the informational channel governed by (γ) generates the largest aggregate output gains.

As shown in Section 6.2, hard integration without accompanying informational integration is not merely less beneficial than the combined programme; it can be counterproductive at the EU aggregate level, with the output-reducing region of the parameter space visible at low γ in Figure 4. The allocative efficiency mechanism behind this pattern is visible in Figures 11 and 5: average EU MPK rises with γ but falls with ω , suggesting that informational integration directs capital toward higher-return destinations while the hard channel does not. The structural interpretation is that the hard channel reduces wedges associated with bilateral financial-development distance. This distance is measured as the absolute difference between

two countries' financial-development indices, not as a directional measure of whether the destination is financially underdeveloped relative to the investor country. Reducing this friction therefore changes allocations across financially dissimilar country pairs, but the resulting reallocation is not necessarily aligned with baseline MPK differences. By contrast, the informational channel appears in the simulation to generate a more MPK-aligned reallocation, which helps explain why γ produces stronger aggregate output gains than ω .

The mechanism can be stated precisely. The hard channel (ω) reduces financial-development distance wedges. Financial-development distance reflects the institutional and regulatory gap between two countries, not capital scarcity per se. Reducing this friction changes allocations across financially dissimilar country pairs, but the destinations receiving capital are not necessarily the most capital-scarce economies: the reallocation is driven by reductions in measured financial-development distance rather than directly by return differentials. The informational channel (γ) reduces news-distance wedges. If the news-distance measure captures investor unfamiliarity, then reducing informational frictions can redirect capital toward destinations that are less visible to investors under the baseline allocation. In the simulation, this channel is more closely aligned with high-MPK destinations than the hard channel, which helps explain why γ produces stronger aggregate output gains. Reducing informational frictions therefore produces a more MPK-aligned capital reallocation than reducing hard frictions in isolation.

This finding has direct implications for the CMU design question. A CMU programme that achieves $\omega > 0$ through regulatory harmonisation but fails to generate complementary progress on informational integration, leaving $\gamma \approx 0$, may reduce EU aggregate output rather than improve it. In the simulated parameter grid, informational integration is not merely a passive complement. Positive aggregate output effects from hard integration appear to depend on sufficient progress in the informational channel. The policy design implication is that CMU should pursue hard harmonisation and informational integration simultaneously if the objective is to avoid the output-reducing region of the simulated parameter space. Sequencing regulatory convergence first with informational spillovers to follow is a design that runs the risk of traversing the output-reducing region of Figure 4 before reaching the output-generating interior.

This conclusion provides quantitative support for the complementarity argument of Véron and Wolff

(2016), but it extends that argument in an important direction. Véron and Wolff argue that hard and soft channels reinforce each other; the present results show that the relationship is asymmetric: the soft channel is the primary output-generating lever, and the hard channel’s contribution to output is conditional on the soft channel operating at a sufficient level.

The distributional implications reinforce the design conclusion. The hard channel, operating alone, generates large capital outflows from core economies and large inflows to selected peripheral destinations; because core economies are large in GDP terms, their output losses dominate the aggregate. The soft channel, by routing capital toward MPK-aligned destinations including smaller peripheral economies, generates a capital-convergence reallocation in which peripheral gains are sufficient to offset core losses in aggregate. The two channels are therefore not simply different routes to the same output outcome; they have fundamentally different allocative properties, and the soft channel’s alignment with the MPK criterion is the source of its aggregate output advantage.

7.2 Theoretical Implications

The results have implications that extend beyond the specific CMU policy question and bear on the theoretical frameworks reviewed in Section 3. Four theoretical claims are directly addressed by the evidence.

7.2.1 Financial Gravity Confirmed Within the EU

The foundational claim of the financial gravity literature is that bilateral portfolio positions can be described by a gravity-like structure in which economic size attracts investment and bilateral frictions repel it. This claim was established on global samples by Portes and Rey (2005) using fourteen countries and by Lane and Milesi-Ferretti (2008) using the full PIP cross-section, and extended within Europe by Coeurdacier and Martin (2009). The present paper finds support for this structure in a sample restricted to EU member states plus closely integrated non-EU comparators, estimated with PPML over a multi-year panel.

Three features of this evidence go beyond the prior literature. First, the specification here includes bilateral financial development distance as a regressor, a friction measure absent from the European gravity

literature but central to the policy question of CMU. Its strong and precisely estimated negative coefficient ($p < 0.001$) suggests that even within an advanced integration bloc, institutional and regulatory divergence is a quantitatively important bilateral barrier. Second, the PPML estimator retains the zero bilateral pairs in the sample that log-linear methods would drop and is appropriate for gravity estimation in the presence of heteroskedasticity, as emphasised by Santos Silva and Tenreyro (2006). This mitigates concerns that the estimates are distorted by the skewed distribution of bilateral portfolio positions, although it does not eliminate the possibility that large country pairs remain influential in the estimation. Third, the year fixed effects absorb common shocks across the sample period, which spans the post-sovereign-debt recovery and the early CMU legislative cycle. They therefore control for time-specific factors affecting all country pairs, but they do not by themselves establish that the estimated relationships represent a stable integration regime.

The result has a broader implication for the use of gravity models in financial applications. The persistent concern in the literature, articulated in Section 3.5, is that the identification trade-off between controlling for multilateral resistance and identifying country-level institutional variables forces researchers to choose between clean causal identification and policy-relevant friction measurement. The present paper illustrates that the country-level-controls specification exemplified by Papaioannou (2009), which substitutes GDP controls for structural fixed effects in order to identify the bilateral institutional friction, delivers signed, significant, and economically meaningful estimates without the fixed-effect structure. The pseudo- R^2 of 0.249 is naturally lower than specifications with full origin-destination effects, but the friction coefficients it identifies are the ones that matter for the policy experiment, and their economic magnitudes are consistent with the broader literature.

7.2.2 Friction Wedge Decomposability: Implications for Policy Simulation

The theoretical foundation of the counterfactual analysis rests on two modelling assumptions: that bilateral friction enters the portfolio allocation rule multiplicatively, so that the total wedge can be decomposed into the product of its constituent components; and that a proportional reduction in one component, parameterised by a power transformation $(\Delta_{ij}^k)^{1-\phi}$, can be simulated while holding the other components fixed. These assumptions are consistent with the trade-cost framework of J. E. Anderson and Wincoop

(2004) and its extension to financial gravity by Okawa and Wincoop (2012). The simulation results do not independently prove that this decomposition is the true structural form of EU portfolio frictions; rather, they show how outcomes behave under this maintained wedge-decomposition assumption.

The key diagnostic is the response of country-level outputs to variation in the policy parameters. As shown in Figures 7 and 8, the country output effects span a wide range, with the largest percentage gains in smaller Eastern European and Baltic economies and the weakest or negative effects concentrated among larger economies. Across the policy grid, the effects vary systematically with ω and γ , but this should not be interpreted as proof of a globally monotone structural relationship. Figure 4 shows that the EU aggregate output effect changes smoothly over the simulated (ω, γ) space. Section 6.9 further indicates that the broad distributional pattern is preserved under variation in the return elasticity η . The wedge decomposition is therefore well-behaved within the simulated parameter space, in the sense that partial reductions in identified friction components produce interpretable changes in model outcomes.

The model structure implies that capital deepening is the main proximate channel through which CMU affects output, since portfolio reallocation changes effective capital and output is then computed through the Cobb-Douglas production function. The TFP contribution, while present, is quantitatively secondary in the reported simulations. Output gains are therefore primarily driven by the movement of capital across destinations, with productivity effects acting as a secondary amplification channel where foreign capital intensity is high enough to activate the spillover mechanism.

7.2.3 The relative Weight of Hard Versus Soft Frictions

The gravity estimates and the simulation results tell two complementary but distinct stories about the relative importance of hard and soft frictions, and distinguishing them is essential for interpreting the policy implications correctly.

From the gravity estimates in Table 8, financial-development distance is one of the largest and most precisely estimated friction coefficients. Its coefficient of -4.215 exceeds the linguistic-distance coefficient of -2.041 in absolute magnitude, and both are statistically robust ($p < 0.001$). However, the news-distance coefficient is larger in absolute value and therefore the hard channel should not be described as the largest

friction coefficient overall. The appropriate interpretation is that financial-development distance is the largest precisely estimated hard-friction channel and an economically important observable barrier within the EU cross-section.

The simulation results tell a different story. From the output decomposition in Figures 4, 5, and 11, the informational channel governed by γ generates the larger aggregate output gain. The hard channel (ω) at low γ can reduce aggregate output rather than increase it.

These two results are not contradictory; they measure different things. The gravity coefficient measures how much the financial-development friction suppresses bilateral investment at a given distance. The simulation output effect measures how much aggregate output increases when that friction is reduced. A friction can be large in terms of suppressing bilateral flows but small in terms of aggregate output impact if the capital it redirects does not flow toward its highest-value use. The financial-development channel operates precisely this way: it generates large bilateral reallocation, but the reallocation is not well-aligned with the MPK criterion at the EU aggregate level, producing the ambiguous aggregate output effect visible in Figure 4.

The informational channel, by contrast, appears to be more tightly correlated with the MPK misallocation between EU countries. If news distance captures investor unfamiliarity, then the informational channel can be interpreted as reducing barriers to destinations that are less visible under the baseline allocation. In the simulation, this channel appears more closely aligned with high-MPK destinations than the financial-development channel, which helps explain the stronger role of γ in the output decomposition. Reducing informational frictions therefore produces a more MPK-aligned reallocation than reducing financial-development frictions, which explains the dominance of γ in the output decomposition despite its smaller gravity coefficient.

This distinction refines the theoretical hierarchy proposed in Section 3.6. The prior classification of hard frictions as the leading policy lever rests on two premises: that hard frictions are among the largest and most precisely estimated observable barriers, and that removing a large bilateral barrier generates the largest output gain. The first premise is supported by the gravity estimates for financial-development distance, while the second is not supported by the simulation. The simulation results show that these

premises can be separated: the largest barrier and the most output-enhancing lever are not necessarily the same friction channel. The CMU programme should therefore be designed not only to address the largest barrier but to address the barrier whose removal most efficiently routes capital toward its highest-value use. The present results suggest those are two different channels.

The geographic-distance coefficient is correctly signed and significant at the 5 per cent level. Because distance is measured in kilometres, the coefficient is numerically small per kilometre but economically meaningful over larger distances. This is not in tension with global studies such as Portes and Rey (2005) and Lane and Milesi-Ferretti (2008); rather, it reflects the fact that the same per-kilometre effect accumulates over distance, while the European sample contains less extreme distance variation than global samples.

7.2.4 The Residual Home-bias Puzzle: What Remains Unexplained

As reported in Section 6.5, domestic portfolio shares fall across EU countries under the main CMU scenario, indicating that the friction reduction operates through the intended financial-integration channel. However, as Section 6.9 notes, the model does not fully resolve observed home bias: even after the wedge reductions, substantial home bias residuals remain, particularly in smaller peripheral economies. This pattern is consistent, within the EU context, with the conclusion of Coeurdacier and Rey (2013): observable bilateral friction variables, even when comprehensively specified, leave a large and economically patterned residual that cannot be explained by the gravity framework alone. The home bias puzzle, in the formulation of French and Poterba (1991), is not that investors hold domestic assets but that they hold them in proportions far exceeding what observable barriers and return differentials can rationalise.

The theoretical significance of this residual lies in what it suggests about the limits of the gravity framework. If observable bilateral friction variables fully explained home bias, the model would account for most of the cross-country portfolio allocation and the simulation would predict much stronger convergence following friction reduction. The persistent home bias documented in Section 6.5 suggests that observable bilateral frictions do not fully account for EU portfolio allocation. The remaining residual points to additional country-level or institutional forces that are not captured by the bilateral gravity framework. Plausible explanations include institutional investor mandates and governance constraints, as surveyed by

Letta (2024), as well as national regulatory channelling of savings, including pension fund domesticity requirements and bank-sovereign loop dynamics. These mechanisms operate at the national or institutional level rather than at the bilateral-pair level, which is why they cannot be separately identified within the gravity specification used here.

Recognising the limits of the gravity framework’s explanatory scope is not a weakness of the analysis but a sharply defined research frontier: the bilateral component of portfolio allocation is partly described by the gravity structure, while the remaining country-level residual appears economically meaningful, and the two require different empirical strategies to disentangle.

7.3 Limitations

Any quantitative model of this scope involves simplifying assumptions whose departures from reality constrain the interpretation of the results. The purpose of this section is not to weaken the empirical contribution, but to make explicit the conditions under which the simulation results should be read. The most important limitations concern the static nature of the counterfactual, the mapping from portfolio claims to productive capital, the sample-denominator convention used to reconstruct domestic holdings, the calibration of the productivity channel, the identification of bilateral frictions, the treatment of financial centres and investable asset supply, and the external validity of the CMU shock.

7.3.1 Static Model and The Absence of Transitional Dynamics

The simulation is a comparative static exercise. It computes counterfactual portfolio allocations and output levels after a reduction in bilateral frictions, but it does not model the transition path from the baseline allocation to the new equilibrium. The results should therefore be interpreted as model-implied static reallocation effects rather than as a complete dynamic growth forecast.

Three dynamic mechanisms are omitted. The first is capital accumulation. In a dynamic setting, capital inflows to under-capitalised economies could raise output, generate retained earnings, and support further capital accumulation over time. This channel could amplify the long-run effect relative to the one-period reallocation effect reported in Section 6.2. At the same time, changing returns, congestion effects, political frictions, or capacity constraints could dampen the realised gain. The model does not estimate this

dynamic adjustment path. The reported output effects should therefore be interpreted as conditional static effects, not as estimates of the full long-run output impact of CMU.

The second omitted mechanism is portfolio adjustment costs. Investors face transaction costs, regulatory mandates, benchmark constraints, institutional frictions, and behavioural inertia when reallocating portfolios. The full steady-state reallocation implied by the CMU shock may therefore take years to materialise. During the transition, the output gains from the new allocation may be delayed or partially offset by the cost of rebalancing.

The third omitted mechanism is anticipation and credibility. As Coeurdacier and Rey (2013) emphasise, financial-gravity estimates are most reliable when applied within a stable integration regime. CMU is a regime change. If investors believe that future frictions will fall, they may adjust portfolios before the formal implementation of policy. Conversely, if the policy is viewed as incomplete or politically fragile, adjustment may be delayed. The comparative-statics framework cannot capture either anticipatory reallocation or credibility-driven delay.

This limitation is common in the financial-gravity and policy-simulation literature. Coeurdacier and Martin (2009) and Lane and Milesi-Ferretti (2008) also study international financial integration using comparative-static frameworks rather than full dynamic transition models. The limitation is therefore not specific to this implementation, but it restricts the interpretation of the output effects to static, model-implied reallocations.

7.3.2 Portfolio-capital Mapping and Asset-supply Constraints

A central modelling limitation is the mapping from portfolio positions to productive capital. The IMF PIP and Eurostat financial balance-sheet variables measure portfolio equity and debt claims. These are financial claims on assets, not additional physical capital stocks. The model therefore does not add portfolio capital directly to Penn World Tables capital. Instead, portfolio reallocations are converted into proportional shocks to productive capital through the bounded exposure weight ϕ_i .

This avoids direct double counting, but the mapping remains a modelling bridge rather than a national-accounting identity. The relevant portfolio object is

$$k_i^{\text{fin}} = \sum_{j=1}^n \pi_{j,i} s_j, \quad (39)$$

where k_i^{fin} is model-implied portfolio capital received by country i . This is not the same object as k_i^{PWT} , the productive capital stock from Penn World Tables. The model uses changes in k_i^{fin} to scale changes in effective productive capital, but it does not treat k_i^{fin} as a separate capital stock that can be added to k_i^{PWT} .

The second issue is the absence of destination-level asset-supply constraints. Country size is proxied by GDP, but actual investable market size depends on stock-market capitalisation, bond-market depth, free float, issuance capacity, legal access, and institutional market infrastructure. This matters especially for smaller economies. A small country may have a high marginal product of capital and low bilateral frictions after CMU, but its actual securities markets may not be large or liquid enough to absorb the portfolio reallocations implied by the model. In such cases, the simulation may overstate the volume of capital that can realistically be reallocated to that destination.

This limitation affects the interpretation of the country-level results. Percentage gains in smaller economies should be read as model-implied proportional reallocations, not literal forecasts of realised investment volumes. It also supports the decision to interpret the GDP results as conditional output effects from a calibrated production mapping rather than as unconditional macroeconomic forecasts.

7.3.3 Domestic Holdings and the Sample-denominator Convention

A further limitation concerns the construction of domestic holdings. Domestic portfolio positions are not directly observed in the bilateral matrix and are therefore reconstructed as the residual between aggregate portfolio wealth and observed foreign portfolio holdings. This is a necessary accounting step given the available data, but it creates a sample-denominator convention in the simulation.

The issue is that the reconstructed domestic position reflects the country's aggregate portfolio position, while the foreign bilateral matrix used in the simulation is restricted to the countries in the sample. Foreign holdings outside the sample are therefore not modelled as bilateral destinations. As a result, the

domestic share used in the calibrated portfolio matrix is conditional on the in-sample denominator rather than on a complete global portfolio matrix.

This can affect the interpretation of home bias. If investors hold substantial foreign assets outside the sample, then restricting the foreign matrix to in-sample destinations may make the reconstructed domestic share appear larger relative to observed in-sample foreign holdings. The calibrated domestic wedge should therefore be interpreted as a within-sample home-bias calibration rather than as an exact measure of total domestic portfolio preference.

This limitation does not invalidate the counterfactual exercise, but it narrows its interpretation. The simulation should be read as a model of within-sample portfolio reallocation under CMU-style reductions in EU bilateral frictions. It is less suited to measuring the full global allocation of each country’s portfolio wealth or the exact level of aggregate home bias relative to all possible foreign destinations.

7.3.4 Reduced-form Portfolio Choice and the Absence of Explicit Risk

The portfolio allocation rule is deliberately parsimonious. Destination attractiveness depends on expected returns, destination size, and bilateral wedges. This structure is transparent and suitable for the gravity-based simulation, but it does not contain a full portfolio-choice problem with risk, covariance, exchange-rate exposure, liquidity risk, sovereign risk, or heterogeneous investor preferences.

This matters because home bias is not only a friction or information problem. Investors may hold domestic assets because they hedge domestic liabilities, consume in domestic currency, face regulatory capital rules, or prefer assets whose risk structure is better aligned with their balance sheet. Pension funds, insurers, banks, mutual funds, and households also face different constraints and investment objectives. A representative-investor allocation rule cannot capture these institutional differences.

The model partially absorbs these forces through the calibrated domestic wedge $\Delta_{i,i}$, which matches observed home shares in the baseline. However, this does not identify the separate structural sources of home bias. The domestic wedge is therefore best interpreted as a residual home-preference or domestic-constraint term rather than as a single observable friction. The counterfactual can show how allocations change when selected bilateral wedges fall, but it cannot decompose all sources of home bias into risk,

regulation, information, and behavioural components.

7.3.5 Portfolio Investment and the TFP Spillover Calibration

The TFP spillover extension in Section 6.8 adds a productivity channel in which finance exposure amplifies the output effect of capital reallocation. The calibrated parameter θ determines how strongly finance exposure enters the productivity term. This channel is not estimated from the thesis dataset; it is calibrated from the existing empirical literature. The interpretation of θ is therefore an important limitation of the model.

The first issue is the distinction between foreign direct investment and portfolio investment. Much of the empirical literature on productivity spillovers from cross-border capital concerns FDI rather than portfolio equity or debt. In the FDI case, the mechanism is relatively direct: foreign firms can transfer technology, managerial practices, production processes, and supplier relationships to affiliates in the host economy. These effects may then diffuse to domestic firms through labour mobility, supplier linkages, and competitive pressure. This provides a clear economic channel through which foreign capital can raise total factor productivity.

The mechanism is weaker for portfolio investment. Portfolio investors typically hold non-controlling financial claims and do not directly manage production. Any productivity effect must therefore operate through more indirect channels, such as improved corporate governance, stronger disclosure requirements, lower financing constraints, market discipline, and a more efficient allocation of capital toward productive firms. These channels are plausible, but they are not equivalent to the technology-transfer channel in the FDI literature. For this reason, elasticities estimated from FDI studies are likely to overstate the productivity effect of portfolio investment.

The second issue is that the model does not identify θ econometrically. The thesis estimates bilateral investment frictions using a gravity model, but it does not estimate a causal relationship between portfolio inflows and productivity. Doing so would require a separate identification strategy, since foreign portfolio inflows are endogenous to expected productivity, institutional quality, and growth prospects. A country may receive more foreign portfolio capital because it is expected to become more productive, rather than

becoming more productive because it receives foreign portfolio capital. This reverse-causality problem means that θ cannot be recovered directly from the same cross-sectional simulation framework without additional instruments or a panel identification strategy.

The third issue is that the TFP channel is applied to portfolio equity and debt together. This is a simplification. Equity investment may plausibly affect governance, disclosure, and risk-sharing, while debt investment is more closely related to financing costs, maturity structure, and creditor discipline. The productivity mechanisms are therefore not identical across asset classes. Pooling equity and debt means that θ should be interpreted as a reduced-form finance-exposure parameter rather than as a pure equity-spillover elasticity.

For these reasons, positive values of θ are best interpreted as scenario values rather than as estimated structural parameters. The case $\theta = 0$ is the conservative lower bound, where all output effects arise from capital reallocation alone. Positive values of θ show how the output effects would be amplified if portfolio-market integration also improved productivity through governance, disclosure, financing-cost, or capital-allocation channels. Since the results in Section 6.8 show that TFP amplification is quantitatively smaller than the main capital-reallocation effect, the headline results do not depend on treating the calibrated productivity channel as precisely estimated.

7.3.6 News Distance and the Gravity Identification Trade-off

The news-distance variable enters the gravity model with a negative and statistically significant coefficient, but its causal interpretation is limited. Two concerns are particularly important. First, the distribution of the news-distance variable is concentrated. Many EU country pairs have values close to one, so the estimated coefficient is partly driven by a relatively thin tail of observations. This makes the coefficient more sensitive to influential bilateral pairs than the coefficients on more smoothly distributed variables.

Second, reverse causality is plausible. Bilateral investment may increase bilateral media coverage, meaning that the coefficient may capture both the effect of information barriers on investment and the reverse effect of investment on information flows. If investors allocate more capital to a country, that country may become more economically salient and receive more media coverage. In that case, the estimated

news coefficient does not isolate a one-directional causal information-friction effect.

The practical consequence is that $\hat{\beta}_{\text{news}} = -7.19$ should be interpreted as an association rather than as a fully causal estimate. This matters because the informational channel is one of the strongest channels in the results. The model therefore treats reductions in informational frictions as a policy scenario rather than as a precisely estimated causal intervention.

There is also a broader identification trade-off. Using origin-destination fixed effects would absorb much of the country-level endogeneity in news flows, but it would also remove the variation needed to identify country-level financial-development distance. The specification therefore follows the financial-gravity approach in which GDP controls are used instead of fully saturated country-pair fixed effects. This preserves the ability to estimate financial-development and informational frictions, but it means that the estimated wedge coefficients should be interpreted with caution.

7.3.7 Uncertainty in the Estimated Wedge Coefficients

The simulation uses the PPML gravity coefficients as point estimates when constructing the wedge matrices. This means that the counterfactual results do not propagate statistical uncertainty from the first-stage gravity regression into the second-stage simulation. In reality, the wedge matrices inherit estimation error from the gravity coefficients, especially for the informational-distance coefficient, which is less precisely estimated than the linguistic and financial-development coefficients.

A fuller uncertainty treatment would draw coefficients from their estimated covariance matrix or use a bootstrap procedure, rebuild the wedge matrices for each draw, and rerun the counterfactual simulation. This would produce confidence bands around the simulated output effects and country-level reallocations. The present model instead treats the estimated coefficients as fixed inputs. The robustness checks over ω , γ , θ , and η address parameter sensitivity, but they do not capture full econometric uncertainty around the first-stage gravity estimates.

7.3.8 Sample Design, Financial Centres, and Country-size Measurement

The sample is restricted to EU member states and a small set of closely integrated non-EU comparators. This makes the estimates directly relevant to the CMU question, but it limits external validity. The sample consists mostly of advanced, geographically proximate economies. The estimated frictions are therefore calibrated to an EU-type investment environment and should not be applied mechanically to a global counterfactual.

Financial centres create an additional limitation. Following Coppola et al. (2021), recorded bilateral positions involving financial centres may reflect intermediation structures rather than genuine ultimate investor-destination relationships. This is particularly relevant for countries such as Luxembourg, Ireland, Cyprus, and Malta. The model partly addresses this by treating major financial centres separately in the diagonal calibration, but bilateral positions involving these countries may still contain pass-through flows that do not correspond to the true underlying investor.

Finally, the model uses GDP as the destination size variable rather than stock-market capitalisation. GDP has broad coverage and is standard in the financial gravity literature, but it is not a perfect measure of investable asset supply. In countries with relatively small or illiquid capital markets, GDP may overstate the amount of securities available to international investors. This could overstate the model-predicted allocation to some smaller economies and therefore exaggerate some country-level output gains.

7.3.9 No External-attractiveness Channel for Non-EU Investors

The CMU shock is applied only to EU–EU bilateral wedges. This is appropriate for modelling the internal integration effect of CMU, but it limits what the model can say about global investor behaviour. In reality, a more harmonised and transparent European capital market could also reduce effective frictions for non-EU investors allocating capital to EU destinations. Deeper disclosure standards, more unified supervision, larger market scale, and lower legal complexity could make EU assets more attractive to investors outside the EU.

The present model does not include this secondary external-attractiveness channel. Non-EU investors therefore do not receive a direct reduction in frictions when investing into EU countries; they only re-

spond indirectly to equilibrium changes in returns and allocations. The model is therefore well suited to analysing intra-EU reallocation and home-bias reduction, but less suited to estimating whether CMU would make Europe a larger global financial hub. Any conclusion about non-EU investor responses should be interpreted as conditional on this shock design.

7.3.10 Base Year and Generalisability

The portfolio simulation is anchored to 2019. This base year is useful because it predates the pandemic dislocations and follows the major post-crisis adjustment in European financial markets. However, it also limits generalisability. The post-2020 period introduced major structural changes, including the COVID-19 shock, the EU Recovery Fund, the inflation cycle, and changes in monetary-policy regimes. The bilateral friction structure of 2019 may therefore not fully represent the current European capital-market environment.

Brexit is also relevant. The United Kingdom is included as a non-EU comparator, but the 2019 data capture the UK during a transitional period after the referendum and before the full long-run post-Brexit financial-market equilibrium was observable. Bilateral positions involving the UK may therefore partly reflect temporary uncertainty and relocation effects.

More broadly, the 2019 baseline reflects a post-crisis integration regime in which Eurozone portfolio integration had recovered only partially from the decline after 2008. The results should therefore be interpreted as the effect of CMU relative to the 2019 European integration structure, not as a time-invariant estimate that would necessarily apply unchanged to later years.

7.4 Future Research

Each of the limitations identified in Section 7.3 points toward a specific research extension. The directions outlined here range from incremental improvements to the existing framework to structurally distinct approaches that would require new data, new modelling choices, or both.

7.4.1 Dynamic Modelling and Capital Accumulation

The most direct extension of the present framework is the introduction of capital accumulation dynamics. A multi-country model with overlapping generations or infinitely lived households, in which bilateral portfolio shares are chosen each period as a function of current returns and frictions, would allow the friction reduction implied by CMU to be traced through an adjustment path rather than a one-period reallocation. The analytical machinery of the present paper (the gravity-estimated friction wedge, the power-form CMU shock, and the Cobb-Douglas production structure) could be embedded in a dynamic general equilibrium as the steady-state mapping between friction parameters and portfolio allocations. The computational challenge is tractable for a model of this scale given standard solution methods for open-economy models with portfolio choice (Engel and Matsumoto, 2005), such as those employed in the Ratto, Roeger, and 't Veld (2008) QUEST framework used by the European Commission and the D. Anderson, Andrle, et al. (2013) GIMF framework used by the IMF. The main payoff would be a time-path of output gains that policy-makers could map onto a legislative implementation schedule, rather than a steady-state comparison whose timing is unspecified.

A further dividend of the dynamic extension would be the ability to compute transition costs. The static model cannot tell us whether a country that loses output in the new steady state faces a smooth decline or a sharp initial adjustment as capital outflows accelerate before the new equilibrium is reached. Transition costs of this kind are politically salient and could be the difference between a CMU programme that is politically sustainable and one that generates resistance from the core economies that bear its short-run adjustment costs. Quantifying the transition path is therefore not merely a technical refinement but a precondition for assessing the political feasibility of the programme.

7.4.2 TFP Spillovers as a Secondary Channel

The TFP channel should be interpreted as an amplification mechanism rather than the core driver of the CMU results. This is consistent with the literature reviewed in Section 3.8, where the strongest evidence for productivity spillovers comes from FDI rather than portfolio equity. Since portfolio investment is typically passive and non-controlling, the relevant spillover elasticity is likely smaller than the FDI-based estimates that motivate the calibration range. For that reason, θ is treated as a sensitivity parameter in

the interval $[0, 0.10]$ rather than as a structurally estimated coefficient.

At the same time, the model already scales down the effective productivity contribution of portfolio capital through the exposure weight ϕ_i . This term measures the share of total capital exposure accounted for by foreign portfolio finance, constructed from equity and debt relative to the broader capital base that includes Penn World Tables capital. The productivity channel therefore does not assume that all foreign portfolio claims translate one for one into productive capital or into FDI-style spillovers. Instead, the impact is explicitly bounded by the share of capital that is actually finance exposed in the model.

The results in Section 6.8 are consistent with this construction. The TFP effect is positive for some countries, but it is quantitatively small relative to the main capital reallocation effect. In other words, CMU primarily works through lower frictions and portfolio reallocation, while the productivity channel only modestly amplifies the output gains. This also means that the main conclusions do not depend on assuming strong spillovers from portfolio equity.

A useful implication is that the TFP channel should be viewed as an informative sensitivity extension rather than as a forecast of the true magnitude of productivity effects. It indicates the likely direction in which gains could be amplified if portfolio integration also improves productivity, but it should not be read as a direct estimate of within-EU portfolio equity spillovers.

7.4.3 Ultimate-ownership PIP and the Financial Centre Problem

The present paper treats the bilateral positions of major financial-centre economies as reflecting intermediation rather than genuine resident portfolio preferences, following Coppola et al. (2021). This is a practical response to a data problem whose methodology for restating bilateral portfolio positions on an ultimate-ownership basis, reassigning PIP holdings from the legal counterparty to the ultimate beneficial owner, would resolve this problem at its source.

Applying the Coppola et al. restated PIP data to the gravity estimation in this paper would allow three improvements. First, bilateral wedge estimates for country pairs involving financial centres could be based on genuine bilateral investment decisions rather than intermediation flows. Second, calibration could proceed for all countries in the sample, including the financial centres, using restated domestic shares

that reflect actual resident portfolio preferences rather than pass-through artefacts. Third, the gravity coefficients themselves would be identified from a cleaner distribution of bilateral positions, potentially altering the estimated magnitudes of the friction variables, particularly for country pairs with substantial financial-centre involvement. The resulting simulation would be a more accurate representation of the genuine bilateral friction structure facing EU portfolio investors.

7.4.4 Extending the Asset-class Decomposition

The present analysis uses aggregate portfolio holdings rather than separately estimating equity and debt positions. This is appropriate for studying broad portfolio reallocation under CMU, but it means that the model cannot distinguish whether the estimated frictions operate differently across asset classes. A natural extension would estimate separate gravity specifications for portfolio equity and portfolio debt, allowing the simulation to distinguish between equity-market and bond-market integration.

The first question is whether the centre-periphery gradient in output effects documented here is driven primarily by equity positions, debt positions, or both. If equity and debt frictions follow similar gravity structures, then the broad distributional pattern may be robust across asset classes. If they differ, for example because sovereign yield spreads, maturity structure, or credit risk dominate debt allocation in a way not captured by the current bilateral friction variables, the distributional effects of CMU could differ across markets.

The second question is how equity and debt integration interact. Lane and Milesi-Ferretti (2008) document that equity and debt positions are jointly determined for many EU country pairs, suggesting that friction reduction in one market may spill over to the other through portfolio complementarity. Separating the two asset classes would therefore provide a more precise account of the channels through which CMU affects capital allocation and output.

7.4.5 Supervisory Convergence as an Explicit Policy Lever

The CMU shock in the present model is parameterised as a uniform proportional reduction in bilateral financial development frictions, governed by ω . This treats all forms of regulatory divergence as equivalent and does not distinguish between legislative harmonisation (the formal alignment of national laws with

EU directives) and supervisory convergence (the actual alignment of how common rules are implemented and enforced by national and EU-level supervisors).

Constâncio (2017) argues that supervisory fragmentation is the binding constraint on EU capital market integration: common legislative frameworks, including MiFID II, AIFMD, and the Prospectus Regulation, are in place, but divergent national implementation and regulatory arbitrage continue to generate material bilateral frictions. Sapir, Véron, and Wolff (2018) make similar arguments, contending that an expanded ESMA mandate with binding supervisory powers would constitute a qualitative step change in integration depth beyond what legislative harmonisation alone can deliver.

Translating this distinction into the present model would require separating the bilateral financial development distance measure into two components: one capturing legislative harmonisation (reducible by ω_{leg}) and one capturing supervisory divergence (reducible by ω_{sup}). If supervisory convergence produces a larger output gain per unit of friction reduction than legislative harmonisation, as the Constâncio-Sapir literature suggests is likely since it addresses implementation rather than formal law, the model would imply larger gains from institutional deepening of ESMA than from further legislative output. This distinction has direct policy implications for the ongoing debate about the appropriate scope of the CMU 2025 Action Plan.

7.4.6 Institutional Heterogeneity and the Political Economy of Friction

The persistent home bias documented in Section 6.5, even after friction reduction, points to a dimension of home bias variation that the gravity framework cannot explain. The most structurally promising explanation lies in the institutional heterogeneity of national savings vehicles: pension systems, life insurance mandates, and retail savings products that direct household wealth into domestic assets through regulatory design rather than through investor choice. Letta (2024) identifies these embedded institutional structures as the primary reason why EU capital market fragmentation persists despite legislative progress, arguing that they reflect national political economies that formal regulation can influence only slowly.

A gravity-augmented model that incorporates national pension fund domesticity requirements as an ad-

ditional determinant of home bias would allow the unexplained variation to be partly decomposed into a bilateral-friction component (captured by the off-diagonal gravity structure), a regulatory-channelling component (captured by pension fund domesticity ratios or insurance regulation indicators), and a residual investor-familiarity component. This decomposition would provide a more precise diagnosis of where policy intervention can reduce home bias and where it cannot, directly informing the prioritisation of CMU measures.

Finally, the political economy of the centre-periphery output gradient deserves attention as a research question in its own right. The present paper indicates that large economies of Western and Southern Europe tend to lose output under the central CMU scenario in regions of the parameter space where peripheral economies gain. These are precisely the member states whose political support is most necessary for CMU legislative progress. Understanding the conditions under which core economies would support a programme that reduces their output relative to the baseline, whether through fiscal compensation, through the distributional benefits of a more stable and convergent EU economy, or through the long-run gains from a deeper capital market for their own corporate issuers, is a political economy question that formal modelling of the kind developed here cannot answer but can precisely frame.

8 Conclusion

8.1 Summary of Findings

This thesis asked how a CMU-style reduction in intra-EU portfolio frictions would affect home bias, simulated cross-border capital allocation, and model-implied output across European countries. The analysis combined a PPML gravity estimation of bilateral portfolio holdings with a calibrated portfolio-allocation model, embedded in a Cobb-Douglas production block and evaluated over an 8,000-scenario grid in (ω, γ, θ) .

Three findings answer the question. First, all four bilateral frictions enter the gravity model with the predicted negative sign. Financial-development distance and linguistic distance are estimated precisely ($p < 0.001$), informational distance is significant but less precisely identified ($p = 0.044$), and geographic distance is significant at the 5 percent level. The within-EU financial-development coefficient is among the largest and most precisely estimated, providing empirical support for the CMU programme’s focus on institutional and regulatory convergence as a measurable bilateral barrier.

Second, the simulated CMU shock raises model-implied EU output across most of the policy space, with a mean change of 1.69 percent across the scenario grid and a maximum of approximately 3.94 percent when both hard and informational channels are fully activated. Domestic portfolio shares fall in every EU country in the sample. The model does not generate gains through an exogenous capital expansion; it reallocates existing portfolio wealth, redirecting capital from core to peripheral destinations along a clear centre-periphery gradient. Smaller economies in central, eastern, and Baltic Europe gain most in percentage terms, while several large western and southern economies experience modest output losses. The TFP amplification governed by θ is quantitatively secondary: capital reallocation, not endogenous productivity, drives the headline output effects.

Third, and most consequential, the two policy levers operate through fundamentally different mechanisms. Reducing hard financial-development wedges in isolation (ω alone) is not merely less beneficial than the combined programme; it can be output-reducing at the aggregate level, reaching approximately -1.05 percent in the hard-only corner of the policy space. Reducing informational wedges (γ) raises average

MPK and compresses cross-country MPK dispersion, which the model interprets as a more efficient capital reallocation. The largest measurable bilateral barrier, namely financial-development distance by the gravity estimates, and the strongest output-generating lever in the simulation, namely informational frictions, are therefore not the same channel.

In light of these results, the thesis advances three contributions. It extends the within-EU financial-gravity tradition by including bilateral financial-development distance as a regressor, a measure absent from the European gravity work of Coeurdacier and Martin (2009) but central to the CMU policy question. It provides a quantitative mapping from estimated gravity coefficients into a structured counterfactual that separates hard, policy-sensitive soft, and immutable structural frictions. And it documents that a CMU programme restricted to regulatory harmonisation, without concurrent progress on informational integration, can traverse an output-reducing region of the policy space, giving quantitative content to the complementarity argument advanced qualitatively by Véron and Wolff (2016).

8.2 Policy Implications

The gravity evidence supports CMU's emphasis on regulatory and institutional harmonisation: financial-development distance is among the largest measurable bilateral barriers within the EU. The simulation adds a sequencing caveat. Hard harmonisation generates positive aggregate output only when informational integration makes concurrent progress; CMU should therefore not be sequenced as legislative harmonisation first, with informational spillovers expected to follow as a secondary effect. Under that sequencing the programme risks traversing the output-reducing region of the policy space before reaching the output-generating interior.

The distributional implication is harder for the programme. CMU is not neutral across member states. The smaller peripheral economies that gain most in percentage terms are not, individually, decisive for EU legislative progress. The larger economies whose political support is decisive are precisely those that experience output losses in the central scenario. The model-consistent case for these economies to support CMU therefore rests on long-run dynamic gains not captured in a static framework, on the system-level benefits of a more convergent EU economy, or on accompanying fiscal arrangements. The present analysis can frame this political-economy problem precisely; it cannot resolve it.

8.3 Limitations and Future Research

The results carry the qualifications set out in Section 7.3. Four merit highlighting in conclusion. The simulation is a comparative-static exercise and reports model-implied reallocation effects rather than dynamic forecasts. The TFP elasticity θ is calibrated from FDI evidence and should be read as a sensitivity range rather than a structural estimate. The sample-denominator convention used to reconstruct domestic holdings narrows the interpretation of home bias to a within-sample object. And the news-distance variable, on which the informational-channel result rests, is identified with the distributional and reverse-causality concerns discussed in Sections 4.6.1 and 7.3.6, a caveat that bears particularly on the relative magnitude of γ versus ω in the policy decomposition.

Four extensions follow directly. A dynamic version of the framework would deliver a time path of output effects mappable onto CMU’s legislative schedule and would make transition costs visible. Direct estimation of portfolio-specific productivity spillovers, for instance using index-inclusion shocks as instruments, would replace the most uncertain calibrated parameter with an identified estimate. Restated look-through portfolio data, following Coppola et al. (2021), would address the financial-centre measurement problem at its source. An explicit external-attractiveness extension would let CMU reduce frictions for non-EU investors allocating into the EU, making it possible to identify global-investor responses rather than only the indirect effects generated through the model’s return updating.

8.4 Concluding Remarks

CMU works in the model primarily as a reallocation policy, not a capital-creation policy. It lowers home bias, redirects portfolio capital toward smaller and less capital-rich economies, and raises model-implied EU output mainly through the informational-integration channel. The gravity estimates and the simulation results identify different objects (the largest measurable bilateral barrier and the strongest output-generating lever), and the central design insight of the analysis is that the two are not the same friction. For policy, this is the substantive point: the question for the Capital Markets Union is not only which barriers most suppress bilateral flows, but which barriers, when removed, route European capital toward its highest-value use.

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A Appendix

A.1 Bilateral flow change from baseline to central scenario



Figure 16: Bilateral flow change from baseline to central scenario.

This figure illustrates the bilateral flow change from the baseline scenario to the central scenario. The matrix is organized with rows and columns representing different countries or regions. Each cell in the matrix shows the change in bilateral flow between the corresponding row and column entities, with the values likely represented in a logarithmic scale to capture a wide range of changes effectively. The colour intensity in each cell may indicate the magnitude of the change, with different colours representing increases or decreases in bilateral flows.

A.2 Wedge matrix baseline (hard and soft)

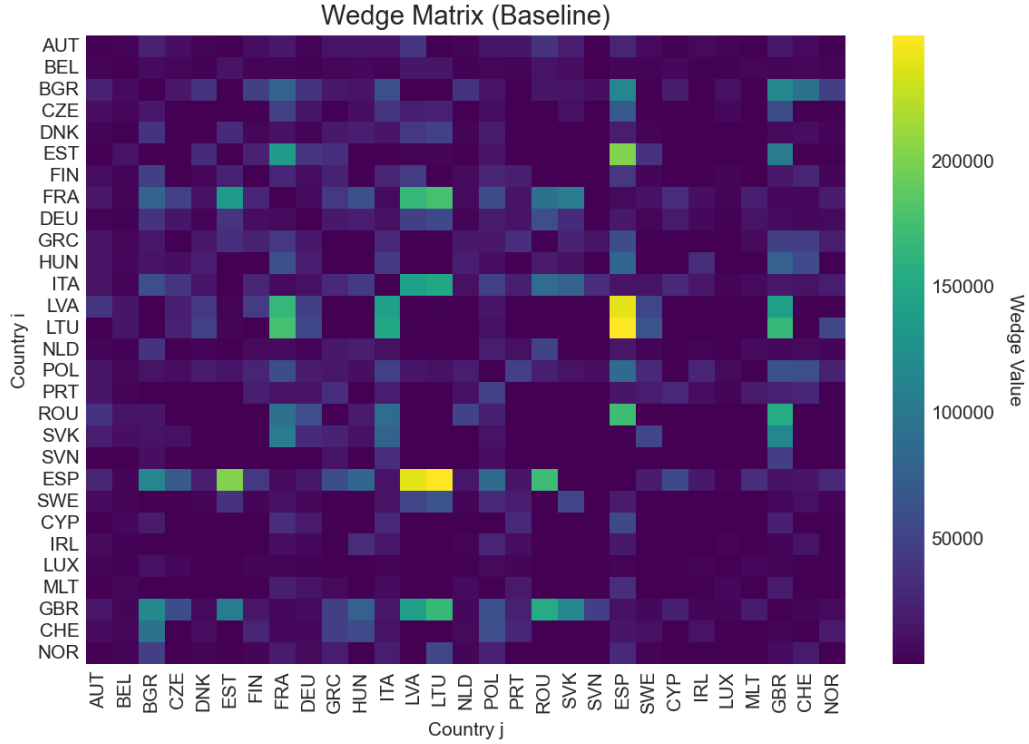


Figure 17: Wedge matrix baseline (hard and soft).

The above heatmap shows the combined hard and soft wedges (ω and γ) for the baseline scenario. And also includes the calibrated domestic wedges as the diagonal elements. The rows and columns of the heatmap represent different countries or regions, and the colour intensity in each cell indicates the magnitude of the wedge for that particular country or region.

A.3 PPML regression output

This subsection reports the full PPML regression output used in the gravity estimation. After removing domestic observations ($i = j$), the dataset contains 21,112 rows. The final PPML estimation sample contains 2,858 observations, of which 2,844 are positive flows and 14 are zero flows. For numerical stability, a_{ij} , Y_i , and Y_j are scaled by 10^6 in the estimation.

Table 9: Summary statistics for the PPML estimation sample

oprule Variable	Count	Mean	Std. dev.	Min	p25	p50	p75	Max
a_{ij}	2,858	0.020828	0.057931	0.000000	0.000161	0.000975	0.012691	0.799689
d_{ij}^{ling}	2,858	0.614230	0.227147	0.006539	0.517260	0.678716	0.776417	0.926690
d_{ij}^{fin}	2,858	0.245209	0.172528	0.001000	0.101000	0.213000	0.368000	0.768000
d_{ij}^{news}	2,858	0.996277	0.012266	0.795394	0.997823	0.999292	0.999764	1.000000
d_{ij}^{geo}	2,858	1401.965031	737.614728	127.004291	821.245494	1303.365994	1851.062053	3768.120503
Year	2,858	2017.455563	1.671359	2015.000000	2016.000000	2017.000000	2019.000000	2020.000000
Y_i	2,858	1.165386	1.372315	0.016859	0.284331	0.516414	1.879422	5.072476
Y_j	2,858	1.183334	1.368936	0.016859	0.291594	0.526893	1.908748	5.072476

Table 10: Full PPML gravity regression output

Variable	Coef.	Std. err.	z	p -value	2.5%	97.5%
Intercept	3.9223	3.7090	1.058	0.290	-3.347	11.192
δ_{2016}	-0.1493	0.0400	-3.748	0.000	-0.227	-0.071
δ_{2017}	-0.0334	0.0340	-0.975	0.330	-0.101	0.034
δ_{2018}	-0.1254	0.0370	-3.405	0.001	-0.198	-0.053
δ_{2019}	-0.0185	0.0440	-0.421	0.674	-0.105	0.068
δ_{2020}	0.2342	0.0470	4.942	0.000	0.141	0.327
d_{ij}^{ling}	-2.0407	0.4470	-4.562	0.000	-2.917	-1.164
d_{ij}^{news}	-7.1898	3.5750	-2.011	0.044	-14.198	-0.182
d_{ij}^{fin}	-4.2154	0.9210	-4.577	0.000	-6.021	-2.410
d_{ij}^{geo}	-0.0003	0.0001	-2.525	0.012	-0.001	-0.00007
Y_i	0.4520	0.0470	9.643	0.000	0.360	0.544
Y_j	0.4348	0.0390	11.103	0.000	0.358	0.512
Observations						2,858
Degrees of freedom (residual)						2,846
Degrees of freedom (model)						11
Pseudo R^2						0.2476
Log-likelihood						-198.18
LL-null						-263.39
LLR p -value						1.429×10^{-22}
Covariance type						Clustered bilateral pair
Converged						True
Iterations						9

Notes: Dependent variable is scaled bilateral portfolio holdings, a_{ij} . The model is estimated by Poisson pseudo-maximum likelihood with clustered standard errors. The optimisation terminated successfully with current function value 0.069341. Year fixed effects are included with 2015 as the omitted reference category.

The GDP coefficients are estimated on scaled variables. Rescaled to original units, the coefficients are 4.520173×10^5 for Y_i and 4.348036×10^5 for Y_j .

Active friction regressors in the simulation are d_{ij}^{geo} , d_{ij}^{ling} , d_{ij}^{fin} , and d_{ij}^{news} , all of which enter with negative estimated coefficients.

A.4 PPML regression predicted vs actual values

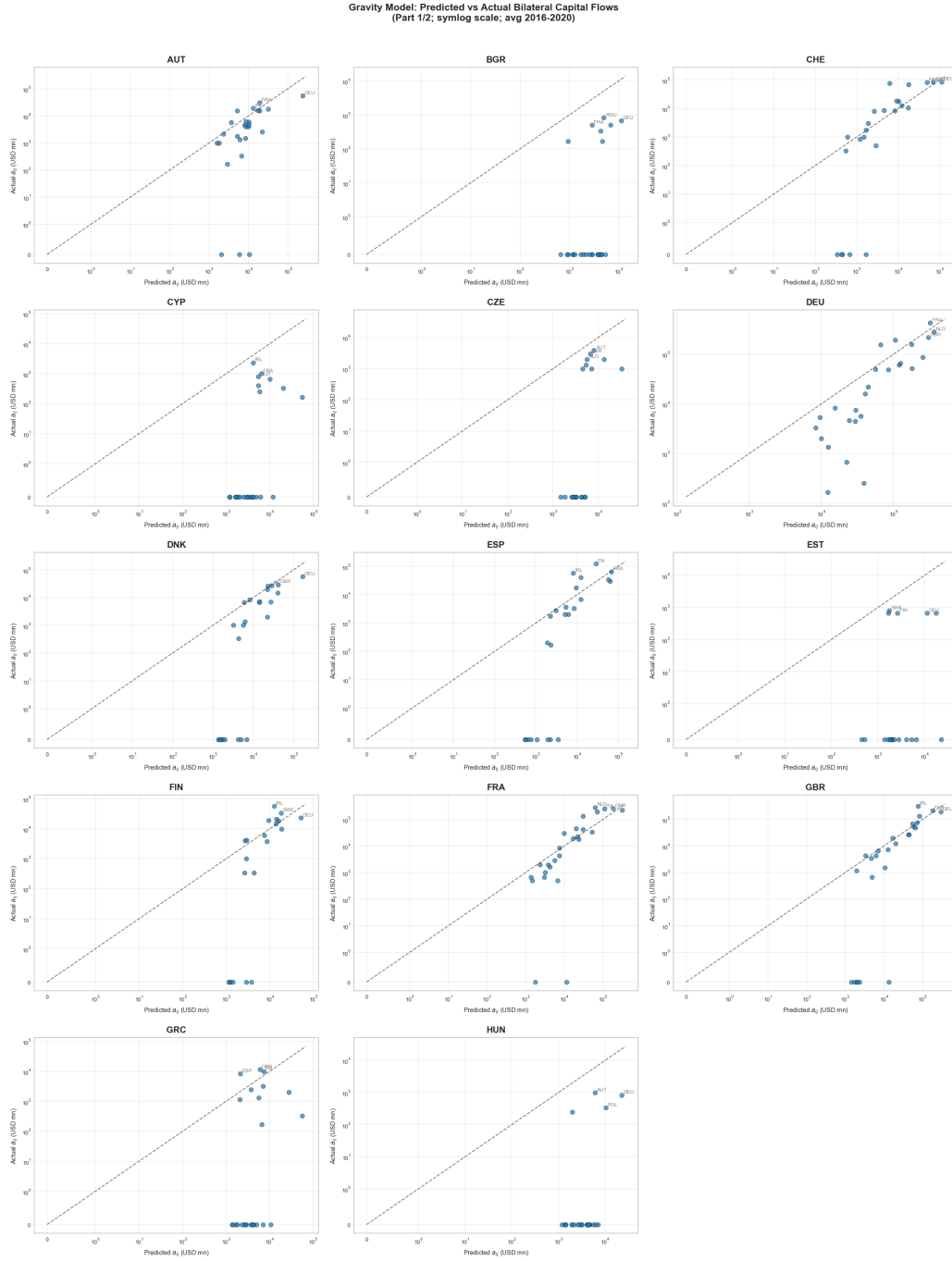


Figure 18: PPML regression predicted vs actual values.

Gravity Model: Predicted vs Actual Bilateral Capital Flows
(Part 2/2; symlog scale; avg 2016-2020)

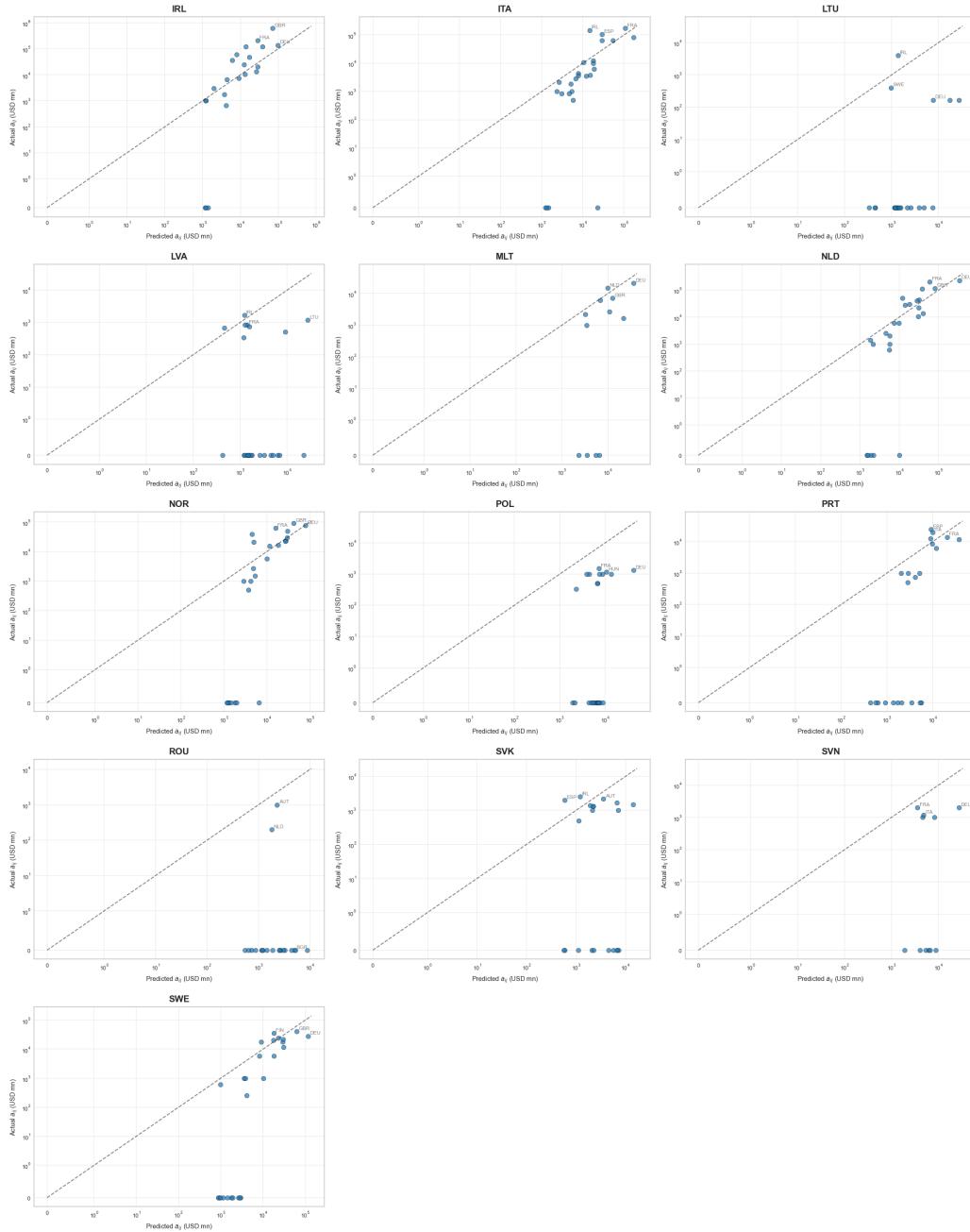


Figure 19: PPML regression predicted vs actual values (continued).

These figures compare the predicted values from the PPML regression with the actual observed values of bilateral portfolio holdings. Each point in the scatter plot represents a bilateral pair of countries, with the x-axis representing the predicted values and the y-axis representing the actual values. The closer the

points are to the 45-degree line (where predicted equals actual), the better the model's predictions align with the observed data.